

INTERSECTIONS

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Michelle Verghese



Elena Slobodyanyuk

It goes without saying that the year of 2020 has been unprecedented. A global pandemic claimed the lives and health of millions of individuals, wrought economic devastation to households, businesses, and nations alike, and dramatically restructured everyday life for most people. The harrowing murders of Black individuals ignited a resonant movement to address deep-rooted issues of racial bias, police brutality, and social injustice. Together, these events largely defined the tumultuous narrative of 2020.

Certainly, these events did not take place in a vacuum. In an increasingly globalized society, we saw the rapid spread of the novel coronavirus across continents and the rapid exchange of information across media outlets. Furthermore, we experienced the synergy of various disciplines and communities coming together to confront these issues. In a period of social distancing, the salience of intersections is particularly clear.

In this issue, our writers explore some of the fascinating intersections that are shaping the modern world. At the atomic level, consider the junction of semiconductor physics with biomedical imaging through the development of quantum dots, or the dawn of bio-inspired smart materials in the form of self-healing polymers. On a broader scale, our writers review the efforts of public health interventions to achieve global salt iodization, and an independent undergraduate research study investigates the effect of urbanization on the abundance of disease vector mosquitoes in French Polynesia. Notably, drawing on the rapidly evolving wealth of expertise at UC Berkeley, we present an interview with several prominent figures in public health and infectious disease research with their insights on the COVID-19 pandemic.

Looking inward, BSJ has reinforced its awareness of the role of science journalism in the contemporary world. Early this semester, our writers engaged in a thoughtful discussion with Caroline Kane, Professor Emerita and BSJ faculty advisor, about the social responsibilities of science communicators. Following the transition to remote learning, our writers participated in a virtual science writing workshop led by Haley McCausland of the graduate journal *Berkeley Science Review*.

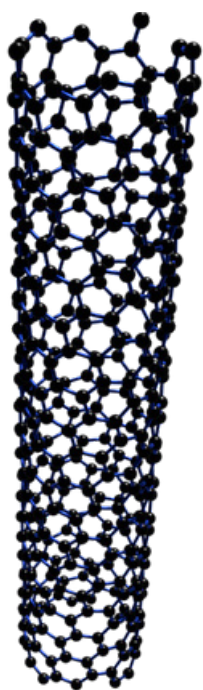
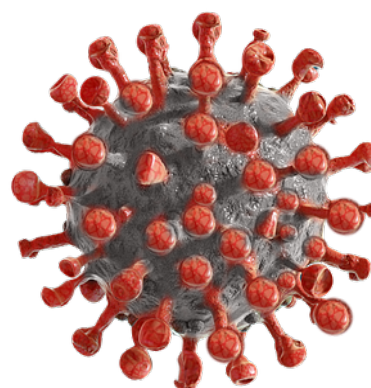
With science journalism at the nexus of scientific discovery and public dialogue, we recognize our duty to uphold reliable, responsible, and accessible reporting. We are proud to present this semester's issue of the *Berkeley Scientific Journal*.

Elena Slobodyanyuk
Managing Editor

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COVID-19: Insights from UC Berkeley's Infectious Disease Experts

By Sharon Binoy, Ananya Krishnapura, Esther Lim, Elettra Preosti, Melanie Russo, Michael Xiong, Katheryn Zhou, Matthew Colbert, and Rosa Lee

COVID-19, the disease caused by the novel coronavirus, SARS-CoV-2, has spread rapidly across our planet and changed our lives in unimaginable ways. In this interview in April 2020, we sought out information from experts in emerging diseases, public health, and epidemiology here at UC Berkeley.

INTRODUCTIONS



SANDRA MCCOY¹

Sandra McCoy, PhD, MPH, is an Associate Professor in the Division of Epidemiology and Biostatistics at the UC Berkeley School of Public Health. She is on the editorial board of the journal *mHealth*. Her research focuses on utilizing technology and behavioral science to advance healthcare access and treatment adherence across different countries, cultures, and populations. Dr. McCoy has led projects in Tanzania and the United States to combat HIV infections and improve reproductive health.

ARTHUR REINGOLD²

Arthur L. Reingold, MD, is the Division Head of Epidemiology and Biostatistics at the UC Berkeley School of Public Health. He has studied the spread and prevention of infectious diseases for over 40 years. He worked at the Centers for Disease Control (CDC) for eight years, serving as the Assistant Chief of the Respiratory & Special Pathogens Epidemiology Branch from 1981-85 and as the CDC Liaison Officer of the Office of the Director from 1985-87. His current research interests include vaccination and vaccine-preventable diseases, outbreak detection and response, and the emergence of infections in the United States and developing countries.

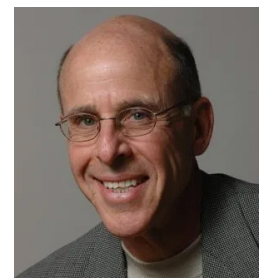


JULIA SCHALETZKY³

Julia Schaletzky, PhD, is the Executive Director of the Henry Wheeler Center for Emerging and Neglected Diseases (CEND). She is also the co-founder of the COVID Catalyst Fund, which aims to provide rapid funding to accelerate COVID-19 research within the Bay Area Virus Network. Before joining CEND, Dr. Schaletzky worked at the biotechnology company Cytokinetics, where she focused on developing first-in-class medicines against heart failure and neurodegenerative disorders. Dr. Schaletzky's current research focuses on treating neglected and emerging diseases, establishing effective collaboration between academia and industry, and translating basic science into new companies and ultimately cures.

JOHN SWARTZBERG⁴

John Swartzberg, MD, FACP, is an infectious disease specialist and Clinical Professor Emeritus at the UC Berkeley School of Public Health. Before becoming a faculty member, Dr. Swartzberg had 30 years of clinical experience working in internal medicine. He is currently the hospital epidemiologist and Chair of the Infection Control Committee at the Alta Bates Medical Center, a chair of UC Berkeley School of Public Health's Health and Wellness Publication editorial board, and a past director of the UC Berkeley–UCSF Joint Medical Program.



BSJ: How does SARS-CoV-2 spread?

REINGOLD: This is a virus that is spread from and to the mucous membranes of the respiratory tract. That can also potentially include the eyes and the nose, in addition to the mouth. So, we basically worry about the virus making it from secretions in my mouth, my nose, or possibly my eye into your nose, mouth, or eye.

SCHALETZKY: It spreads through surfaces and droplets, and it is also aerosolized, meaning it floats in the air for a little bit. So, if you are in a room with a lot of people and shared air, like an airplane, it can stay in the air for a while. Also, initially we thought that when you cough, you only make droplets fall within a six-foot radius around you. However, the virus can be detected from even 23 feet away because it is aerosolized and can waft in the air, although at lower concentrations. So, spread is not just caused by droplets. However, one particle is not going to infect you; you have to reach a certain critical mass of particles in order to get an infection.

BSJ: How does its rate of spread compare to the seasonal influenza or other diseases?

SWARTZBERG: We use this term called R_0 , which is the reproductive number. The R_0 for SARS-CoV-2 was around 3 in March. This means one person, on average, is going to infect three people. That compares to an R_0 for seasonal influenza from this year of about 1.8, and we see how many people got influenza. But, the R_0 is an elusive number. By late April, it was below 1.0. It does not take an R_0 that is very high to get an awful lot of people infected. The most infectious, contagious disease that we deal with currently is measles, and the R_0 for that is around 15. These R_0 values show you how much less contagious SARS-CoV-2 is than measles, but how much more contagious it is than seasonal influenza.

McCoy: An issue that helps determine mitigation and control measures is the level of transmission that occurs before symptoms develop or from people who never develop symptoms. For influenza, people become contagious about one day before they develop symptoms. This means that infected individuals are going through their daily lives and could be unknowingly transmitting to others. For the 2003 SARS coronavirus, people were not able to transmit to others until after they had symptoms. This meant that a combination approach of case finding, isolation, contact tracing, and quarantine was enough to interrupt the 2003 SARS epidemic. However, some people infected with SARS-CoV-2 have peak viral shedding before they develop symptoms, and potentially up to 25% of all people infected never develop symptoms at all. This means that we have to take much more aggressive control strategies.

BSJ: How many people do you estimate have actually been infected as compared to the number of documented cases [as of April 18, 2020]?

SWARTZBERG: We know that this virus did not enter the United States until sometime in early January. We also know that if you have a positive antibody test, you have been infected sometime in

the last three and a half months. So, initially, I would have told you we have 10 times more cases than are reported. However, a study came out of Stanford recently, surveying a few thousand people in Santa Clara County. There are a lot of valid criticisms in terms of how well their antibody tests work, whether they used the best representative sample, and so on. But, putting those criticisms aside, what they found was that the number of people who had antibodies was 80 to 85 times greater than the number of cases reported. So, if their data is accurate, it means we have between 50 and 85 times more cases than we thought we had. This paper tells us that there are a lot more cases out there than we know about. How many more is still a question, but I think a conservative number is 10.

BSJ: With over 900,000 confirmed cases and 50,000 deaths⁵ [as of April 26, 2020], how effective have the U.S.' attempts been to "flatten the curve"?

SWARTZBERG: You cannot speak of the United States in the aggregate because, like any pandemic, it flares up in a community, dies down, and then flares up in another community. You have to look at specific communities to answer your question. If we look at the San Francisco Bay Area, there is no question that the number of cases we had in March and April is far fewer than were predicted. We never got a spike or a peak; we got a hill. We never exceeded the medical resources in the Bay Area that we needed. Clearly, something that we did here worked. New York was not successful in flattening the curve; they have had an incredible spike in cases. What is the difference? One hypothesis that I think is really tenable (but, is not sufficient in itself) is that the Bay Area was the first large metropolitan area in the United States to shelter-in-place. New York followed between five to seven days after California or almost 10 to 12 days after the Bay Area. Although five to seven days does not sound like a lot, in the rate of spread of this virus, it is a lifetime.

BSJ: What are the current preventative measures against COVID-19, and what are some long-term projections?

REINGOLD: Ways to prevent spread include various social distancing measures such as covering your mouth when you cough and sneeze, staying home when you are sick, washing your hands, and wearing a mask. Once a virus like this starts to spread, the sooner you introduce prevention and control measures, the more likely they are to be effective.

McCoy: Right now we are implementing a very aggressive and intensive community-based approach. However, our goal is to eventually transition to a case-based approach where we can ease social distancing policies and rely on intensive case identification, isolation, contact tracing, and quarantine. One proposal is a series of phases spanning 12 to 18 months where we repeatedly tighten and then loosen physical distancing directives according to the level of virus transmission in the community. We would have smaller and smaller epidemic curves as we gradually dampen the epidemic. But until we have a safe and effective vaccine or therapeutic, we cannot go back to life as we previously knew it.

SCHALETZKY: For the pandemic to stop spreading, we need people

to either get it and become immune, or we need to have a vaccine or a therapy. It is also important to remember that the regulating step in this is really hospital capacity. We do all of this social distancing because we do not have the hospital capacity to deal with a scenario where everybody gets infected at the same time. Even if everyone got better in two weeks, it would not be possible. For that reason, we want to flatten the curve. It increases the survival of patients. But, although quarantine is a good method, we cannot do this forever. If we were all on lockdown and nobody was interacting with each other, we could in theory eradicate the virus, but that is impractical. So ultimately, people need to build immunity by either getting the virus and getting optimal care so the survival rate is maximized, or we get a vaccine out really fast. It is a balancing act.

BSJ: Currently, how accurate are the diagnostic tests for COVID-19?

REINGOLD: There are basically two types of tests people are interested in: do you have the virus now and have you had the virus in the past. The “do you have it now” tests you hear about are polymerase chain reaction (PCR) tests that detect the viral RNA. But, RNA might still be present when the virus itself is not there. So, there is a slight problem with using PCR tests because it could be that you are no longer infectious, but still have the RNA detected. Another problem is that most PCR tests currently take a day or two to get the results back. The third problem is that they are not very sensitive, so there may be a lot of false negatives. You can also do viral cultures, but they are much more expensive, difficult, and laborious, so they are not routinely done out in the real world. Thus, people are currently working really hard to do point-of-care diagnostic tests where, just like a pregnancy test or a test for group A strep pharyngitis, I take your sample and 15 minutes later, I tell you the result. That is what we need.

People are also developing and starting to use antibody tests, called serological studies, where I take some of your blood and I look for an immune response. If there is an IgM response, that means it is a recent infection. If it is an IgG response, it means that the infection occurred longer ago, and we can test for antibodies, immunity, or both.

BSJ: What does the timeline look like with respect to potential vaccines or treatments?

SWARTZBERG: There are quite a few therapeutic drugs that are being tested, but there is nothing that looks like it is going to be a panacea. There are also quite a few vaccines being developed in the United States and over 70 internationally. Two major things have to be established in vaccine trials: safety and efficacy. There is currently one vaccine in a Phase 1 trial testing for safety in a small group. If it passes the Phase 1 trial, then we have to determine whether it is efficacious, first in a small group, then in a larger group. Let us say we find one or more of these vaccines to be efficacious. We then have to expose larger groups of people to also make sure they are safe. To do this, you give a group of people the vaccine and follow them for as long as you possibly can. If it is efficacious, but not safe, it is no good. If it is safe, but not efficacious, it is no good. If we are

lucky, we can have a vaccine in 14 to 18 months. It will be the fastest we have ever developed a vaccine by years.

BSJ: Of the different types of vaccines—live attenuated or inactivated (including conjugate, subunit, toxoid, or recombinant)—which one is most suitable for combating COVID-19, and why?

SCHALETZKY: The most well-known or original way to make vaccines is to attenuate the virus so it cannot be fully infectious. But in the SARS epidemic, we saw morbidity associated with the live attenuated vaccine, possibly due to the lung-specific pathology the SARS coronavirus causes in the immune system. We do not know for sure if that will also occur for COVID-19, but that is why people are a little wary of live attenuated vaccines. The next most frequent approach is to make a recombinant vaccine where you purify an antigen [a foreign substance that elicits an immune response in the body] in some other system like yeast or plants. Johnson & Johnson is working on one and has already done work on mice. They estimate that they may be able to manufacture a vaccine by summer 2021 in the best case scenario. The mRNA vaccine is a little different. Instead of making the antigen already in a system, you inject the genetic information into you so that your own body can make the antigen. The advantages are that you can make it a lot faster because a nucleotide synthesis is in principle simpler. Moderna is developing an mRNA-based vaccine. They already dosed people weeks ago with the first dose, and so far we have not heard anything about adverse effects.

REINGOLD: We do have a live attenuated vaccine for influenza and for some other viruses, but I do not think that it is very likely for COVID-19 because you could potentially end up not attenuating it enough and causing the disease instead. The vaccine is also pretty unlikely to be a killed, inactivated whole virus vaccine; nowadays, there are much more modern approaches, such as using DNA or RNA vaccines. My guess is that the vaccine that an Oxford group is working on, which is a carrier virus that has coronavirus antigens being expressed, may work. You take a virus that you know cannot harm people and put certain genes from the virus you are trying to protect against into that backbone. These genes are for antigens that will produce the antibodies we want to protect you with.

BSJ: What are some of the challenges faced by scientists who are trying to create a COVID-19 vaccine?

SCHALETZKY: There are a few issues surrounding the mRNA vaccine. One is does it actually elicit an immune response? If your body for some reason does not make the protein very well, or the mRNA is degraded or is not taken up by your cells in the proper way, you might not actually make enough antigens. The other is if you make enough antigen, do you actually make any antibodies and are those antibodies actually protective?

One of the biggest non-scientific obstacles in our fight against COVID-19 is having too many regulations. Now we have testing capacity, yet there is so much over-regulation in the US that makes it impossible to do simple community testing of who has

SWARTZBERG: The major challenge we have with COVID-19 is that we do not know what correlates with protection. There are infectious diseases where our bodies respond to the infection by producing antibodies, but they do not correlate with protection. We do not know whether the humoral immune response and antibody production will be the major way to protect us or if it is going to be the cell-mediated immune response or some combination of the two.

REINGOLD: The \$64 question—that is an old saying—is do these tests tell you anything about your immunity? An assumption made by many people is that if you have antibodies, then you are now immune. But, it is going to take some work to figure out if the antibodies that we are measuring correlate with clinical protection or not. For example, some people are talking about using a positive antibody test as proof of immunity. We also do not know how long immunity will last. You might not still be protected after some time.

BSJ: In your opinion, when will it be safe for students and professors to return to campus?

SWARTZBERG: I do not know what is going to happen in the fall semester because other respiratory viruses, especially influenza, start circulating in our area right around late October through April. So, we are going to add on the annual influenza epidemic that occurs every year along with whatever else starts coming to us at that time. However, when we do attempt to bring people back to campus, we will need to get some of the researchers back to have some of the labs open up. But, who should those researchers be? Once we start reintroducing people back on campus, then who would be the students to be reintroduced if we are not going to let all students come back once? Where will students live? Everybody wants to be back there, so how are you going to prioritize that?

REINGOLD: There will definitely be a fall semester. I sit in on the calls with the Chancellor and Vice Chancellor three mornings a week. We are definitely planning to have school in the fall. The question is how much can be on-campus, in-person education and how much will be online. I do not know the answer to that and neither does anybody else, including the Chancellor and the Vice Chancellor. I am working with the Tang Center and the City of Berkeley on the issue of tracking who might have become infected and preventing infected people from transmitting the disease to others once we do start letting people be more out and about. All that preparation is underway. Anybody who says that he or she can predict what is going to happen three or four months from now, I think is delusional. They might get it right by chance, but I am not smart enough to know the answer to that question.

1. Sandra McCoy [Photograph]. Retrieved from <https://publichealth.berkeley.edu/people/sandra-mccoy/>
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MARINE SNOW: THE LARGEST SNOWFALL ON EARTH

BY LILIAN ELOYAN

Imagine if you could dive straight into the ocean and just keep swimming down, unrestrained by physical limitations. At first, you'd be in a world of wonder, surrounded by a plethora of unique shapes and colors: vibrant fish popping out from every corner of coral, anemones and seaweeds drifting in the current. The water would be warm, saturated by the sun's rays, and for this reason the pelagic zone of the ocean is bursting with life. Once you pass two hundred meters underwater, however, things start to look a little different. Entering the mesopelagic zone, also known as the twilight zone, the sunlight begins to fade and all of a sudden you hit a wall of cold water, the permanent thermocline, where water temperature rapidly drops below 43 degrees Fahrenheit.¹ Somewhere between 200 and 1,000 meters you'll enter the oxygen minimum zone, where oxygen saturation in the water column is at its lowest. The reduced amount of light means fewer photosynthetic organisms like plants and cyanobacteria. No photosynthesis means no oxygen, and no oxygen means no life. A ghost world.

The farther down you swim, the further away the concepts like night and day and seasons begin to drift. But then, to your surprise, you see—with eyes equipped with night vision, of course—a burst of life, movement in the darkness. Passing a thousand meters, you reach the bathyal layer of the ocean, or the midnight zone; life begins to thrive again despite the absence of even a single ray of sunlight. Even deeper, at 4,000 meters, you'll reach the abyss where life is still abundant despite the seemingly uninhabitable conditions. The increase in life here is due to thermohaline circulation, or the ocean conveyor belt, a system of deep-ocean water circulation through which oxygen-rich water from the surface sinks down to the ocean floor, supplying animals with life-sustaining O_2 .¹ And lastly, you'll enter the hadal layer of the ocean or the trenches. True to its name, it is the underworld where strange is the norm.

LIFE WITHOUT SUNLIGHT

In the deep sea, biomass, or the total combined mass of individuals per species, is relatively low compared to that of the surface regions, but the biodiversity is unmatched in the ocean.¹ Creatures of the deep have evolved over billions of years to perfectly suit life over a thousand meters beneath the ocean's surface. Since photosynthesis is impossible here, some fascinating

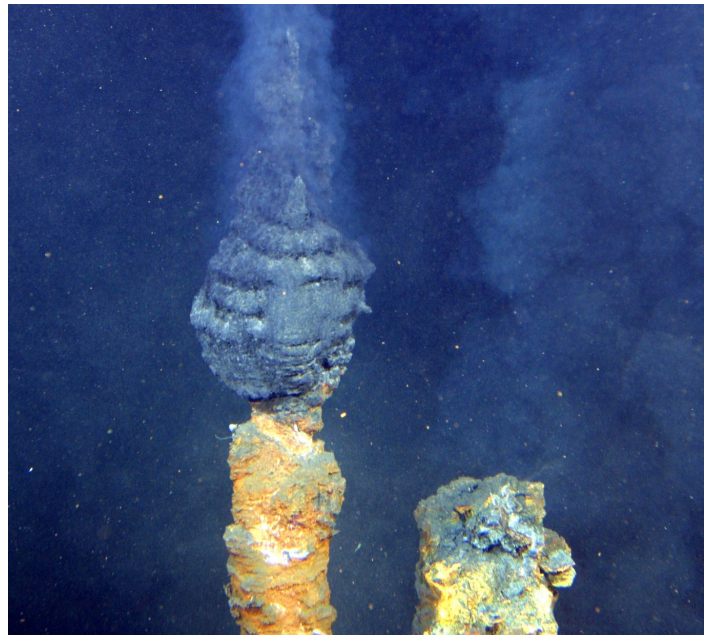


Figure 1: Black smoker hydrothermal vents located in New Zealand's southern caldera secreting hot water full of minerals such as iron sulfide, which chemosynthetic bacteria can use to extract energy.

"A few billion years ago, one bacterium's mutation allowed it a much more efficient means of deriving energy: from the sun."

creatures use another method of extracting energy from the world around them: chemosynthesis. Before cyanobacteria came non-photosynthetic bacteria.² In the early days of the Earth, primitive lifeforms used the concoction of gases in the Earth's atmosphere to extract energy from chemical reactions (Fig. 1).² Just as baking soda and vinegar create energy for a volcano science project, chemical energy was used by some of the first-ever life forms on Earth. A few billion years ago, one bacterium's mutation allowed it a much more efficient means of deriving energy: from the sun.² The rest is history. Today, sulfur chemoautotrophic bacteria still extract energy from the hydrogen sulfide pockets found in the Earth's crust that erupt through hydrothermal vents at the seafloor.³ These bacteria form the basis of unique food webs where species richness is low, but biomass is teeming. They provide food for species such as hydrothermal vent shrimp and tubeworms that are endemic to these nutrient-rich areas.³

MARINE SNOW

However, such evolutionary adaptations as those apparent in chemosynthetic bacteria are few and far between as most life in the deep depends solely on what drifts down from above. Marine snowfall occurs in the ocean year-round, but unlike snow on land, it is not composed of soft, beautiful ice crystals. Rather, it is composed of biological debris, mainly the remains of plankton and fecal pellets that clump together to form particulate organic matter (POM) as they drift down.⁴ This phenomenon may not sound important to us, but for creatures in the deep, it is indispensable. At the surface of the ocean, photosynthetic organisms form the basis of the food web, but in the deep sea, most animals rely on marine snow for food.⁴ Many animals like the giant larvacean have adapted unique strategies for catching this marine snow to eat.⁵ Contrary to its name, the giant larvacean, a tadpole-like filter feeder, is only about four inches wide, yet it casts a mucus net that spans over a meter, collecting marine snow as it falls (Fig. 2).⁵ Once the net gets too heavy to bear, the larvacean discards it and the net then acts as food for other deep-sea dwellers like the vampire squid.⁵

POM is part of the biogeochemical process known as the carbon cycle (Fig. 3). The plankton or microscopic crustaceans that make up most of the POM contain carbon in their bodies due to the ocean's absorption of carbon dioxide from the atmosphere.⁶ Once these organisms are consumed by predators or begin to decompose, the carbon is released back into the atmosphere as CO₂.⁶ Any unused or unconsumed particles of POM, however, drift to the seafloor, sometimes taking weeks to reach the bottom. The POM piles up on the seafloor and forms a thick, carbon-dense topsoil layer of sludge.⁶ This sludge hardens into sediment deposits of limestone where carbon is locked away through a process called mineral carbonation, eventually covering over one billion square miles of the seafloor and becoming several hundred meters thick.⁷ Thus, marine snow is a natural form of carbon sequestration. By



Figure 2: A giant sea larvacean (*Bathochordaeus charon*) and the mucus net it constructed to collect the falling marine snow, its main food source.

effectively dragging carbon down to the ocean floor, it can be tucked away in rock for millions of years, only returning to the atmosphere through periodic volcanic eruptions.⁷

Over time, marine snow stores more and more carbon in the oceans' depths. According to a 2019 modeling study, in the Cretaceous period about 80 million years ago, roughly one million tons of carbon in the form of carbonate were stored each year in deep-sea sediments.⁸ These carbon stores were responsible for cooling the Earth's climate enough to sustain life for a number of animals, most notably dinosaurs.⁸ Today, scientists estimate that over 200 million tons are deposited and stored annually.⁸ The underwater sediments formed by marine snow are the world's largest means of carbon storage, and if it weren't for this natural cycle, the amount of carbon dioxide in the atmosphere would make Earth too warm for habitation due to the greenhouse effect.⁹ Carbon dioxide in the atmosphere acts as a greenhouse gas, absorbing and trapping heat, making the Earth's climate warmer.⁹ Today, by burning fossil fuels and cutting down carbon dioxide-absorbing forests, we reverse the natural cycles that have helped make Earth a comfortable home for all life forms. The ocean soaks up the excess carbon dioxide produced by humans, effectively absorbing 90% of the planet's excess heat.¹⁰ With more carbon dioxide in the atmosphere, however, oceanic carbon sequestration is disrupted as increased oceanic CO₂ levels raise the water's acidity, harming the plankton community and decreasing the amount of falling marine snow.^{10,11} Although we may feel utterly disconnected from what lies beneath the waves, what occurs up to 11,000 meters deep in the ocean affects us in more ways than most of us realize.

As you start to make your ascent, swimming past angler fish and other various bioluminescent beasts, avoiding some sharks, then finally back to warm, sunny, waters full of clownfish and turtles, you'll eventually be met with a refreshing burst of cool air once you break through the surface of the water, partly thanks to marine snow.

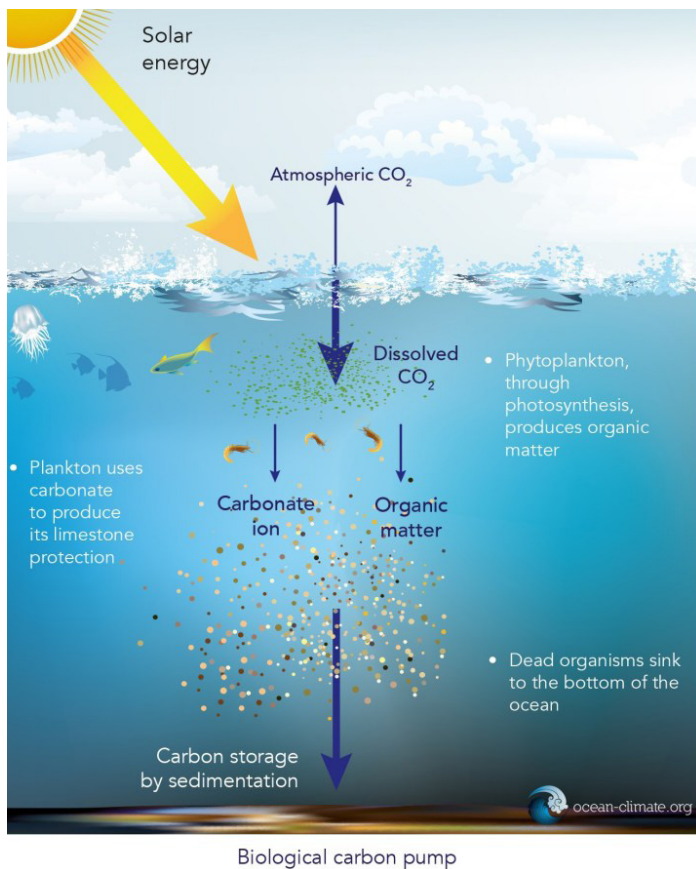


Figure 3: The ocean's biological carbon pump is a key piece of the Earth's carbon cycle as carbon-rich particulate organic matter sinks to the seafloor.

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GLOBALIZING IODINE THROUGH PUBLIC HEALTH

BY JESSICA JEN

By early spring of 2020, the SARS-CoV-2 outbreak, also known as COVID-19, was reaching an unprecedented scale. Amidst the stressful unease, massive efforts were undertaken to contain and combat the pandemic. In-person contact was decreasing, events were cancelled, and it seemed that to handle the outbreak, the world was on pause. Researchers were scrambling to develop vaccines and hospitals were at their tipping points, evidence of the lengths being taken to preserve as many lives as possible. Numerous public health interventions were taking effect worldwide, with nations executing a number of distinct responses.

Public health is the union of science, policy, and community acting in harmony to minimize the onset of disease in the largest number of people.¹ In other words, public health is about protecting people's health at the population level.² This union becomes evident during a widespread outbreak when containment, epidemiological analysis, pharmaceutical development, economic effects, and even policy changes become familiar aspects of the outbreak response.

But beyond outbreak responses is the cooperation between disciplines to tackle issues and implement solutions. Such responses have been crucial in managing public health crises for a number of centuries, becoming more complex with advances in scientific thought such as epidemiology in the nineteenth century.³ The global introduction of iodized salt to resolve iodine deficiencies in particular illustrates the complexity and ramifications of public health interventions.

IODINE SUPPLEMENTATION

Iodine is an essential mineral in the human diet. Those with iodine deficiencies experience thyroid disruptions manifesting as impaired brain development in the fetal stage and otherwise as goiter, an abnormally enlarged thyroid (Fig. 2).^{4,5} The link between iodine deficiencies and goiter was first hypothesized in 1852 by French chemist Adolphe Chatin, leading scientists to begin advocating for iodine supplementation over the following decades.⁶ The introduction of iodized salt beginning in the early

20th century resulted in a sharp drop in cases of iodine deficiency disorders (IDD) and the rise of salt as a primary source of iodine.

The success of salt iodization, however, was in part colored by geographical location: programs were more effective in some regions of the world than in others. Ensuring the success of any preventative measure, after all, involves more than creating a product (such as iodized salt). The political agreement that public health programs rely on to be successful is dependent first on recognition of the problem, and second, on

"The global introduction of iodized salt to resolve iodine deficiencies in particular illustrates the complexity and ramifications of public health interventions."

epidemiological data and community backing, according to prominent IDD researcher Basil Hetzel.⁷ Hetzel studied this process of developing effective public health initiatives in Indonesia where the use of steady community feedback to pull continued political interest was widely successful.

Indonesia's salt iodization efforts first emerged for a brief two decades under Dutch colonization in the early 1900s, but the end of Dutch administration in the 1940s caused these efforts to phase out.⁸ In the 1970s, renewed efforts at widespread iodized salt distribution and iodized oil injections appeared promising. Levels of detected iodine in individuals increased to a healthy range in two years; the percentage of children born with thyroid-related neurological disorders fell from seven percent to zero; and Indonesian legislators proscribed non-iodized salt in 1983. Nevertheless, the program ultimately failed due to a lack of internal government coordination and accurate population data.^{7,8} A third attempt in the 1990s also showed some success—with assistance from UNICEF and the World Bank, the Indonesian government established a nationwide program to increase access to iodized salt and monitor iodine levels in the population. As a result of this program, child goiter rates fell by 30 percent between 1980 and 1998, and the consumption of iodized salt at the household level rose by three percent from 1995 to 1999. However, despite clear improvements in people's health, this initiative's effects were limited by inadequate

"The overall failure in achieving nation-wide salt iodization was most likely due to both a lack in coordination between trade and health departments and a lack of incentive for salt producers to iodize their products, as it was more profitable to sell raw salt directly than through a middleman who could iodize it."

enforcement of salt legislation and a financial crisis. The overall failure in achieving nation-wide salt iodization was most likely due to both a lack in coordination between trade and health departments and a lack of incentive for salt producers to iodize their products, as it was more profitable to sell raw salt directly than through a middleman who could iodize it.

Indonesia was not alone in losing iodization efforts to political and economic stumblings, as evidenced by a drop in Brazil's salt iodization in 1974.⁹ A lack of widespread awareness, funding, and poor control over proper salt production led to a sharp rise in endemic goiter rates, forcing Brazil's Ministry of Health to renew its salt iodization programs in 1983. In an effort to combat this increase in disease, salt producers were offered proper equipment, inspectors were trained to manage the process of iodization, and laboratories were set up to monitor iodine levels; as a result, av-

erage iodine levels in the tested population increased to a healthy range over 10 years.

The late 20th century saw similar salt iodization programs emerging across the globe. The International Council for Control of Iodine Deficiency Disorders, now known as the Iodine Global Network (IGN), has assisted countries in eradicating IDD since its inception in 1985 by communicating research on IDD's severe health impediments to health ministries.¹⁰ Once governments realized that suitable iodine levels boosted productivity, quality of life, and therefore the economy, many established national standards and enforcement practices for salt iodization.

The IGN has frequently partnered with organizations such as the WHO and UNICEF to act as a bridge between scientific research on IDD and the implementation of global solutions like iodized salt.⁹ In China, salt iodization programs have also shown positive results.⁸ China's health of-



Figure 1: *The Khewra Salt Mine, a major producer of rock salt and brine, is a popular tourist destination and the second largest salt mine worldwide.*

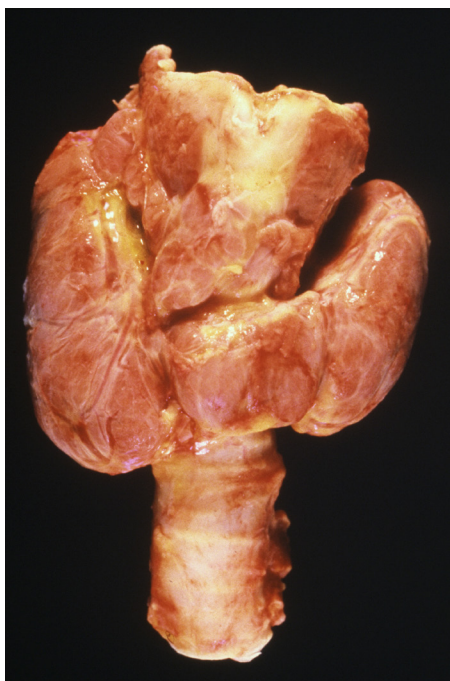


Figure 2: An enlarged thyroid that would have resulted in a goiter, extracted during an autopsy.

ficials and salt producers advocated IDD awareness in tandem with local governors monitoring iodine production and enforcing sales of proper salt. As a result, both the quantity and quality of iodized salt consumption increased from 1995 to 1999. National consumption of iodized salt rose to nearly 95 percent, parts per million of iodine in salt nearly tripled, and goiter rates were slashed in half in the four-year period. These positive results were largely due to improved iodine availability, centralized salt production, and close surveillance by health officials and local governments. The various methods of salt iodization efforts in Indonesia, Brazil, and China have shown the adroitness required to implement public health interventions and the coordination necessary for their successes.

BEYOND IODINE

Global efforts to provide adequate access to iodine have overcome a myriad of obstacles to reduce IDD. Most programs established in the 20th century have shown some measure of success, yet there is always work to be done. Two billion people worldwide still suffer from IDD as of 2017.¹¹ However, the stories of preventing

IDD using salt iodization provide uplifting testimonials of success, evidence that large-scale execution of preventative public health measures can be effective with political, technical, and societal changes.⁸

While the ongoing COVID-19 pandemic is more urgent than IDD, similar tactics are being applied to current public health interventions. Just as governments endorsed iodized salt, they are encouraging citizens to self-quarantine; instead of salt refining, funds are being directed towards hospital equipment. Although the pressing nature of the viral outbreak garners more media coverage than a chronic nutritional deficiency, both crises demonstrate the intricacy and scope of public health measures necessary for effectively preserving global health.

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UNDERSTANDING HUMAN COGNITION: THE COGNITIVE NEUROSCIENCE OF MOTOR LEARNING

Interview with
Professor Richard
Ivry



Professor Richard Ivry¹

By Sharon Binoy, Ananya Krishnapura,
Esther Lim, Michael Xiong, and Rosa Lee

Dr. Richard Ivry is a Professor of Psychology and Neuroscience in the Department of Psychology and the Helen Wills Neuroscience Institute at the University of California, Berkeley. He directs the Cognition and Action (CognAc) Lab. The CognAc Lab employs a diverse array of approaches, from classic psychological experiments to computational modeling and fMRI, to explore the cognitive neuroscience that underlies skilled movement. In this interview, we discuss his findings on the influence of task outcome in implicit motor learning and the neural signatures of prediction errors.

BSJ: What is cognitive neuroscience?

RI: Well, I'm trained as a cognitive psychologist. Cognitive psychology is building models of the mind: how do we think, how do we use language, and how do we perceive objects? These questions are usually addressed in terms of psychological processes. Cognitive neuroscience was a real new push in the early 1980s which recognized that we can use neuroscience methods not just to describe what part of the brain does something, but actually use that neuroscientific information to help shape our psychological theories.

So, whereas behavioral neurology is solely interested in what part of the brain does what, cognitive neuroscience is the idea that not only can we use insights from our psychological experiments to understand how the brain works, but we can also then take the insights from studying the brain to build psychological theories. That's the essence of cognitive neuroscience—it is a bidirectional interest.

BSJ: What initially drew you to the field?

RI: I was in the right place at the right time. I entered graduate school in 1982 at the University of Oregon. One of the pioneers in the cognitive neuroscience movement, Mike Posner, was there (and is still there) as a professor. He was just starting to do research with people who had brain lesions, and he worked with stroke patients who had disorders of attention. He was interested in showing how you could use sophisticated cognitive experiments to build models of attention.

I was in the right place to not only learn from him, but to realize that we could apply a similar strategy in studying movement disorders to understand how people perform skilled movements. I just landed happily in a graduate program as things were taking off.

BSJ: We read your paper on the influence of task outcome on implicit motor learning. Could you define implicit versus explicit learning for our readers?

RI: Tracing it all the way back to Freud, one of the most fundamental distinctions in psychology research is the notion that much of our mental life is occurring subconsciously,

as opposed to consciously. Freud tended to frame this distinction in terms of a battle between the conscious and the unconscious. The unconscious was all your desires, and the conscious was the way to keep a check on things.

But I don't think it's two armies battling each other, although I think it's quite obvious that we're only aware of a limited amount of our mental activity. Right now, I'm sure that everything in this room is activating sensory systems. Probably percolating somewhere in my brain is the thought of getting ready for my first Zoom class at one o'clock this afternoon. I'm only aware of a limited amount of that information. So that's the fundamental question: why is there a limit on what we're aware of?

Setting that aside, our interest in human performance, at least in terms of motor control, is recognizing that when we learn a new skill, much of the learning is implicit. For instance, we can certainly benefit from coaching. In baseball, the coach tells you how to orient your shoulder to the pitcher, how to hold the bat, and so on. This information is the essence of something being explicit—I'm aware of it. If I'm aware of it, I can tell someone else about it. But a lot of that skill learning is really implicit, as in, "I can't quite put my finger on it." The classic example here is bicycle riding. It's really hard to tell a person how to ride a bicycle. You can try to coach someone, but in the end, it basically comes down to giving them a shove and letting them go across the playground. The body figures it out.

Lots of our memory is implicit. I'm not aware of all the different memories that are being activated; they're all just churning around and away in my brain. As you say a word, all the things I associate with it get activated, but I will only become aware of some of that information. So whatever domain you study, whether it's tension, memory, or motor control, there are always things happening at both the explicit and the implicit level. What we observe when someone performs a skilled behavior is the sum of those processes. Our research has been aimed at trying to dissect these skilled behaviors to determine the characteristics of what you can learn implicitly versus explicitly, as well as the brain systems that are essential for one type versus the other and how they interact.

BSJ: Some previous studies concluded that reward has no effect on the rate of learning, while others concluded that reward does have an effect. What do you think led to these inconsistencies?

RI: In the classic case, experiments that study these effects perturb the world in some way. In the laboratory we would certainly like to study real, natural skill development, but that's a pretty difficult process. So, we usually try to make more contrived situations where we can accelerate that learning process, to do it within the confines of a one-hour experiment. In any learning situation, it's quite likely that you could have both explicit and implicit processes operating, and it may be that reward only affects one of those two. So one account of the inconsistency is that in the older literature, they didn't really have methods to separate the different contributions of the different learning systems.

BSJ: What is clamped visual feedback, and how did you use it to isolate implicit from explicit learning?

RI: If I reach for your phone, I'm getting feedback. I see how close my hand is to the phone. We can easily replace the hand with some proxy. This is what you do every time you're on your computer and move your mouse. You move your mouse to click someplace. You're using feedback because you recognize that your movements of the mouse are corresponding to the movement of the cursor.

It's very easy for the human mind to accept that a moving cursor is representative of their hand position. If we want to create a perturbed world to study learning, we change this so that whenever you move your hand, the cursor becomes offset by 45 degrees. Then we can ask, "How do people adjust their behavior to compensate for that?" It's what you're doing all the time with your mouse anyhow. In this "game," sometimes a small movement with the cursor takes a big movement of the mouse, or a big movement of the mouse is a big movement of the cursor. You very quickly adapt.

Clamped feedback is fixed feedback. It's the idea that we can fool the implicit system to think that the cursor is where the

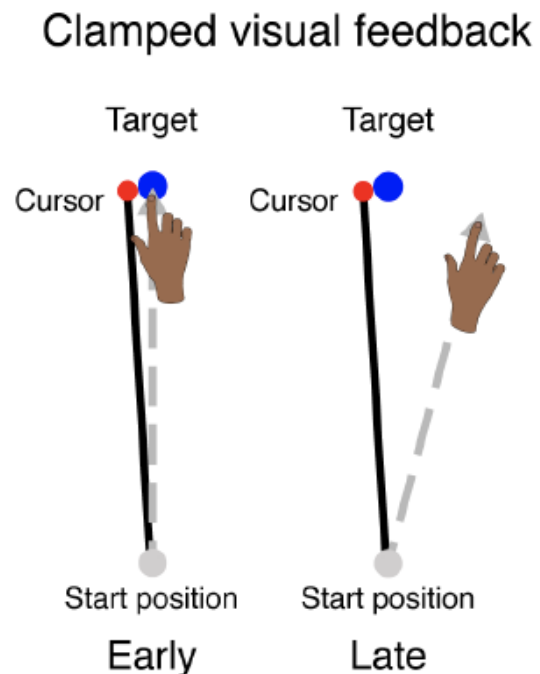


Figure 1: During clamped feedback, the angle of deviation between the cursor and target remains constant. The difference in hand angle to the target depicts adaptation to the clamped feedback over the course of the experimental period (early to late). If the cursor is consistently to the left of the target, adaptation will result in hand angle increasing to the right (Kim et al., 2019).²

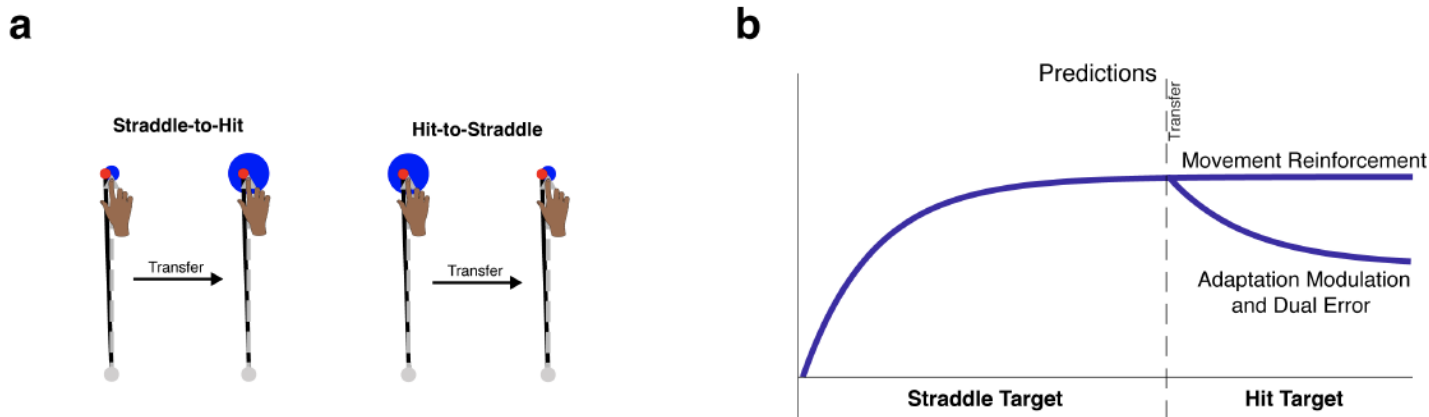


Figure 2: To distinguish between the three models, after an initial period of adaptation, subjects were “transferred” to a different target size. The size of the target is either increased (straddle-to-hit condition) or decreased (hit-to-straddle condition). As depicted in part (b), where the y-axis represents hand angle, the movement reinforcement makes different predictions for this experiment than the adaptation modulation and dual error models (Kim et al., 2019).²

hand is, even when you are explicitly aware that it isn’t; no matter where you reach, the cursor’s always going to do the exact same thing. This very primitive implicit motor learning system might not have access to that information, and it will respond as though it is the feedback. If I reach somewhere but the cursor tells me that I’m somewhere else, then I’m going to gradually correct my movement. The clamped feedback seems to be picked up by the motor system to recalibrate, even though I have no control over it (Fig. 1).

We set up our experiment so that motor error would vary. The motor error is the discrepancy between the dead center of the target and where the cursor goes. We assume the motor system demands perfection. It really wants you to be right on the center of the target, so the motor system is going to respond to this error. The reward manipulation happens by using a big target or a small target. If it’s a small target, the clamp lands outside. That looks like an error. Not only did the cursor not go where you expected it to, but it missed the target. For the big target, it still didn’t go where you expected it to, because you’re probably aiming for the center. It goes off to the side, but it’s still within the target. So you have this contrast between “Did you hit the target? Or did you miss the target?” Then you can ask how that changes how much learning is observed, how much adaptation occurs as the result.

What was surprising to us was that the amount of adaptation was really reduced in trials with a big target. For a variety of reasons, it was previously thought that the implicit system didn’t care about reward. And yet, the output of that implicit learning was attenuated when you hit the target. So we thought that reward might be attenuating how much you learn.

BSJ: Can those results be generalized to different conditions? For instance, would you expect to see similar results if variables like hand angle were altered?

RI: I think the results can be generalized. Since our paper came out, other groups have picked up on this question and have been testing different manipulations. The favorite one these days is that I reach towards a target, but as I’m reaching, the target jumps to where the clamp is, which the person knows they have no control over. You see a similar attenuation under those conditions.

It’s like how the study of illusions has always been very useful to help us understand how perceptual systems work. Even when we know about them, we still see the illusions, and that is because they tell us something fundamental about how the perceptual system is organized. We’d like to think that the same thing is true here—we’re able to isolate a system that’s constantly happening. Every time you put on your jacket, it’s a little heavier when you have to reach for something than when you don’t have that jacket on. The motor system has to constantly be recalibrating, right? So our belief is that this system is always operating at this implicit level, commanding perfection and making subtle changes to keep yourself perfectly calibrated.

BSJ: How do the movement reinforcement, adaptation modulation, and dual error models differ in their explanations of how reward and error affect learning?

RI: So we came up with hypotheses about different learning processes, each one subject to its own constraints. We have to specify what we really think is happening, and then by writing the computational models we can make quantitative predictions based on results. Sometimes, unexpected things come out of modeling.

The first model, the movement reinforcement model, just says that there’s one learning system driven by errors and another driven by rewards. It basically is the classic sort of reinforcement

learning model in the brain where if I do something, I get rewarded to do it again. So the first model says there are two different systems operating, one regarding independent learning and one that just reinforces rewards, and that actions and behaviors are the composite of those two. But that model makes predictions that don't hold up very well.

The second model, the adaptation modulation model, shows

a direct interaction between learning systems—I have an error-based system, and I can turn the strength up and down depending on reward. The third model, the dual error model, says there are two separate implicit learning systems that independently operate, where one system cares about whether my hand went where I wanted it to go (that's the sensory prediction error), and another system cares about whether I achieved my goal. Performance is the sum of their two outputs.

BSJ: What did your experiments suggest about the accuracy of each of these three models? How were you able to differentiate between them?

RI: Well, we weren't able to differentiate between the last two models. We saw that once you start hitting the target, the amount of adaptation decreases. That's how we can rule out the first model, due to a qualitative difference between the prediction and the result (Fig. 2, Fig. 3).

It's more of a quantitative distinction between the dual error and the adaptation modulation model. We just didn't have good enough data yet to distinguish between the two, so that's why we continue on the project, having to come up with new experiments to differentiate between those models, or find out that they're both wrong and find some other alternative.

BSJ: We also read your paper on neural signatures of reward prediction error.³ Could you explain for our readers what reward prediction error is, and its relevance in understanding human cognition?

RI: Sensory prediction error is what we typically think of as motor system error. I expect that when I reach for this phone, I'm going to grab it. If I miss, I call it a sensory prediction error. I have an expectation of what I'm going to experience, and what I'm going to feel. If what I feel is different than what I expected, that's a sensory prediction error. It is used to recalibrate the motor system to improve your movements in a very fine-tuned way.

A reward prediction error is when I have an expectation of how rewarding something's going to be. For example, I go to Peet's Coffee and order a latte, and I have an expectation of what a latte tastes like. But there's a lot of variability in those baristas. Say I get one of the bad ones. I take a sip of the coffee, and it's a badly-made latte. That's a negative prediction error. I had my expectation of what it tastes like, and it wasn't as good, so I didn't get the full reward I cared about. If it happens consistently, I'm going to use those prediction errors. Sometimes I get a great coffee, so I'm going to figure out which coffee shops I like and use those reward prediction errors to help me make choices in life. The reward prediction error influences our choices, while the sensory prediction errors are more of, once I made a choice, whether I actually succeeded in accomplishing the desired action. It's a distinction between selection (the reward prediction error) and execution (the sensory prediction error).

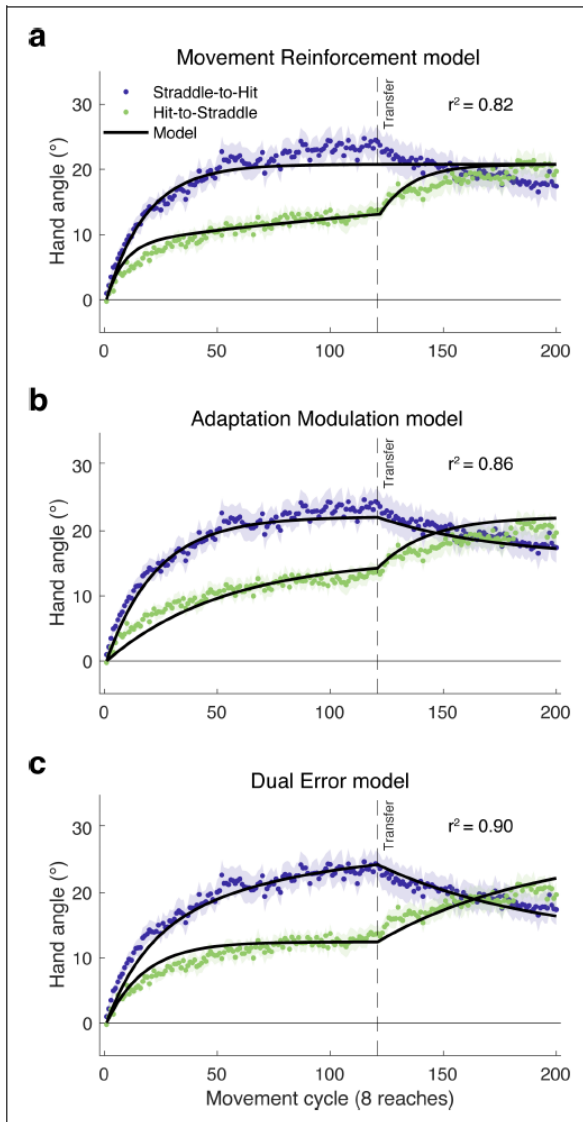


Figure 3: Black lines are predictions made by the movement reinforcement model (top), adaptation modulation model (middle), and dual error modulation model (bottom). Purple and green dots are experimental data, representing the straddle-to-hit and hit-to-straddle conditions, respectively. As demonstrated by the discrepancy between the black line and the dots in part (a), experimental data fails to align with predictions made by the movement reinforcement model (Kim et al., 2019).²

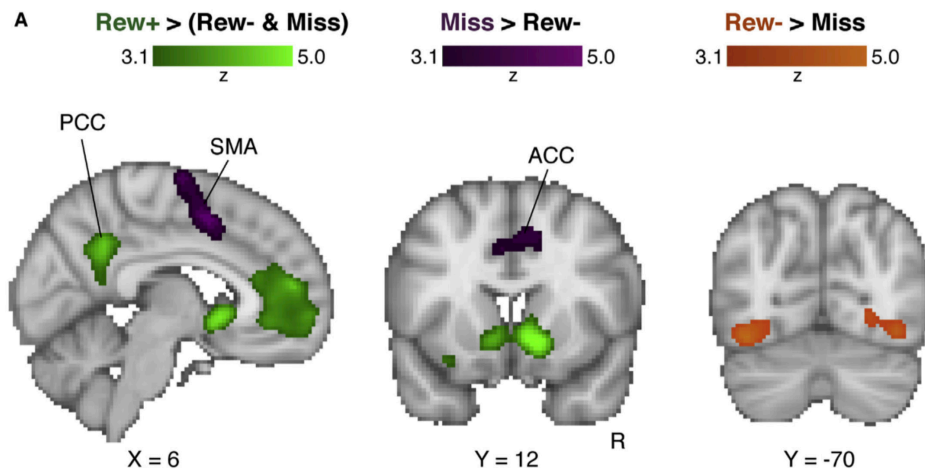


Figure 4: Regions of the brain implicated in reward prediction error, determined by fMRI. Note that many of these regions are localized to the frontal lobe (McDougle et al., 2019).³

BSJ: What parts of the brain are thought to be involved in reward prediction error, and how do their roles differ?

RI: To have a reward prediction error, I have to have a reward prediction. So what's my expected outcome? Staying with the Peet's Coffee example, I have an expected reward for getting a latte at one Peet's, and I have a different one for getting it at another Peet's. That represents what we call the value; I think one Peet's location is more valuable than another. Then we need a system that actually processes the feedback, to recognize if I got a tasty coffee or not. Finally, we have to compare the two. That's the reward prediction error. So I need the prediction, I need the outcome, and I need the comparison to generate the reward prediction error.

What are the neural systems involved? There isn't a simple answer, but evidence suggests that a lot of the frontal lobes, especially the orbital frontal lobes, are important for long-term memory, or at least having access to our memories of the values of things (Fig. 4). That may also be tied up with an intersection of goals and memory systems. If we said the orbital frontal cortex has a big role in the value, then it's going to be our sensory systems that are going to give us information about the feedback: the taste of coffee, the sound of the music I hear, or the experience I have when I try to hit a baseball. That feedback can come from very different systems depending on what kind of reward we're talking about. The activity level of dopamine neurons is, in a sense, best described as a reward prediction error. It used to be thought that dopamine was for reward only. Rats would press levers to get dopamine even if they'd starve to death, so it was thought that dopamine was a reward reinforcement signal. The subtle difference is that rather than just the reward signal, it's actually more about the reward prediction error. So if I expect tasty coffee and I get a tasty cup of coffee, that's a good thing, but I don't get much of a reward signal. I don't really get a strong dopamine

signal because there was no error. I got the reward that I expected. It's not a negative prediction error signal, but it doesn't strengthen things.

BSJ: Could you briefly describe what multi-armed bandit tasks and button-press tasks are? What is the distinction between execution failure and selection error, and how did you modify the setup of the classic 2-arm bandit task to distinguish between the two?

RI: To study human behavior, one of the things that economists like to do is to set up probabilistic reward tests. The bandit tasks come from the idea of slot machines, which are frequently called one-armed bandits because they're stealing your money. Classic behavioral economics experiments present three different slot machines, but the experimenter controls the payoff and probability for each of those three machines, which change over time. If I'm smart, I'm always going to go to the machine with big payoffs and big probabilities because I have greater expected value. I might choose a slot machine with big payoffs but low probabilities, or one with little payoffs but big probabilities. Then it's a matter of whether the person is risk-seeking or cautious.

But in the classic way that these studies have been done, you just press buttons and there are no reward errors. We basically repeat that, but we now make people reach out and touch the bandit. They don't have to pull the slot machine, but they have to reach out, because if you're just pressing the buttons, there isn't any action or execution error. We're making a more realistic situation. Actually, this project got started partly by us watching big ospreys on the east coast. An osprey is like an eagle, but it's a seabird. It swoops around and then suddenly, it does a dramatic dive. From my informal observations, the osprey's hit rate is maybe about 25%, so it's diving a lot but coming up with no fish. It can't be all that pleasant to slam your face into the water. So the osprey has to make a decision. It wants that fish, it values

it, and it has to make the right choice. Then, it has to make a perfectly timed dive. Afterwards, the osprey has to figure out, “Did I actually dive at the right thing, and I just missed because I didn’t dive well? Or was I diving at something that didn’t really exist?” The latter question is a choice problem. The osprey has to decide: “Did I really make the right choice to think that that little flash I saw was a fish?” And then another problem: “If it was a fish, then what did I do wrong in my motor system that I can learn from to do better in the future?” So that’s why we thought it was important to bring in the action component to these choices. How do you decide if it was a problem of choice (selection error) or a problem of execution (execution failure)? It’s a fundamental distinction we always have to make.

BSJ: What is functional magnetic resonance imaging (fMRI), and what are its applications in cognitive neuroscience?

RI: Well, fMRI has definitely taken over cognitive neuroscience. An MRI is a way to get a picture of internal anatomical structures. It takes advantage of the fact that the body is composed of a lot of water, and molecules can be made to vibrate at certain frequencies by exposing them to a magnetic field. That allows you to get these beautiful pictures of the structure of the body. We take an MRI scan to look for things like tumors. Functional MRI (fMRI) resulted from the insight that as parts of the brain are active, their demands for oxygenation change because they use up the oxygen and they have to be resupplied. The molecules in oxygenated versus deoxygenated blood are different. So we can then set up a magnetic field to perturb those molecules and then measure the signals emitted when we remove that magnetic field. Basically, it’s an indirect way to measure how oxygen is being utilized in the brain, or how blood supply is being distributed in the brain. We’re only measuring metabolism, not neurons. But we make inferences because we know that the parts of the brain that are more active are going to require more blood.

BSJ: Through fMRI, you demonstrated how specific brain regions respond to the distinct types of failure we previously discussed. How do neural signatures differ for execution versus selection failures, and what is the significance of these differences?

RI: There’s literature from doing standard button bandit tests in the fMRI scanner which shows that when people play the slot machine and get a big payoff, there is a big positive reward prediction error in the dopaminergic parts of the brain, implicating those regions in reward prediction error. So if I don’t get that reward, but it’s because of an execution error rather than a reward prediction error, do I still see that dopaminergic signal?

When I don’t get the payoff from the slot machine because of an execution error—say I didn’t pull the slot machine arm properly—we don’t see much of a reward prediction error or the corresponding dopamine signal. These results suggest that the reward system has input from the motor execution system.

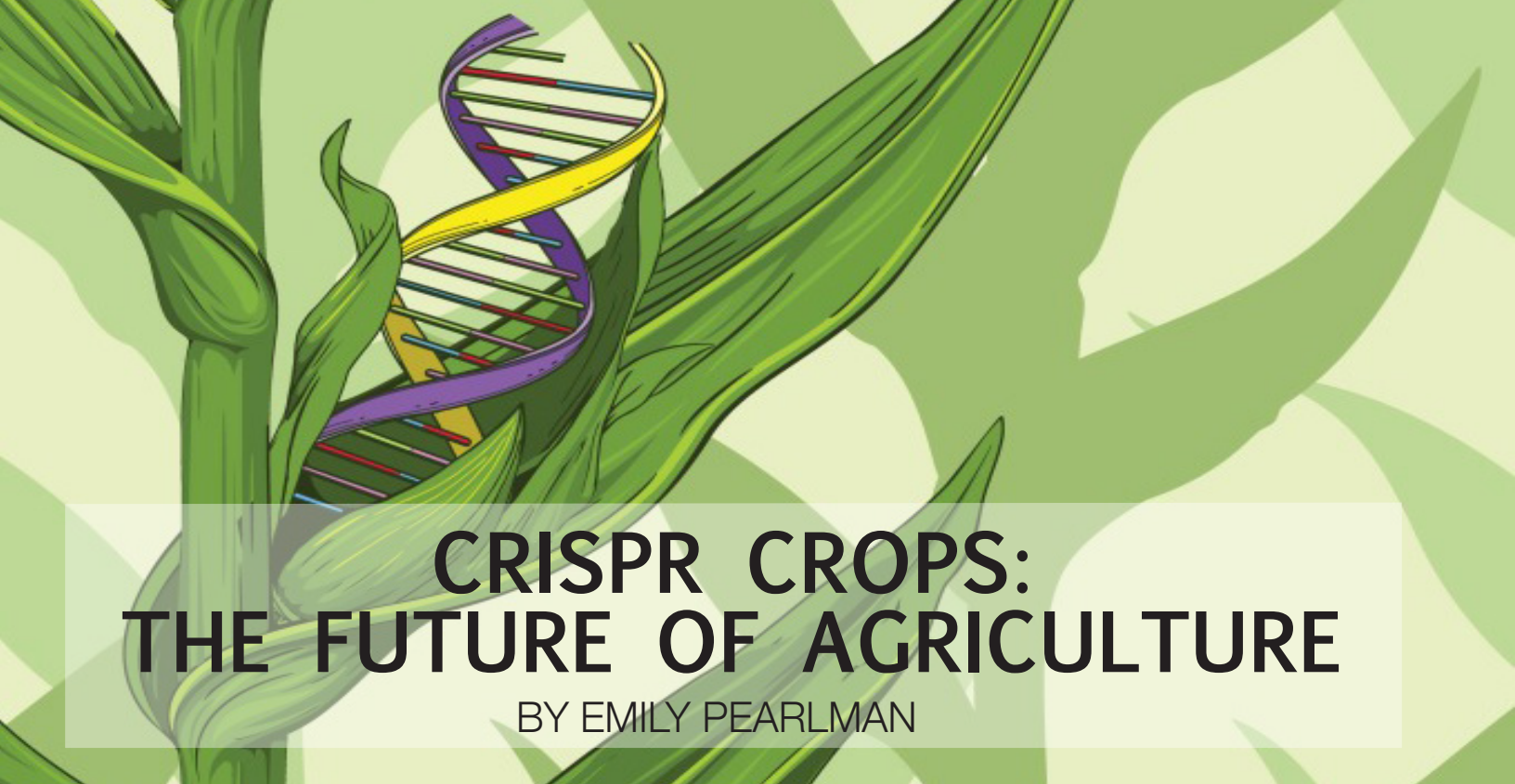
BSJ: You are on the editorial board for *Cerebellum*, and you have been an editor for many scientific journals in the past. What do you foresee for the future of scientific publication?

RI: There’s a financial challenge because everyone wants everything to be available online and open access. But then how do you pay for the costs of the publication process? Of course, there are people who say that we shouldn’t have the publication process anymore, we should just post the articles and then word of mouth will help spread the good ones, that we should just let natural selection operate and get rid of the journals entirely. Others think that the journals serve a useful purpose by facilitating the peer review process, by having some insiders evaluate the merits of the paper. But again, people also think, “Why should we allow two or three reviewers to have such power over whether something gets published or not?” So journals are experimenting with different techniques.

I’m actually one of the editors at *eLife*, [a peer-reviewed, open access journal]. One of the papers we talked about in this interview was published through a new experiment at *eLife*. We send them the paper, and they look it over and decide whether it’s worthy of review. Then, if they invite you to have it reviewed, they have a policy where the reviewers don’t decide whether it’s published, but the authors do. That’s pretty radical. So we get the feedback, and we then decide how we want to change the paper in response. We could publish the paper as it is or retract the paper. Or, we could modify the paper and say we’d like the reviewers to comment a second time. So you can see both the author’s view along with some commentaries on the value of the paper. I think more experiments like this are going to come along because there is a fundamental question: should editors and a very small number of reviewers be making the big decisions about publication, or should there be a way to actually get all the information about the process out there?

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CRISPR CROPS: THE FUTURE OF AGRICULTURE

BY EMILY PEARLMAN

Corn: the crop America knows and loves... or so we think. Modern corn looks so different from its original form, a wild grass called teosinte, that you would be hard-pressed to recognize the two are related. Over thousands of years of directed evolution, teosinte's tiny ears and indigestible kernels morphed into maize's large ears, each with as many as 500 juicy kernels (Fig. 1).¹ Corn is just one example of human-directed evolution of plants; we have been domesticating and cultivating crop species since the beginning of civilization. Although the technology used to selectively breed plants has grown more advanced over time, the basic principle remains the same—harnessing existing variation in a species to increase the prevalence of traits we consider “desirable,” such as larger ears on corn. Historically, this was achieved by successive rounds of breeding; today, with powerful genome editing tools at our fingertips, we have the ability to attain the same (and more drastic) results with less time and effort.

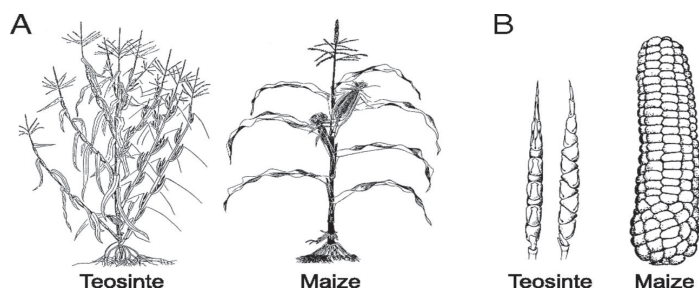


Figure 1: Thousands of years of selection for traits amenable to human consumption has resulted in the evolution of teosinte into maize. Now, with genome editing, we can accomplish the same changes in a much shorter time.

The power of genome editing is unprecedented. Never before have we been able to direct evolution in the blink of an eye. What once would have taken years of selective breeding can now be accomplished in a lab in a matter of months. This is not only possible, but also becoming necessary—with a growing human population and increasingly challenging environmental conditions, the use of genome editing to augment agricultural production will be essential to feeding the world. Current research focuses on using CRISPR-Cas genome editing to enhance crop species by increasing yields and bolstering resistance to abiotic (e.g., drought) and biotic (e.g., bacterial and viral pathogens) stresses.²

WHAT IS CRISPR-CAS?

CRISPR has received a lot of attention in the scientific world lately for its applications in genome editing, but it has actually been around for millions of years as a bacterial immune system. Just as we have an immune system that defends us against the viruses and bacteria that make us sick, some bacteria have immune systems of their own that protect them against viruses called bacteriophages. As the name suggests, CRISPR-Cas consists of two components: clustered regularly interspaced short palindromic repeats (CRISPR), and Cas, a DNA-cutting enzyme. You can think of CRISPR as “molecular memory” and Cas as “molecular scissors.” When a bacteriophage infects a bacterium, the bacterium can steal sections of the phage’s genetic information and store them within its own genome, filing them away in a library of other phage-derived sequences. By retaining phages’ genetic information, bacteria can essentially “remember” phages and mount more efficient attacks against them during future encounters, using their Cas enzyme to cut the phage’s genome and prevent infection (Fig. 2).³

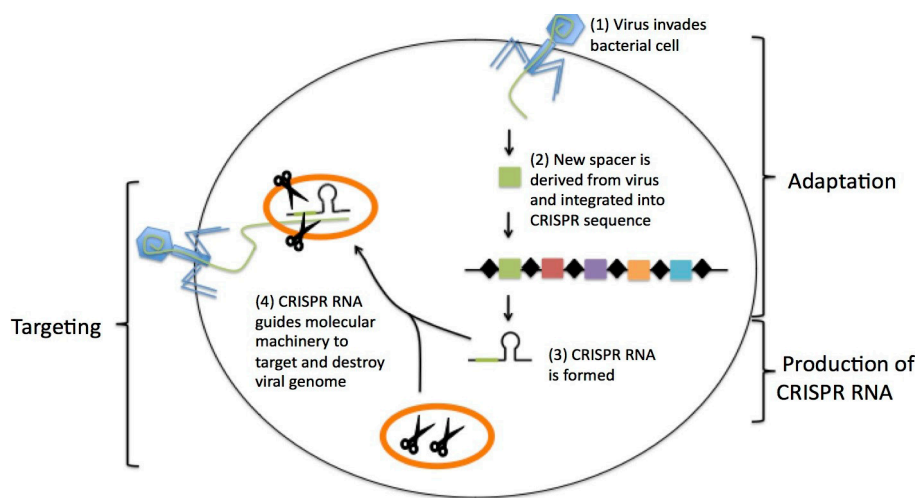


Figure 2: CRISPR is a type of bacterial immune system. When a bacterium encounters a virus, it can steal part of the virus's genome and store it in a CRISPR sequence. If it encounters the same virus again, the bacterium can use this stored information to target and destroy the viral genome.

The power of Cas enzymes lies in their sequence specificity; this means that they can be directed to cut specific DNA sequences. This makes them the perfect tool for targeted genome editing. Researchers can program Cas9, a commonly used Cas enzyme, to target a particular gene and create a break in the DNA, which the cell will subsequently repair via one of its natural DNA repair pathways. One option for the cell is to directly attach the two ends of the break, creating large mutations which often result in a loss of function of the gene, effectively “knocking out” the gene. If researchers provide the cell with a piece of DNA, however, the cell will repair the break using this piece of DNA as a template. Knocking out a gene can be helpful when trying to determine its function, while using a repair template allows researchers to make specific edits to a gene.³

“With a growing human population and increasingly challenging environmental conditions, the use of genome editing to augment agricultural production will be essential to feeding the world.”

INCREASING DROUGHT TOLERANCE IN MAIZE

Drought causes substantial agricultural losses each year and is only expected to worsen with warming global temperatures, so one major focus of current research is increasing crop yield under drought conditions.⁴ Jinrui Shi and his colleagues at DuPont, Inc., used CRISPR-Cas9 to do just this. By replacing the regulatory elements of a drought resistance gene with those of another gene that is expressed at much higher levels, they increased the overall expression of the drought resistance gene. When Shi and his team grew the genome-edited plants alongside non-edited plants in drought stress locations, they found that their experiment was a success—the edited plants produced significantly more bushels per acre, showing that targeted genome editing is effective in increasing maize yield under drought conditions. Although this result had been achieved by introducing foreign DNA into plants through a process called transgenesis, Shi's CRISPR-edited maize plants con-

tained no non-maize DNA. Furthermore, all of the genes used for editing (e.g., Cas gene) were removed through backcrossing, leaving no trace of the team's work in the genome.⁵ This distinction explains the disparity in regulation of CRISPR-edited plants and classical transgenic GMOs—CRISPR-edited plants contain no foreign DNA, they pose fewer safety concerns, and aren't regulated as strictly.²

BOLSTERING VIRAL RESISTANCE IN CASSAVA

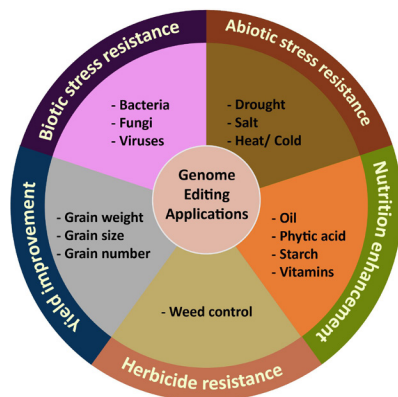
Viral diseases also pose a major threat to crop yields worldwide. Cassava, the fourth most important staple crop worldwide (after rice, maize, and wheat),⁶ is threatened by Cassava brown streak disease (CBSD), caused by Cassava brown streak virus (CBSV). CBSD, which renders the cassava root inedible, poses a major threat to food and economic security in East Africa, causing losses of \$175 million every year.⁶

Brian Staskawicz, Professor of Plant and Microbial Biology at UC Berkeley, used CRISPR-Cas9 to increase CBSD resistance in cassava by selectively mutating multiple genes. These genes encode receptors that the CBSV viral proteins must interact with in order to successfully enter the cassava cell. Staskawicz's team used CRISPR-Cas9 to create a range of knockout mutations in the receptor genes, effectively preventing them from being expressed. When the genome-edited plants were inoculated with CBSV and grown in a greenhouse, they displayed reduced CBSD symptoms: their roots were significantly less rotted and contained less virus than the roots of control plants.⁷ Developments like this demonstrate the potential of CRISPR-Cas to effectively engineer pathogen resistance in major crop species and avert sizable losses in yields.

CONCLUSION

Although CRISPR-Cas is a powerful genome editing tool, some obstacles to its widespread implementation still remain. One limitation of genome editing is its reliance on sequence and functional genomic data; in order to alter a certain trait of a plant, it is necessary to know which genes are responsible for the trait, a determination which can be difficult. For instance, the drought response in plants is complex and relies on a multitude of genes,

Figure 3: Genome editing in crop species has many applications, ranging from abiotic and biotic stress resistance to yield improvement and herbicide resistance.



not all of which are known.⁸ But despite these drawbacks, genome editing remains a burgeoning field of research that will be essential to the future of agriculture. The studies outlined here represent a small subset of the research being conducted on CRISPR-Cas genome editing in crop species; this technology has been used to generate pest resistance in wheat, enhance herbicide resistance in tobacco, and increase yield in rice, among many other traits (Fig. 3).^{2, 8} Exciting new research is using CRISPR to explore the complex symbiotic relationship between legumes and their rhizobia, with the goal of recreating this relationship in other crops and reducing our dependence on harmful chemical fertilizers.⁹

Since the beginning of civilization, humans have been transforming plants to suit our needs. Genome editing is merely the latest chapter. Climate change has already had a major and multifaceted impact on agricultural productivity, and its effects are only expected to worsen.^{10,11} Additionally, the human population is expected to increase to nearly 10 billion by 2050, placing further pressure on us to dramatically increase agricultural productivity.¹² In the face of a rapidly growing human population and increasingly difficult environmental conditions, the use of genome editing tools such as CRISPR-Cas to enhance crop species will be vital to feeding the world.

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The Rise and Demise of Software

BY CANDY XU

The field of information technology bloomed in the late 20th century. With the continuous advancement of hardware during this period, accompanying new software products raced into the market. Many of them, such as Lotus 123 and the Pascal language, became crucial features of their systems and took their respective companies to new heights. However, many of these software products did not last long. Fleeting like sparks, they were quickly replaced by their competitors. Now, all that is left behind are their creators' innovative ideas, ideas that have set up the foundations for our current generation of software products.

RISE

The development of computer applications revolutionized both technology-based and non-technological industries. "Tech" was no longer a mysterious term used by researchers and computer enthusiasts, but a product accessible for use in consumer's daily lives. Lotus 123, a spreadsheet program developed by Lotus Software, was one such application (Fig. 1).

As the spreadsheet standard in the 1980s, it was widely utilized in various fields including biology and economics.¹ In one scenario, running statistical significance tests across large datasets in biological research was made more efficient by using Lotus 123; in another, multiple reference peak identification methods for certain chemical techniques were developed using the software.^{2,3} The Lotus file compatibility structure even became the de facto industry standard for the exchange between database and spreadsheet programs.¹

Aside from commercial softwares, computer scientists also made user-friendly innovations in programming languages. The Pascal language, first released in 1970, is a programming language that was also highly influential in a number of different industries. Due to its structured design philosophy, Pascal—more specifically, its extension Pascal-FC—became very popular among educators because of its ability to demonstrate running functions concurrently and give students hands-on experience with code.⁴ Exciting and unprecedented products like Pascal quickly became extremely widespread.

DOWNFALL

These applications' prevalence, however, did not last long. Lotus 123's popularity decreased as it was soon replaced by Microsoft Excel in the early '90s. Although there are many hypothesized reasons for Lotus 123's downfall, the singular reason for its demise remains unclear. One hypothesis for Lotus 123's downfall is its failure to match the strong economic modeling features of Microsoft Excel. Excel's Binomial Option Pricing Model, for example, is an extremely powerful tool that streamlines mathematical work that analysts would otherwise do manually, and is a feature that Lotus 123 was unable to effectively mirror.⁵ Furthermore, Excel's ability to create

"The Lotus file compatibility structure even became the de facto industry standard for the exchange between database and spreadsheet programs."

modeling tools such as large decision trees, hugely supports the needs of the financial industry, and thus outperforms Lotus 123 in this domain. Aside from the technological side of the story, Excel also had a huge strategic and marketing advantage. Microsoft DOS, later Windows, was the dominant operating system in the market. It was thus much easier for Microsoft to market Office to their existing customer base and to develop applications that aligned better with their own system. What's more, Lotus 123 is a 32-bit program, making it incompatible with 64-bit Windows 10 unless installed in a complex process using a CD, further contributing to its current elimination from the spreadsheet market.⁶

Similarly, Pascal was eclipsed by the C language. The TIOBE index, an indicator for the popularity of programming languages, indicates that Pascal ranked number 5 in popularity in 1985, but had decreased to 229 in 2020, while C remained consistently ranked in the top 2 spots.⁷ The differing design philosophies of Pascal and C likely caused the eventual decline in Pascal's popularity. Pascal is more secure than C, making it harder for outsiders to hack into. C, on the other hand, has greater flexibility and can complete more dynamic tasks, thus giving it appeal to a broader audience.⁸ Notably, C is also a lower level programming language, meaning that it is better at directly communicating with the hardware of a system. These factors, however, are only some of the reasons for Pascal's downfall.

LASTING EFFECTS

Despite their decline, applications like Lotus 123 and Pascal remain foundational to future generations of software. Products based on these applications continue to support a variety of subject fields, both in academia and in industry. For instance, Turbo Pascal, released over a decade after Pascal, is a famous system that compiles the Pascal language and provides an integrated development environment to the users (Fig. 2).⁹ It directly connects the rather technical programming language to a wide range of users, increasing the accessibility of Pascal to educators. And while it has faced competition from other Pascal compilers, Turbo Pascal stands out due to its low price and short compiling time.^{9,10,11} Bruce Webster, famous software engineer and Adjunct Professor at Brigham Young University, also noted

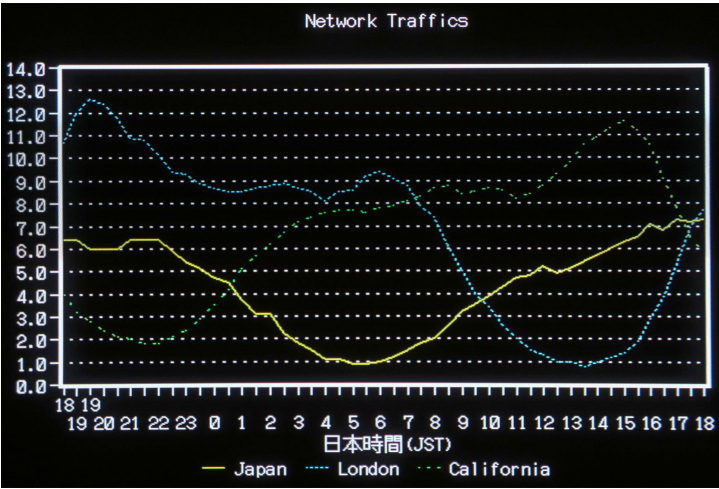


Figure 1: Lotus 123 running on PC-98 DOS in its Japanese version. It displays the network traffic of Japan, London, and California throughout a day.

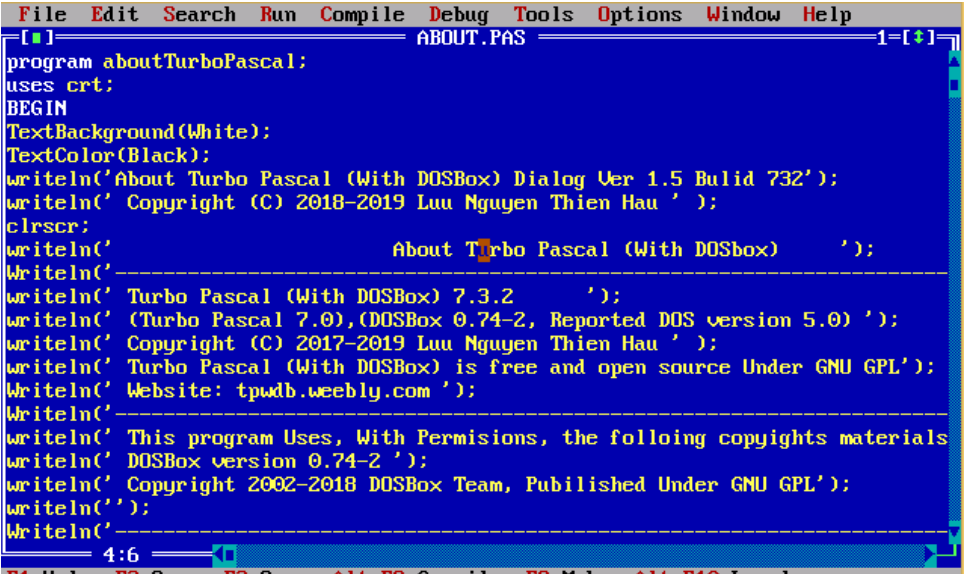
in *Byte Magazine* that Turbo Pascal has a price that is unbelievably low considering its quality.¹⁰ These aggressive pricing and direct marketing strategies further resulted in its widespread use. Newer editions of Turbo Pascal have also been included in packages with other libraries, solidifying its presence in the market. It has not only demonstrated the potential of the Pascal language itself, but also shows a successful product model. Similarly, Lotus 123 set the precedent for spreadsheet softwares in the decades after its demise.

FUTURE

Developments in both business and scientific fields emerge frequently, with some ideas becoming popular only for short periods of time, and others remaining in the

mainstream for decades. The timing of a product's release and its architecture's flexibility both seem to be important factors in determining how long it lasts in the market. Turbo Pascal was well timed to set up a bridge between Pascal and its growing user base. Although the product lost market share over time, this loss wasn't necessarily due to poor functionality, but rather because of the decrease in popularity of the Pascal language. Lotus 123, when first released, similarly seemed pressed to become the de facto spreadsheet norm. By taking full advantage of the PC's improved memory and display, Lotus 123 could outperform other applications in its field. Furthermore, Lotus Development, the company that owns Lotus 123, sought to eliminate their competition while bringing their own

Figure 2: The Turbo Pascal 7.0 interface running on MS-DOS, an antiquated operating system. It displays some basic information about the software and its copyrights.



"Notably, many popular softwares do not fight alone in their marketing wars."

product to fame. They purchased VisiCalc (Fig. 3), the first ever computer spreadsheet program and their biggest potential competitor, replacing it with Lotus 123. Unfortunately, since its underlying structure didn't allow it to match enhanced hardwares that other newer applications could fully utilize, the demise of Lotus 123 was inevitable.

Notably, many popular softwares do not fight alone in their marketing wars. They are often accompanied by other applications or features. For example, Microsoft Office overtook the markets of document processing, spreadsheets, and presentations by packaging the three applications together. Here, besides the powerful features of Excel alone, Word and Powerpoint sweetened the deal for consumers trying to choose between different products. As technology advances, developers will be able to embed more complicated features into a single application.

Yet broadly, the contemporary "tech" moment isn't too different from historical ones. With Google offering a new set of collaborative "office" features organized under Google Drive, it's competing with both Microsoft Office and Apple iWork for market share. Google Drive's architecture is fashioned to offer real-time collaboration, a feature harder to achieve with Microsoft Office. For programming languages, Python is the new star. In fact, our very own Berkeley campus has chosen Python as the first language to teach in its introductory Computer Science course, CS 61A. Other applications, such as those based on cloud databases and blockchains, are also

interesting innovations—each of which will likely create new eras in software markets. As we all continue to move from one software to the next, it will be worthwhile to keep our attention on these trends and witness the rise and fall of new technologies before our very eyes.

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ITEM	NO.	UNIT	COST
MUCK RAKE	43	12.95	556.85
BUZZ CUT	15	6.75	101.25
TOE TONER	250	49.95	12487.50
EYE SNUFF	2	4.95	9.90
SUBTOTAL			13155.50
9.75% TAX			1282.66
TOTAL			14438.16

Figure 3: A VisiCalc spreadsheet running on Apple II. It displays a simple price calculation of four goods plus tax.



THE HUMAN BRAIN AND THE EVOLUTION OF LANGUAGE

Interview with Professor Terrence Deacon

BY ELETTRA PREOSTI, MELANIE RUSSO,
KATIE SANKO, KATHERYN ZHOU, AND
MATTHEW COLBERT

***Dr. Terrence Deacon** is a current Professor and former chair of the Department of Anthropology at UC Berkeley. He is also a member of the Helen Wills Neuroscience Institute. Professor Deacon's work combines human evolution, semiotics, and neuroscience to examine how humans evolved the capacity for cognition and language. In this interview we discuss his findings on what sets humans apart cognitively from other species, and how those differences and further evolutions allowed for the use of language.*



Professor Terrence Deacon¹

BSJ: You originally did your PhD in biological anthropology at Harvard, and now your research is in neurology, linguistics, information theory, and even the origin of life. How would you describe the path your research has taken, and how would you describe your focus now?

TD: My PhD at Harvard was actually about the neural connections in monkey brains that correspond to areas in the human brain associated with language. Even though it is a biological anthropology PhD, my work was primarily in the neurosciences. I studied neurosciences at MIT as well as Harvard before getting my PhD, and I also got a Masters degree in Education from Harvard, so I had multiple backgrounds. One of the reasons I studied the neurology of language and worked in education was because of my interest in

Charles Peirce, the father of semiotic theory, and his way of thinking about language and communication. From there, I became a professor at Harvard for eight years, where I went on to do other studies of how brains evolved. Then, I moved on to Boston University, and then to Harvard Medical School. There, I worked on transplanting neurons from one species' brain to another to understand how species differ in the way their axons make connections to other parts of the brain during development in order to create neural circuits. Using this technique of transplanting neurons from one species to another, we were able to ask questions about what mechanisms were being used to guide those connections to their destinations. I left Harvard Medical School in 2000 and came here to Cal in 2002. I had basically moved on beyond my transplantation research, and

most of my research then moved to more abstract questions. We were interested in how brains grow, how they develop across species, and much more generally language and its evolution.

BSJ: A lot of your work focuses on human evolutionary biology and evolution in the brain. What is encephalization and its significance in human evolution?

TD: Encephalization is a way of talking about how big brains are for a given body size. We think that our big brains for our body size is something very important to our species and that it is clearly something that enlarged during our evolution. If you know the size of the mammal's body, you can do a pretty good job of predicting how big its brain is (Fig. 1). One of the exceptions to this is primates, and anthropoid primates in particular: monkeys, apes, and us. In fact, we have brains about three times larger than you'd expect for a typical primate of our size. The rest of the non-anthropoid primates have about twice as much brain for the same sized body as almost any other mammal. That suggested that something was special about primates and us.

Now, the problem with encephalization is that it's not about how big brains are. If you were to look at mice, about four percent of a mouse's body size is made up of its brain. But for you and me, it's two percent; we actually have a smaller ratio of brain to body size than mice. It turns out that as bodies get bigger in different species, brains enlarge to the two-thirds power, sort of like surface and volume relationships. When you enlarge a sphere, the surface increases to the square, and the volume to the cube. Well, that's sort of the way the brain expands. Over the course of the last century, people have tried to figure out why that's true. One of my hypotheses that we've tried to test, but can't get a really definitive answer to, is that brains develop from the same primordial part of the embryo that will turn into skin. Since our body's surface has to enlarge to the square of our body mass, you might think of that as a reason why brains should do the same. However, that doesn't explain why primates deviate and why humans deviate further. We've found that in primate development, it's not brains that are chang-

ing; it turns out the bodies of primates grow slower than the brain. At the same time, we've found that all brains of mammals grow at the same rate. That means that humans' unusual brain sizes can't be the result of our brains growing a lot faster. Instead, they grow for a longer period of time. Humans are like other primates in the womb, but we keep growing our brains a little bit longer, so we end up having a much larger brain. So, there are actually multiple ways that brains and bodies can be out of proportion, and this affects the connection patterns in the brain. Simply changing the relative sizes of brains changes how axons find their targets and how connections get formed. If you change the total proportion of the brain in relation to the body, you'll also change the way these axons compete for connections. This is almost certainly what accounts for some of our own differences.

BSJ: What aspects of the human brain allowed us to develop the unique cognitive and linguistic abilities lacking in our closest homologues?

TD: One of the things that's really obvious about us is the connection between our cerebral cortex, the motor area that controls skilled movement, and our larynx muscles. Because of this, I can articulate the movement of my mouth, tongue, and lips really precisely and consistently with the timing of vocalizations from my larynx. Other species that don't have this direct connection to the larynx can't coordinate the production of sound with the articulation of their mouth, tongue, and lips. That's clearly one of the things that makes speech possible. My work has suggested it's because of a change in the prefrontal part of the brain, which plays a big role in dealing with multiple possibilities at once. When answering a multiple-choice question on an exam, you have to hold multiple options in your head at once and then sort through them. These functions are pretty important in language. Now, does that mean that we're more intelligent? Well, it turns out there are some tests that we would think correspond to intelligence that we don't do as well on as chimpanzees. This is interesting because it means that some of our capabilities have been traded off for other capa-

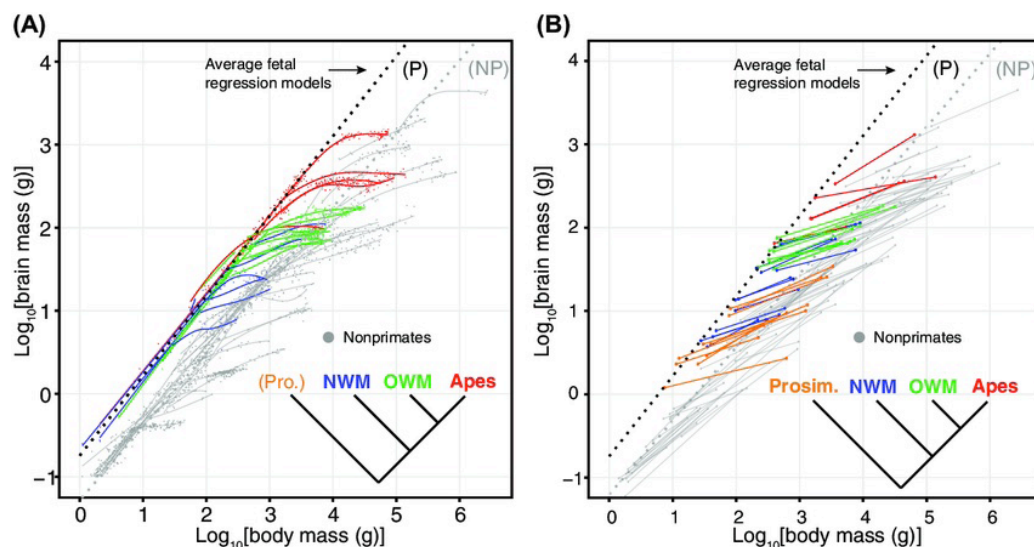


Figure 1: Logarithmic graphs showing primate brain size versus body size in grams.² The mammalian average can be seen in the left graph. (Reproduced from Halley and Deacon, 2017.)

"We remember things as stories, as theories, as ideas, as histories; even our own personal identity is defined by the way these kinds of memories are linked to each other. Language has allowed us to use our memory systems in absolutely novel ways."

bilities. For example, language uses our memory in very unique ways. Other species have a memory system like ours in the following sense: when you learn a skill, you learn by repeating it again and again. However, what you had for breakfast two days ago isn't something you're able to repeat; it happened once. What you did on your seventh birthday is maybe something you could go back and recover with a bit of effort. This kind of memory is the result of connectional or associational redundancy: because you are this many years old, you can make the connection to what town you were living in when you were seven years old, what school you were going to, and so on. We can use all these things to zero in on that memory. It turns out those two memory systems are controlled by two different parts of the brain. A structure called the hippocampus is involved in the memory of one-off events, sometimes called episodic memory, because it has to do with episodes in our life. The other is procedural memory, which is controlled by structures called the basal ganglia. It turns out language is the result of both. Language involves things we repeat over and over again. But when you want to remember something that's complicated in terms of its associations, you need this other kind of memory. We remember things as stories, as theories, as ideas, as histories; even our own personal identity is defined by the way these kinds of memories are linked to each other. Language has allowed us to use our memory systems in absolutely novel ways. So, we think about and remember the world differently from other species precisely because of our language abilities.

BSJ: How did these findings inspire your theory of coevolution in humans?

TD: In the 1990s, I was beginning to look at how human activities changed the world, which then will change humans. Today, we call this niche construction: a niche is the environment that an animal adapted to live in and has specializations for. Beavers are a great example of an animal that creates its own niche by building dams to create an aquatic environment that it's adapted to. Well, in a real sense, humans have constructed this niche that we call language and culture that we have adapted to. My argument about coevolution concerns the fact that changes of the brain have changed the possibilities of communication and social organization, which have created this specialized niche of culture and patterns of activity that are shared socially. In turn, this has changed our brains so we can learn language more easily and interact with each other. Over the course of a bit more than two million years, we've become adapted to the world that we've created, and one of

the ways we're so different from other species is that we've adapted to this niche.

BSJ: What exactly is language in terms of your research and how does that differ from its traditional colloquial definition?

TD: We all know what language is because we're using it; exactly how it's different from other forms of communication has really been a challenge to define. My approach came early on from studying the semiotic theory of Charles Sanders Peirce, who said we communicate in lots of ways that are not linguistic. Just think about laughter and sobbing. A laugh doesn't have a definition, it's something we do. We can describe the context in which we produce it—say it's associated with a particular emotional state—but it doesn't have a definition. Our brains have effectively evolved the ability to do other kinds of communication besides language, what Peirce called iconic and indexical communication. Iconic communication is pictorial-like and it has to do with things that look, sound, or feel like something else. We have lots of knowledge that is based upon this kind of relationship. In fact, after this conversation is done, you won't remember hardly a single sentence that I've talked about, but you'll have a mental picture of what it was about. That's very different from language. On the other hand, we use a lot of indexical communication. Human beings are specialized for pointing, and children learn this well before they learn language. Unfortunately, linguistics sort of takes language out of the context of communication in the body and asks about things like the structure of a sentence. This led to one of my teachers in the distant past, Noam Chomsky, arguing that the meaning of words is totally independent from the grammar and syntax. A lot of my work had to always place language in the context of non-linguistic communication, particularly because it was about how language came about in the first place. The evolution of language in our species had to grow out of non-linguistic communication. So, iconic and indexical communication is the basis of our linguistic capacities. If you only focus on the grammar and syntax of sentences, you miss the infrastructure on which it's built. It's the icing on the cake. Unless you focus on this rest, you won't understand how the icing got there, so to speak.

BSJ: We were looking at your work on language development and how words get their meanings and were wondering if you could talk to us about the symbol grounding problem?

TD: The symbol grounding problem derives from the fact that you can think about symbols like words and imagine that

"Over the course of a bit more than two million years, we've become adapted to the world that we've created, and one of the ways we're so different from other species is that we've adapted to this niche."

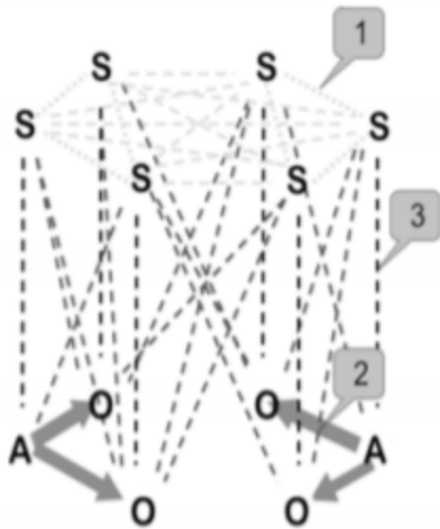


Figure 2: A demonstration of the possible relationships between a sign (S), an object (O), and an action (A).⁴ Line 1 denotes a sign-sign relationship, line 2 denotes a sign-object relationship, and line 3 denotes a sign-action relationship. (Reproduced from Raczaszek-Leonardi and Deacon, 2018.)

they're arbitrary. They don't carry any clue inside them, and that knowledge isn't in the context of other kinds of communication. Then, you have to wonder how you ever have symbols map to what they refer to. In our work, we recognized that children begin with symbols that have no clues carried within them. They come into a world where other people are communicating with these things and they have to figure out what's going on. By the time they start learning the language, children have a lot of experience communicating iconically and indexically: pointing, gesturing, reaching, exchanging objects, and all kinds of things that go on during the first year of life. An icon is grounded because the sign vehicle itself has some likeness to what it refers to; an index like pointing has to be associated with what's being pointed at (Fig. 2). As a result, iconic and indexical communication are grounded—you can look at the sign vehicle itself and know what it refers to. We have to use grounded communication to figure out how to use ungrounded sign vehicles. So how does a child do this? This is what we call the symbol grounding problem. They've got to use signs they're already familiar with and figure out how to throw away that groundedness and yet not lose the reference. If the grounding is not inside the sign vehicle, it has to be in the relationship of the sign vehicles to each other, which is grammar and syntax, which have basically internalized a lot of more grounded sign relationships. Those relationships are used differently in different societies, cultures, and languages.

BSJ: In regard to your work at Harvard Medical School with Ole Isacson on neuro-transplantation and Parkinson's Disease, you found that fetal porcine dopaminergic neurons were able to survive and restore these nonfunctional pathways in rats.⁵ Is it surprising that neurons from one species can function in another?

TD: It is surprising in two senses. Number one, when we transplanted rat neurons into the rat brain, it didn't work. You would expect that rat neurons should be the ones that really work, but they didn't because they couldn't quite find their connections. There turns out to be a reason for this. In adult brains, axons are impeded from growth. It turns out that rat neurons take only a week or so to mature, which wasn't enough time for them to find their connections. Interestingly enough, pig neurons take months to mature. As a result, when transplanted into the rat's brain, even though it was a different species, the same molecular signals that direct neurons to grow in a specific place turned out to be in both rat and pig brains, and even human brains. Because the pig neurons were slower to mature, they made connections that rat neurons could not, so the pig neurons improved the Parkinsonism of these rats. It was surprising that the cross-species transplantation worked better than the transplantation within the same species. But it showed that the time course of maturation of neurons really mattered in this process. It showed something else surprising as well. Even in adult brains, the rat neurons that would guide axons to their targets were still there. We even did electron microscopy to show that synapses formed directly between pig and rat neurons that affected the formation of working connections. Now, pigs and rats have almost certainly been apart in evolution for more than 65 million years. Even though they are this far apart in evolutionary time, they are using the same neuronal guidance mechanisms. This says that brains are based on really conservative mechanisms, even though the size of the brain, the body, and the structures have changed a lot. In one sense, it's good news because this means that brains are generic enough that you can use these techniques. But



Figure 3: The first neuron transplantation patient for Parkinson's Disease in the initial study.⁶ Photo courtesy of Professor Deacon.

there is bad news here as well. If you were to place human fetal neurons to repair the brain, their axons wouldn't grow far enough to reach their targets. You have to get neurons from a brain that grows much slower. I doubt that we're ever going to raise whales for this purpose. In Parkinson's disease, a whole class of neurons have died,

"In a computer, none of the elements have any kind of variation of electrical activity. None of the components are worried about staying alive, nor is their activity affected by changing the amount of metabolism there. In brains, neural activity and metabolism are coupled in a really complicated way."

and it turns out that we can replace them by transplantation. The bad news is that the repair will take years in order for the neurons to find their targets.

It turns out that some of the work we did has been replicated in humans. In fact, I did some of the first autopsy work showing pig neurons transplanted into human brains in Parkinson's cases. This was a set of studies done by a company called Genzyme, who effectively took the technology and ran with it, thinking that they could make this work. There were as many as 12 patients in the initial study, and there was improvement for a couple of the patients. In fact, the very first patient was significantly improved. He had spent his recent years mostly confined to a bed or chair. After the transplantation, although it took a couple of months before he had any kind of improvement, he actually was able to get up and walk around. He even tried to play golf at some point (Fig. 3). Unfortunately, all of the grafts were eventually rejected. Now, there are many other techniques that we are beginning to be used for Parkinsonism, and transplantation is probably not going to be the approach. In this process, we learned a lot about the differences between species, how axons find their targets, and that brains are very conserved in the mechanisms they use to make connections happen.

BSJ: What aspects of the human brain make emulating it so difficult in artificial intelligence (AI) systems? How close do you think today's AI systems are to capturing human brain activity?

TD: One major part of the story is that neurons are trying to keep themselves alive. Techniques we have for looking at brain function *in vivo* all use metabolism: both PET scans and fMRI look at metabolism. We know that the metabolism of certain areas goes up and down depending on what we're thinking, and this tells us brain function and energy use in that region are coupled. In a computer, none of the elements have any kind of variation of electrical activity. None of the components are worried about staying alive, nor is their activity affected by changing the amount of metabolism there. In brains, neural activity and metabolism are coupled in a really complicated way. The other important difference is that units of information in the computer are states. In my computer I don't have a hard disk, just memory chips, which are little transistors that can hold a charge. This pattern of charge is the unit of information, and then I run electricity through it and it

does things. The problem in the nervous system is that the unit of information is probably not just a state or structure. Connections are structures that change, and thinking, acting, and perceiving are dynamical processes. It's not a state, and it's not the relationships between one state to another state. In some respects, artificial intelligence is very much like the theory of behaviorism from a half century ago. Behaviorism came out because people wanted to make psychology scientific, and the only way to do it if you couldn't look at what brains were doing was to look at the inputs and outputs. That's exactly what we do with neural net artificial intelligence. We give it lots of inputs, we strengthen some relationships between the inputs and the outputs, we punish others to eliminate some of those connections, and eventually we get the input-output relationship that we want. Imagine we give a neural network thousands or hundreds of thousands of pictures of cat faces, and eventually it recognizes cat faces and distinguishes them from dog faces or automobile pictures or human faces. But, a child of three could see three cat faces, and from that point on be able to recognize what's a cat and what isn't a cat. For our neural net research, it takes thousands, oftentimes millions of trials, to get that kind of capacity with artificial intelligence. This also tells us that what we're doing computationally is something fundamentally different from what brains do. That doesn't mean we couldn't make devices to work like brains, it just means we're not doing it yet, and we maybe don't have an idea of how to do it because we have components that don't work like neurons work. Will it ever happen? Probably so. Should we want to do that? Probably not.

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How to Win: Optimizing Overcooked

BY NICK NOLAN

Overcooked—the sensational group cooking video game—made its smash appearance in 2016. The premise is simple: prepare as many of the requested meals as possible before the time runs out by cooking meat, chopping vegetables, or boiling soup; combining the ingredients; and cleaning the plates you just put out for consumption (Fig. 1). The innately chaotic nature of the game unfurls in the floor plan: some levels are coated with ice that make it easy to slip off of the stage; others, a rotating center stage. When four people gather to try flipping burgers, it becomes difficult to navigate the unruly layouts while following occasionally-complex cooking instructions in time to satisfy the customer. Strategy is crucial to fulfill orders quickly and proceed to the next level.

Naturally, this idea prompts the question: how can we develop the best strategy to play Overcooked? How can we find the way

to make the greatest number of food items in the time allotted for each level?

First and foremost, it seems the solution to our quest may not lie in our own hands—recent advances in computer science have demonstrated that machine learning algorithms have been able to outplay human experts. Specifically, these methods have bested human players on some of the first games developed on the Atari 2600, a gaming console from 1977.^{1,2}

Machine learning is likely the way to go to play the best game of Overcooked, then.

These Atari algorithms, developed in 2013 by DeepMind Technologies' Volodymyr Minh, were able to learn how to play a host of Atari games, eventually exceeding the highest scores of some of the most skilled human players. To accomplish this task, Minh utilized a class of machine learning known as reinforcement learning, in which the computer is a sort of “player,” learning about the environment with which it interacts. The “player” eventually encounters good or bad rewards, and learns how to act to maximize the good rewards it receives.

Figure 1: Gameplay footage of a standard level of Overcooked. The dropoff location, on the left, is where players need to submit completed meal orders, in the top left. The rest of the level is home to treadmills that move the player around, and resources are spread far apart from each other to make the level more challenging.



This amounts to a massive amount of trial-and-error-type learning, and takes minutes to simulate a single second's worth of game-play; the Atari system was slowed down by multiple orders of magnitude, and it was not until late 2014 when IBM's Xiaoxiao Guo developed a more efficient, real-time algorithm that—while slightly less effective—was able to play live.³

We are met at a crossroads: we can either develop an algorithm that takes a long

time and plays a slowed-down version of the game, *or* we can prioritize an algorithm that plays real-time, albeit possibly suboptimally. Here, we will focus on the former to develop a “best” strategy, due to the generally messy nature of real-time strategy development.

In order to properly tackle this problem, we must first consider our approach. To procure the greatest number of meals in a set time frame, we will make some simplifying assumptions, and establish *how* we will frame the data to be inputted into this system:

The game board will be split into discrete tiles in a grid over the entire board, small enough that only one unit—either player, plate, or other utility on the board—can fit into each tile.

The board will be split into layers of the above grid, corresponding to the distinct types of entities on the board: one layer corresponding to tablespots; one for non-walk-

able hazards; one for each player; one for tomatoes; and so on. For each of these items, its corresponding grid copy will contain 1's in every position in which the item is contained; every other position, will be filled in with 0's (Fig. 2).

One time step will be defined as the amount of time it takes for a player to move from one grid tile to any adjacent tile.

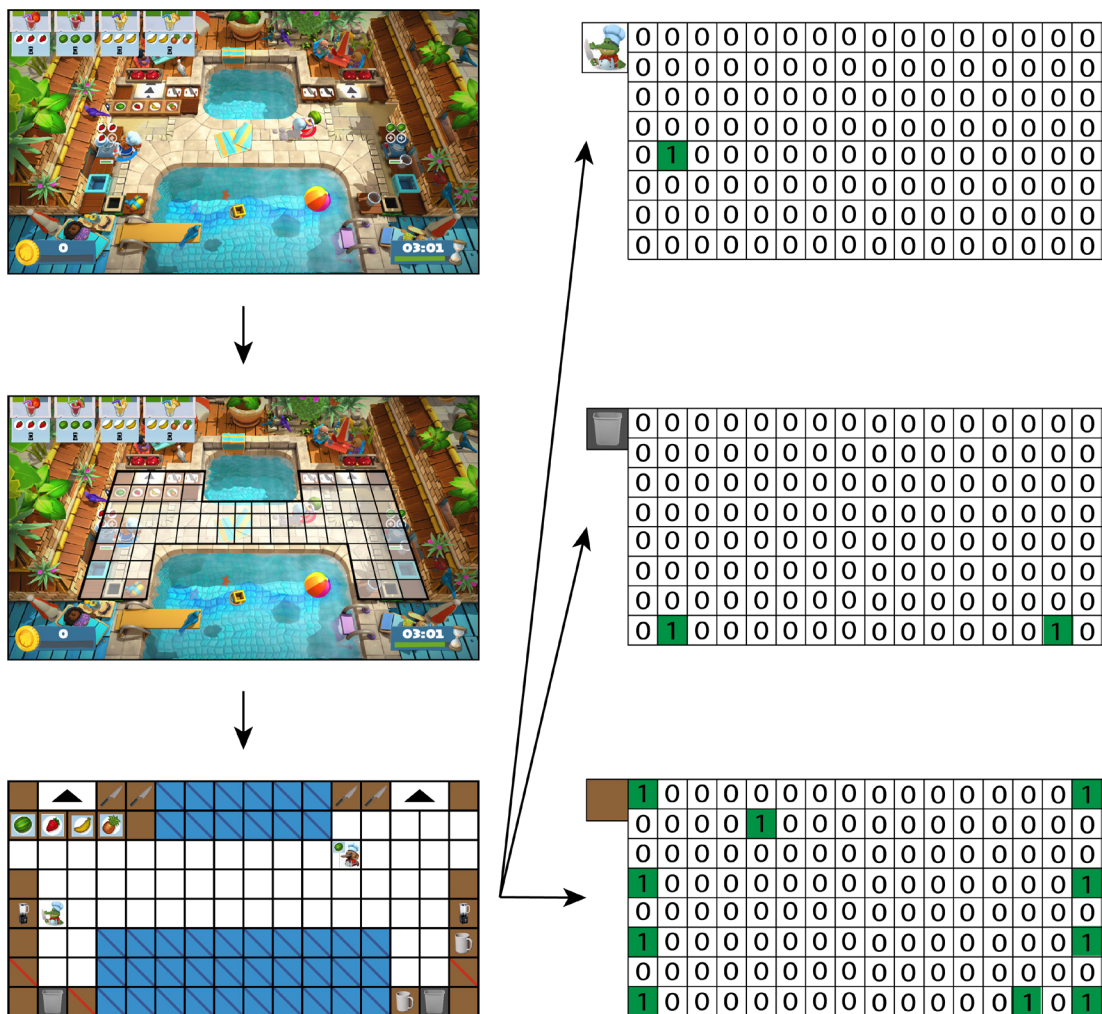
So, great. We've created a model for the system which speaks to where everything is on the board, and we've discretized the system such that players have a finite number of moves until the game is finished. However, the computer does not know what the objective of Overcooked is—it has been pre-programmed exclusively with knowledge of how to move and use other moveable items. So how can a computer choose the best ordering of actions, when it doesn't even know what it should be doing?

Simple: it won't. Not for a *long* time.

We have to start by making a guess of

“Naturally, this idea prompts the question: how can we develop the best strategy to play Overcooked?”

Figure 2: Each game board can be roughly modeled as a grid of tiles; to format this for input into our machine learning algorithm, we simply need to break the game down into its gridlike structure. From here, all that's left is to create a new copy of the grid for each type of tile, and label each tile that has the item with a “1,” and each that doesn't with a “0”.



"To accomplish this task, Minh utilized a class of machine learning known as reinforcement learning."

how to proceed from each possible configuration of positions, called a state, and updating our guesses as we proceed through simulations. Enter Q-Learning, an algorithm proposed in 1989 which does precisely this.^{4,5} More recent advances from 2002 by the University of York's Spiros Kapetanakis expanded Q-Learning to work with multiple players, allowing for this approach to work with Overcooked.⁶

Q-Learning is a standard reinforcement learning algorithm which identifies and determines the *value*, denoted with a Q , of every *action*, a , at every possible *state*, x . From here, we can determine the optimal action to take at any state, which is known as a *policy*. The value of each state-action pairing for a policy π is defined as follows:

$$Q^{\pi}(x, a) = \mathbb{R}_x(a) + \gamma \sum_y P_{xy}[\pi(x)] V^{\pi}(y).$$

By no means is this pleasant to look at, but the general takeaway here is as follows: the first term, $R_x(a)$, speaks to the reward obtained, purely from taking an action a at a state x . For example, dropping a filled order down into the dropoff location would yield a large reward, while cutting carrots or cooking meat would not achieve the same reward. This is not to say that these latter actions aren't valuable, though—their value shines in the second term of the equation. This term—which is admittedly complex—equals the potential of any given action to result in a downstream success. Note that this success is not guaranteed; each order that comes in is random, and so each action may be more or less successful based on the randomly generated requests. Regardless, this is how preparing food is useful—it sets up for future completed dishes to get even more points.

This is all splendid, but how do we use

this? In essence, it amounts to making random or slightly-educated guesses of how valuable each state-action pairing is, testing these out, and updating them according to whether we found them to be as useful as we initially thought they would be. For example, we might have initially thought that it would be very valuable to chop lettuce, but if our simulations reveal that there's only one very uncommon dish on that level that requires lettuce, we can safely say that we overestimated the value of chopping lettuce, and update accordingly.

So, there we have it—under the model we have constructed, we will be able to play the optimal game of Overcooked. It will take some time, but by playing several thousands of simulations of a given level, a computer will be able to roughly determine the values of each state-action pairing for the optimal policy. From here, we simply need to choose what Q-Learning believes to be the most valuable action for each state we're in until the end of the game.

This concludes our quest; from constructing our model, we have simulated several thousand rounds of play, developed an approximation for the value of every move, and used these approximations to make the best strategy in Overcooked. Armed with that knowledge, replacing your own player with the optimal computer may even leave your teammates happier without you.

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LABELING TUMORS WITH QUANTUM DOTS

THE FASCINATING WORLD OF NANOCRYSTALS AND QUANTUM DOTS

BY NACHIKET GIRISH

The prefix “nano-,” the word “quantum”—both are overused tropes in science fiction. Perhaps writers guess that they instantly make anything they are attached to sound futuristic and cutting-edge.

While in the movies these scientific-sounding prefixes are used to describe portals, time machines, and other fantastical objects, the real world is often even weirder than fantasy, and lends surprising plausibility to the existence of those clichés. Perhaps no single entity provides greater evidence of this weirdness than the baffling world of quantum dots.

DISCOVERING CONFINEMENT IN COLLOIDS

Louis Brus, born in Cleveland in 1943, might have been destined to be a chemical physicist. He attended Rice University on a Navy scholarship, where during his freshman year, a major in chemical physics was offered for the first time ever—an opportunity he pounced upon. After his graduation, he was commissioned to the Navy, but was given a special dispensation to go to graduate school instead. Later, having obtained a PhD in chemical physics

from Columbia University, Brus joined Bell Labs, one of the premier research centers in the world.¹

It was here in 1983 that Brus made the accidental observation which would become his most significant work. Brus was working on the redox reactions of organic molecules, reactions taking place on the surfaces of semiconductors. A crucial factor in these reactions was the semiconductor’s band gap, the energy the electrons need to begin conduction and moving charge. After testing various models, Brus realized that the organic reactions would occur faster if the molecules were suspended in a colloidal solution (a mixture of evenly dispersed particles suspended in a medium) so that they could bind to the semiconductor surface from all directions.

Upon performing the experiment in the colloid, however, Brus noticed something odd—the band gap of the semiconductor particles in the colloid seemed to be shrinking with time! Upon some more analysis, Brus was able to link this effect to an increase in the particle size inside the colloid over time. Particles would agglomerate and naturally get larger inside a colloid, and this increase in size was causing a decreasing band gap and, consequently, increased conductivity of



the semiconductor. Brus had just observed a system whose energetic properties depended on its physical size—a system which we now know as a quantum dot. “We realized,” says Brus in his autobiography, “that this was (and is) an excellent research problem.” In 2006, Brus, along with the other fathers of quantum dots, scientists Alexei Ekimov and Alexander Efros, won the R. Wood Prize of the Optical Society of America. He probably would not have dreamed, however, that his excellent research problem would, among other things, find applications in cancer research and renewable energy.²

BUT FIRST—WHAT ARE QUANTUM DOTS?

A quantum dot at the most fundamental level is just a particle bouncing around in a box. That’s it. Apart from size, there are few other special conditions to their existence. This, however, is the puzzling, unintuitive beauty of quantum mechanics—just going down a few scales of size takes us to a whole new world. In this case, it is a world where we can make that box emit different colors of light just by squeezing it. To understand the particle in a box, we need to understand the basic tenets of quantum mechanics.

The development of quantum mechanics was first motivated by the observation that particles often behave

surprisingly like waves, and vice versa. The particle in a box model can be understood if we consider the particle to not actually be a particle, but a wave reflecting back and forth in a box. In most cases, a wave bouncing around inside a box would be a confusing mess—think of water sloshing around in a bathtub. There is one special condition, however, which brings order to this chaos: if the waves have wavelengths such that the crest of the incoming wave overlaps exactly with the crest of the reflected wave, they add up quite neatly to form a stable configuration known as a standing wave.

Now, our box can only manage to exactly overlap incoming and reflected waves if it is large enough to fit a whole crest, or an integer number of whole crests exactly within it (Fig. 1). And since the wavelength of the particle depends on its energy, for a given size of a box we thus have a set of special energies for which the associated wavelength (and an integer number of them) matches the size of the box. The particle inside the box can only ever possess one of these special energies as its energy value, no other.³

If a particle gains some energy, it will emit the same energy, in the form of a definite wavelength of light. And since we know that for different sizes of the box we have different corresponding sets of special energies, the upshot of this becomes that for

“Brus had just observed a system whose energetic properties depended on its physical size—a system which we now know as a quantum dot.”

different sizes of the box, we can make the particle emit light of different wavelengths. This is the fundamental property which makes quantum dots so fascinating—by tuning the sizes of particle boxes, we can control their color!^{4,5}

BIO-INDICATOR DOTS

The discovery that nanoparticles show quantum confinement (the technical term for the effect discussed above) has led to a wide spectrum of applications. Probably the most fascinating is the use of quantum dots for biomedical imaging.

Biomedical imaging is the technique of looking inside the human body without cutting it open. It’s not as exotic as the name might suggest—the X-ray and CT scan are both types of biomedical imaging techniques.

Biomedical imaging is extremely useful for targeted uses, such as when we need to spot tumors among healthy tissue. The name of the game is to find a molecule that has an affinity to the tumor cells and attach to it a label dye, a compound which gives off radiation we can detect. Once released into the body through one of several possible delivery systems, these labeled molecules reach tumor cells and bind to them, thus indirectly attaching to them the radiation-emitting label dye. Doctors can then analyze the radiation being emitted to get a real-time display of where the tumor cells are in the body.⁶

Organic label dyes have traditionally been used for quite some time now; however, most of these dyes emit optical radiation, i.e., visible light, which gets easily blocked/absorbed by the tissues surrounding them.

This is where quantum dots come in. Since they offer dyes with emission ranges in the Near Infrared (NIR) range, which

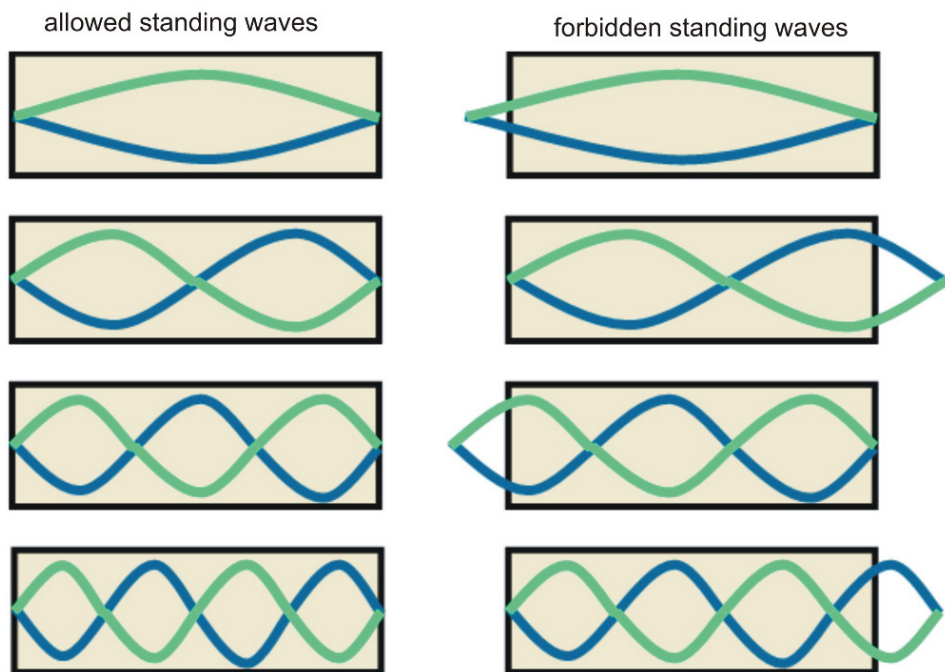


Figure 1: Notice that all the allowed standing waves fit perfectly inside the box, while all the forbidden ones have a partial crest sticking out.

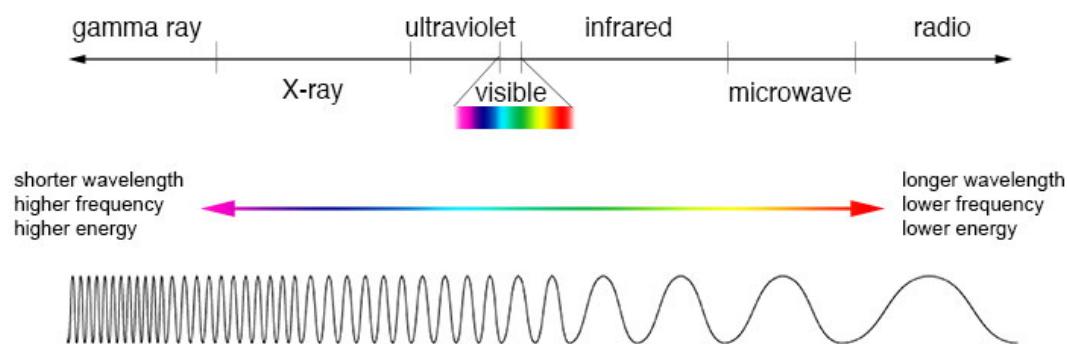


Figure 2: Light of higher wavelength, as shown here, has lower energy and is closer to the red end of the spectrum. Shorter wavelength (and hence higher frequency) light has more energy and is more towards the blue end of the spectrum.

is sparsely blocked by tissue, they are transmitted through living cells, enabling easy detection by instruments (Fig. 2).⁷ Moreover, since the emission wavelength of quantum dots is so closely linked to their size, by controlling the size of the quantum dots we can very finely tune the emission wavelength to our desired value. They also fluoresce for several hours and do not disperse away from the tumor into the body as quickly as conventional label dyes, so that surgeons have much more time to carry out operations.⁸ Of course, this technology is still relatively new, and there are still challenges to overcome. Universal protocols involving the use of quantum dots have still not been developed, making their present use challenging. Moreover, their relatively large size compared to conventional label dyes raises potential complications in accessing cellular targets. Nevertheless, their myriad advantages mean that we can expect their use to only grow in the future.⁹

AN EXCITING TIME AHEAD

There are simply far too many exciting possibilities for quantum dots to cover in a few pages. They are revolutionizing solar panel technology, they are being used to develop effective indoor farming, and they are at the center of Samsung's fight for dominance against LG in the television industry—an industry where quantum-dot-enabled technologies such as emissive quantum dot LED displays and MicroLED promise to bring in a new era of display technology.¹⁰ What we can certainly be sure of, however, is that the future of quantum dots holds nothing but exciting possibilities.

Acknowledgements: I would like to thank my physics professor Matthias Reinsch for reviewing the physics of this article.

"The name of the game is to find a molecule that has an affinity to the tumor cells and attach to it a label dye, a compound which gives off radiation we can detect."

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CARBON NANOTUBE FORESTS: THE NEXT STEP IN ENERGY STORAGE?

INTERVIEW WITH DR. WAQAS KHALID

BY SHARON BINOY, ANANYA KRISHNAPURA, ESTHER LIM,
ELETTRA PRESOTI, MELANIE RUSSO, AND ROSA LEE



Dr. Waqas Khalid¹

***Dr. Waqas Khalid** is a research scholar at the Lawrence Berkeley National Laboratory (LBNL), a project manager at the UC Berkeley College of Engineering, the founder and CEO of the nanotechnology startup company Jadoo Technologies, and a lecturer in the UC Berkeley Department of Physics. In this interview, we discuss Dr. Khalid's work on optimizing and employing carbon nanotube forests for energy storage, biosensing, and beyond.*

BSJ: How did your early experiences growing up in Pakistan shape your scientific career, and what led you to continue those studies in the U.S.?

WK: I was a very smart kid and a big time mama's boy. The goal was to be the best in school, athletics, and extra-curricular activities. I did the British education system, the A levels, when I was in Pakistan. After that, I got into a very elite high school. There, the goal was to go to the U.S. for higher education. I was actually asked to apply, but my mom flipped out. She said, "Oh, no, you're not gonna go there. You're never going to come back." My dad was okay with it. It took a whole weekend to convince my mom to allow me to go to the U.S. Eventually she agreed, and I flew here. So my drive for excellence and scientific and academic innovation was always part of my being, so to speak.

BSJ: Why did you go into nanotechnology?

WK: I always wanted to do something really cool and fantastic. I got into a school called Wayne State University in Detroit, Michigan. After the first semester, I got a Global Spartan Scholarship and ended up at Michigan State University. My commute was 200 miles a day, every day, for two semesters. I bought a \$1,000 car and I used it to commute, but after two semesters, the car broke down and I ended up having to move back to Wayne State.

While I was at Wayne State, a lot of things happened. Serendipity, I guess. A company called Delphi, which had been a part of General Motors (GM), got kicked out of GM. They spun out as a company of their own, and they built a state-of-the-art cleanroom for micro-fabrication at Wayne State. Then, a new professor from Caltech joined the faculty at Wayne State. He brought his projects from Caltech on making smart skins and biosensors using nanotechnology. I took a MEMS [micro-electromechanical systems] class with him. I ended up working with him for my PhD. I did my undergraduate, master's, and PhD degrees in five and a half years, all three degrees.

BSJ: What are carbon nanotube (CNT) forests, and what are their current applications in nanotechnology?

WK: Graphene is the magic material that everybody knows about. Carbon nanotubes were the predecessors of graphene technology. A sheet of single carbon atoms in a straight format is called graphene. If you roll that sheet into a tube, you will create carbon nanotubes. They have been around for almost 30 years now. They were the promised material before graphene came along, and they have really great electrical properties. They're super good conductors, and depending on how they're folded, they can be semiconductors as well. They can come in an onion-like format, with multiple layers of nanotubes around each other forming multi-walled carbon nanotubes. They can be grown in populations, called "forests," and they also have great optical and thermal properties and a huge surface area.

BSJ: How did you fabricate the CNT forest on chip?

WK: The standard state-of-the-art process is to grow the nanotubes on a surface, peel them from that surface, mix them in a solution, and then use them. I wanted to do an experiment where we grow two pillars of nanostructures, charge one pillar with positive charge and one pillar with negative charge, and electrostatically actuate them. I did some of this work at the University of Michigan, and then I went back to Saudi Arabia to test these devices. They short-circuited through the non-conducting substrate that we had on our devices, the silicon wafer. That was a failed experiment. So from Saudi Arabia, I moved to Chalmers University of Technology in Sweden. There, I built new devices, and then I started a company based on this technology. After four hours of really gruesome discussion with folks at Chalmers Innovation, an incubator for technologies and startups, we realized that nobody in the world has actually ever grown nanostructures that short-circuit through the substrate, and there is no literature about it. I'm the only person who has done it and who can do it. Nobody knew how to grow nanotubes with a good metal contact. The discussion started with questions like, "How do you do it? Why does this happen? What is going on?" I got extremely curious. I thought, "Why don't we do some more experiments and figure out why this happens?" That was the bulk of my work in Sweden. That's where we filed the patents on this innovative idea, detailing how we could build and apply this technology. All great technology is developed by failed experiments and accidents. This is a prime example.

BSJ: What is the purpose of chemically functionalizing CNTs? What are the benefits of functionalizing CNTs on chip, as opposed to conventional methods where CNTs are modified in solution before being redeposited on chip?

WK: When nanotubes grow on a surface, the surface usually contains some metal catalysts. To functionalize carbon nanotubes, people use acids like nitric acid or sulfuric acid to damage the carbon in the nanotubes and make holes. On these

holes, -COOH or -OH groups can attach to the nanotubes and functionalize them (Fig. 1). But in this process, the acid actually etches the metal layer and peels the nanotubes from the surface, causing them to float in solution. So this is why most materials that are made of nanotubes are grown on a surface, mixed in a solution, and then redeposited again; it's so they can be functionalized. This was the only way they could be used, technically. While I was in Sweden, this was a huge problem for us as well. At Chalmers University, I worked with Bengt Nordén, who is the chairman of the Nobel Prize committee in Chemistry. He had become a good friend of mine; I met him in Saudi Arabia, and I ended up coming to Sweden to work with him. We came up with a process where we could actually functionalize the nanotubes while they were still on the surface, without peeling them off. We filed a patent on that as well. This protocol now allows you to not only grow the nanotubes in a unique fashion, but also maintain electrical contacts, all in a very cost-effective and simple fabrication process. After growing them on a surface, you can then functionalize them directly on that surface without them peeling off. It allows you to create a very powerful component that can then be used as a powerful device for a lot of different applications in energy storage, biosensing, field emission, or energy harvesting.

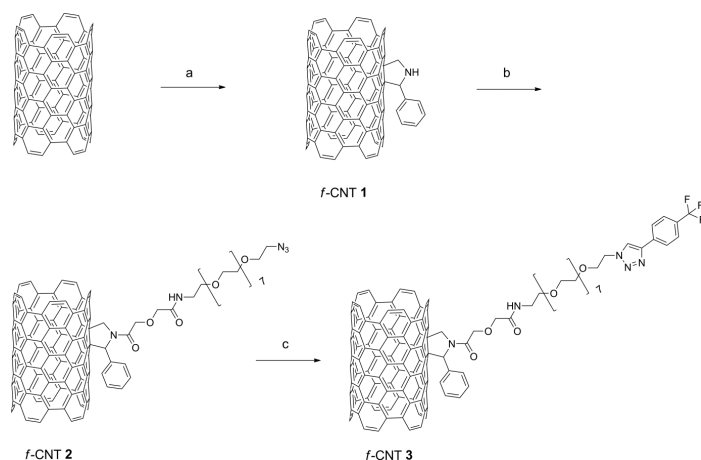


Figure 1: Carbon nanotubes functionalized with different chemical groups.²

BSJ: You confirmed functionalization by using scanning electron microscopy (SEM), equipped with energy dispersive X-ray spectroscopy (EDS). Could you briefly explain for readers what SEM and EDS are and how they were used to confirm functionalization?

WK: When you go to the nano- and micro-scale, things are so small that light cannot really give you a good image. So we use electron microscopes. They're pretty big, and they have a vacuum chamber where you suck all the air out and then introduce an electron beam. The electron beam hits the surface of your sample and reflects from the surface in a way that is then picked up by specific detectors. You then have a software that creates an image of the surface. Because electrons are super small, you can get really fine resolution of the surface (Fig. 2).

Once we have an image, the question becomes, “How do we know what materials are on that surface?” To answer this question we can use spectroscopic techniques. We can detect chemicals by the interaction of different waveforms on the surface. Then, different analyzers take the data, and a powerful software can then produce an image or a spectrum to let you know what is going on. EDS does all of that. X-rays fall on the surface, and that interaction with the surface gives you the material properties. If you have a compound with carbon, nitrogen, oxygen, iron, or any other kinds of elements, you will get a spectrum with a peak for each of those particular elements, signifying their presence. This was the technique that we used to confirm that we were actually able to functionalize the nanotubes specifically on a surface.

BSJ: You noted from your SEM data that the technique you employed to functionalize CNT forest on chip preserves the morphology of the CNT forests. What implications does this hold for potential technological applications of your method?

WK: I’ll give you an example. You have a lawn outside. You mow the lawn, take the grass out, and put it on your living room floor. Does that grass have the same morphology and alignment as it did when it was outside in the lawn? Probably not, because grass grows very directionally. It grows vertically, perpendicular to the ground, and it’s nicely organized in a range. But when you chop it off and put it on a surface, the leaves fall flat, and they’re parallel to the surface. That’s what we mean when we say “morphology.” So when you grow nanotubes on a surface, mix them in a solution, and re-deposit them, it’s like the grass from your lawn on your living room floor. In our fabrication process, we can define where to grow these nanostructures—where to grow these grass patches. We provide essential electrical contact with these small patches; it’s like having specific plumbing to water each of these patches individually, wherever they’re grown. Now, this allows you great morphology because you have structures which do not get destroyed and are exactly where you want them to be, since you can define where they are. This fabrication process gives you a platform where you have individually-addressable structures that are very well-defined. Then, you can tap them electrically and chemically to use them in various applications.

BSJ: Continuing our discussion on applications of CNT forest on chip, we wanted to talk to you about your work on energy storage using nanotechnology with Jadoo Technologies. To start, what is the mission of your startup company, Jadoo Technologies?

WK: I started this company a while ago. How do you take a cool concept and build an actual product out of it? Given that I finished school so quickly and developed this technology while I was really young, I was ambitious and enthusiastic. I thought that we could change the world by doing things in a new way. While I was in Sweden, I was offered 15 million Swedish crowns, which is around \$3 million, to build this idea and show a working prototype of the technology. But I was of the idea that we could do this in collaboration with academia. Why did we need to burn all this investment capital to get data and lose so much equity early on? We could get the scientific feasibility testing done in a more efficient and effective fashion by grants and collaboration with academia. We could publish papers, distribute the science, and share it with students, rather than having a venture capitalist hold a gun to our head and tell us, “You can only do this, you should not look at any other applications.”

I did a lot of this work while I was in Sweden. I ended up coming to Lawrence Berkeley National Labs to work with Paul Alivisatos, who was the Director of LBNL and is now the Executive Vice Chancellor and Provost at Berkeley. I raised a little grant money here and there and worked with NASA, because NASA wanted to use this technology for monitoring human health in space applications and the human mission to Mars.

While at Berkeley, I started a URAP [Undergraduate Research Apprenticeship Program] project. Since everything was patented, we could use the learning process of rapid prototyping to teach students while getting things done. I became a part of a center called COINS [Center of Integrated Nanomechanical Systems], where a part of my duties was to develop an ecosystem to take technologies developed in an academic environment to an entrepreneurship and industry level.

So, I built the interface for my technology. But to test these things, we had to use these huge commercially available instruments, about the size of a microwave. Okay, you have this very

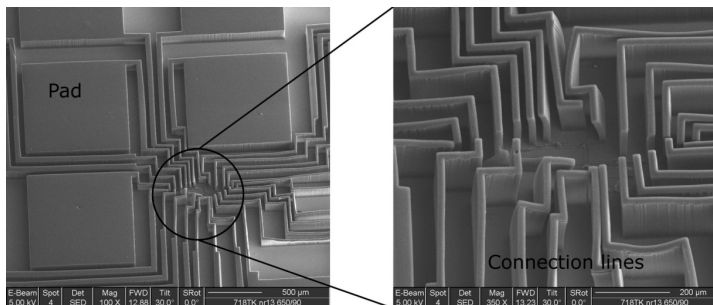


Figure 2a: SEM micrograph of CNT forest on chip prior to functionalization. The CNTs have been grown in a pattern containing pads and connection lines.²

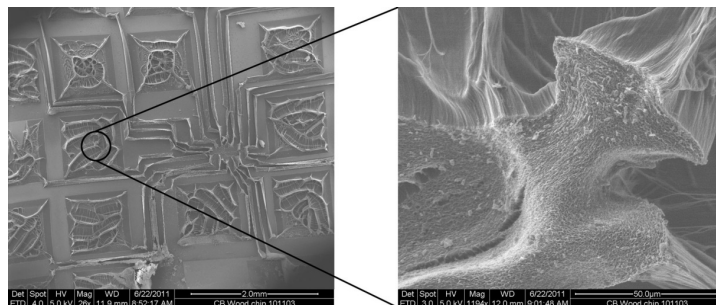


Figure 2b: SEM micrograph of CNT forest on chip after functionalization with an amine group (f-CNT 1 from Fig. 1). Note the preservation of overall morphology: the connection lines to the pads remain well-defined.²

small sensing array which can fit on the tip of a hair and has hundreds of sensors. But to use it you need to have something the size of a microwave attached to it, which does not make sense. So, we embarked on the journey of building our own electronics. We also built our own analytical software, using artificial intelligence and machine learning. So now, we have a complete system: the nanotechnology, the interfaces, the electronics, and the software.

This was all done in collaboration with the university, and it's all open source—all this work is shared with researchers at NASA and Berkeley. We are trying to promote the idea that you should share the knowledge, so that rather than everybody doing the same thing over and over again, we can be more efficient. This is one of the many aspects I included when building this new ecosystem, where you could actually mitigate technological and financial risks and create pathways on how to take your product to market. That's the crux of Jadoo Technologies.

BSJ: What are the advantages of using nanocapacitors to store energy as opposed to more conventional modes of energy storage such as lithium ion batteries or supercapacitors?

WK: The idea is the following: energy storage is a huge problem that humans are facing in the modern world. The evolution of clean energy, like solar panels and wind power, are transforming the energy generation landscape. But to store energy, we all use the recently Nobel Prize-awarded technology of lithium-ion batteries. The problems with this technology are that it's slow to charge, it has a limited number of cycles of charging and discharging, and lithium is a limited resource. So, the question becomes: how do you create a new technology platform where you can store charge in a much safer, faster, and cleaner fashion? Once these batteries die out, we'll be left with a huge amount of e-waste that we'll have to clean up again.

I taught a class at Berkeley where students reviewed the different potential applications of our technology. Energy storage was a winner. Our technology can provide a very unique storage solution. You can create these capacitors, called nanocapacitors, and you can put a lot of them together on a smaller chip. That will give you a capacitance unlike anything else previously built. We were very enthusiastic, performing a lot of calculations and simulations to show that our technology could actually work. Then, I gained access to Lawrence Berkeley National Labs through a proposal process, where I built a few prototypes and showed that the technology really works. We also got some preliminary data, which looks extremely promising. On the size of a postage stamp, we can store enough charge to power toys and electronics, more than a 9V battery. So, if successful, we could actually create an energy storage solution that could charge instantly. You can create wireless energy charging protocols with this technology. The problem is, many people think our technology is too good to be true. Everybody wants to see a working prototype and more data. Who knows what will happen? The best thing is to show a working prototype.

There are also other interesting electric materials coming out in the market called colossal dielectric materials. These materials

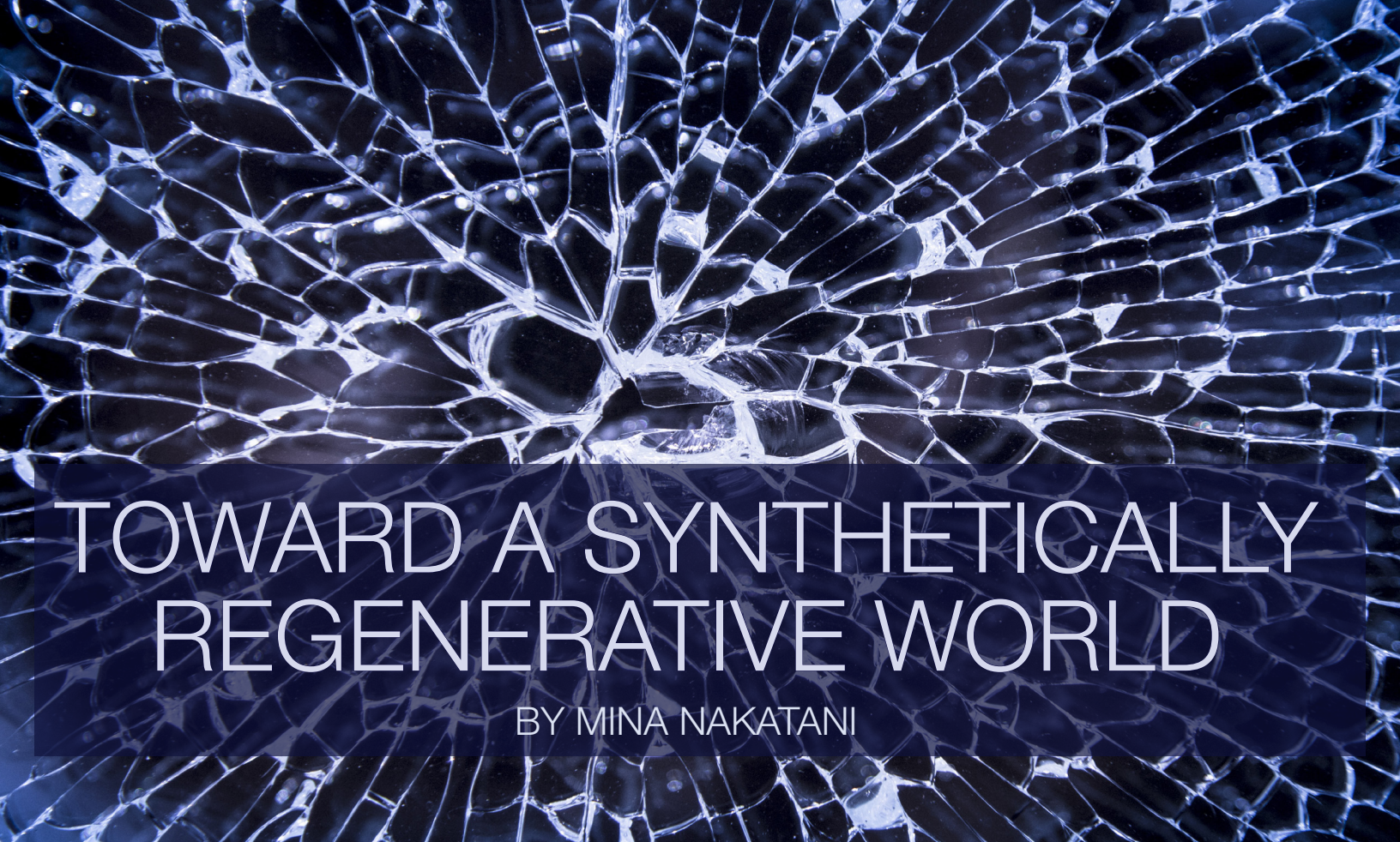
don't really exist in nature. What we do is put atomic layers of these materials on our nanostructures and build different materials on top of each other, creating a new kind of material that has really tremendous energy storage properties. Combinations of these cutting-edge technologies allows us to build an energy storage solution that can revolutionize the future of energy storage if successful.

BSJ: Could you tell us about your journey as an entrepreneur? What unique challenges have you faced along the way?

WK: Life has been full of interesting twists and turns. Once I finished my PhD, I moved to California and started working at a company called Brocade Communications. I went to Mexico to get my H-1 visa situation sorted out. Somebody at the U.S. border detained me and put me in detention for 37 days. They deported me to Pakistan and gave me a five-year ban from the U.S. I was lucky enough that I had a visa for Canada. I ended up in Canada. The U.S. government later said somebody made a mistake, and they fixed the situation and removed my ban. I lost my job during this process, but while I was in Canada, I started working with the University of British Columbia and got exposed to carbon nanotubes. Bringing in my knowledge of MEMS, we developed some biosensors and got some really cool data. This led to me moving to KAUST [King Abdullah University of Science and Technology] in Saudi Arabia and continuing my entrepreneurial endeavors, trying to see how we could build this new idea that I had. These events led to me interacting with Bengt Nordén, to my work in Sweden building up my IP portfolio, coming here to Berkeley and LBNL, meeting NASA folks; life just took its own turns.

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TOWARD A SYNTHETICALLY REGENERATIVE WORLD

BY MINA NAKATANI

Everyone is familiar with the experience of dropping their phone—the all-too-relatable moment of fear that one will pick it up to find a spiderweb of cracks across the screen. We fear seeing our own possessions broken, whether that be through careless accidents or deterioration over time. This process of deterioration is intrinsic to many other man-made objects, ranging from the parts holding together a car or the internal metal beams bolstering bridges and skyscrapers. At some point, these structures will inevitably require replacement and maintenance. Although traditional engineering has worked to lengthen the life of such materials, biology offers new ideas for further improving their properties, one of which is the biological process of self-healing.

Typically, science and engineering have been focused on creating parts and products that can withstand damage, whether by the use of more intrinsically robust materials or through design informed by the evaluation of material properties.^{1,2} Biological systems, however, offer an elegant way

to keep materials viable through the use of self-healing. In the body, self-healing is a ubiquitous process; a cut will seal itself with a blood clot allowing the skin to mend, and a broken bone will fuse back together after it fractures.³ Granted, these organic systems and tissues are inherently different from those typically used in engineering—on a chemical level, organic tissues rely on largely non-metal elements such as carbon and nitrogen, while building materials tend to utilize metals like iron and aluminum

(Fig 1).⁴ But even though the actual mechanisms of biological self-healing are rather complex, its general concept can still guide advancements in the development of synthetic materials.³

Self-healing, in terms of synthetic materials, refers to the ability of a material to return to its initial state with its original properties.⁵ Moreover, an ideal self-healing material would be able to carry out this process without any human intervention.³ It would be, for instance, as if the cracks on

Figure 1: Materials like iron constitute most of the elemental content in support beams for large structures. Currently, no means of self-renewal for such materials/structures exist.



"As the cracks mend, most properties of the original material are restored, even over the course of multiple tests."

a phone screen closed of their own accord, restoring the touch screen to its original state. Although such a material has not yet been designed, a similar phenomenon has been observed on a smaller scale with polymers.

Polymers encompass a large range of compounds that make up various plastics and rubbers, among other materials. Structurally, polymers are similar to a beaded bracelet: a long chain of small, linked components, each contributing its own unique properties to the whole structure. Due to the many ways in which these components can be organized, polymers can be easily tuned to fit specific applications.⁴ Furthermore, some polymers have exhibited self-healing mechanisms, with cracks in pieces of stretched polypropylene (PP) or poly(methyl methacrylate) (PMMA) healing after being heated up, restoring the elastic properties of the original material (Fig. 2).⁵ This instance of a polymeric property is a promising indicator of the potential development of self-healing technologies.⁵

These observations have led to the synthesis of materials meant to heal themselves in response to cracks or tears. One such system utilizes tiny capsules embedded within

a polymer such as poly(dimethyl siloxane), more widely known as silicone.⁶ These tiny capsules are filled with liquid silicone and a catalyst which, when mixed together, polymerize and harden.⁶ When the silicone polymer is torn, the capsules break and release their liquid components, filling in the gaps caused by the initial damage.⁶ As the cracks mend, most properties of the original material are restored, even over the course of multiple tests.⁶

However, not all polymers need capsules to self-heal. Instead, if self-healing is understood as merely the breaking and reforming of chemical bonds, then bonding mechanisms can be manipulated to create polymers that can repair themselves constantly.⁴ Take, for instance, a single bond connecting two sulfur atoms within a crosslinked polymer, constructs with additional linkages connecting polymeric chains together. This bond can be severed, splitting the original polymer in half.⁷ The polymer, however, is not necessarily beyond repair. Rather, the split polymer is akin to two sheets of Velcro that have been ripped apart; the two ends—analogueous to our two sulfur atoms—may no longer be in contact, but they can be easily reattached

by bringing the two Velcro strips back together. Reforming the sulfur-sulfur bond is conceptually similar: bringing the atoms into proximity results in a polymer chemically identical to its initial state, satisfying the basic requirements of self-healing. Of course, in practice this process is more complex—the sulfur-sulfur bond cannot reform in the presence of oxygen, and other similar processes may require manipulated environmental conditions in order to occur.⁷ Additionally, the two ends are not guaranteed to find each other quickly, in which case they will be unable to reform the cleaved bond.⁷ This problem can be

"But while this particular type of polymer is still in the early stages of its development, its current abilities demonstrate a significant step forward in the creation of self-healing materials, putting synthetic materials closer to the functional scope of their organic counterparts."



Figure 2: Rubbers such as polypropylene (PP) are considered polymers. The cracks in a large piece of rubber were found to seal themselves when heated, a basic form of self-healing.

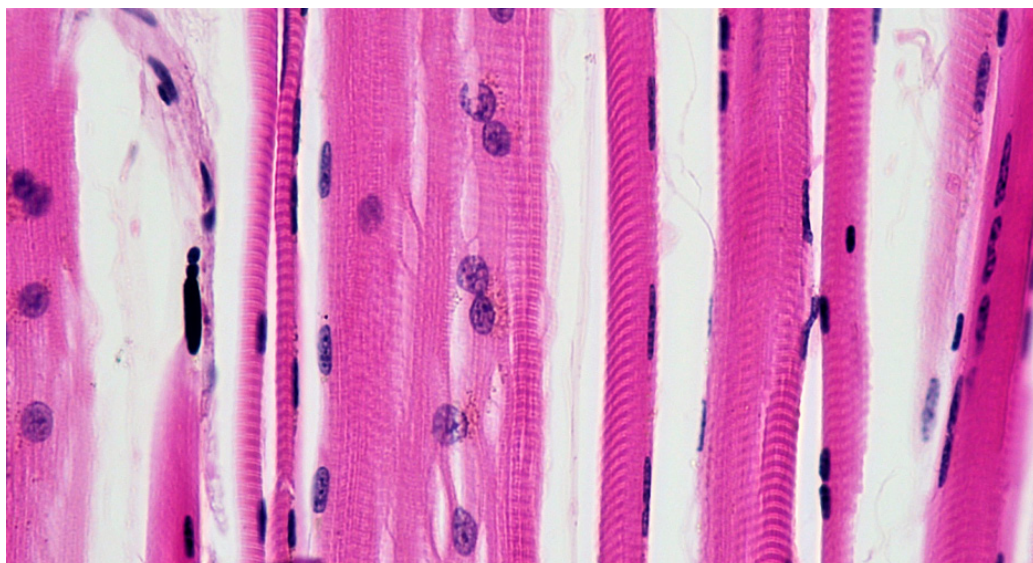


Figure 3: Image of muscle fibers, the biological tissue in which titin plays an important role due to its ability to heal upon fiber stretching and breaking.

solved, however, by adding different types of molecules with which the polymer ends can react and recombine. Such an addition increases the chance that two compatible atoms will meet, facilitating the self-healing process.⁷

Coming full circle to their biology-inspired roots, self-healing polymers have also been used in the creation of artificial muscle fibers. In particular, polymeric materials have demonstrated their capacity to act in ways similar to titin, an important molecule in muscle fibers (Fig. 3).⁸ Titin has been studied due to its self-healing properties in muscles as they absorb energy and break down during physical exertion.⁸ In order to replicate this process, researchers have created crosslinked polymers containing bonds that are able to undergo cleavage and subsequent restoration.⁸ These bonds break when stretched—absorbing energy in the same way as titin—and subsequently repair themselves.⁸ The cycle is then repeated, in the same way that muscles are constantly being used.⁸ But while this particular type of polymer is still in the early stages of its development, its current abilities demonstrate a significant step forward in the creation of self-healing materials, putting synthetic materials closer to the functional scope of their organic counterparts.

Although self-healing materials are unlikely to become ubiquitous, they offer many possibilities for the development of stronger and more damage-resistant materials in the future. As self-healing materials continue to be developed, the lessons learned through their synthesis pave the

way towards a future where dropping your phone no longer needs to be a terrifying prospect.

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URBANIZATION DETERMINES THE ABUNDANCE OF DISEASE VECTOR MOSQUITOES IN MOOREA, FRENCH POLYNESIA

By: Jason Mark B. Soriano; Research Sponsor (PI): Dr. Brent Mishler

ABSTRACT Mosquitoes are important disease vectors of infectious diseases. The effect of anthropogenic habitat modification is essential to the abundance and dissemination of mosquitoes. However, little is known about anthropogenic habitat modification's effect in the context of remote, oceanic, and tropical islands. This study assessed the impact that habitat modification, specifically urbanization, had on different mosquito species on the island of Moorea, French Polynesia. We measured the abundance of three mosquito species across a gradient characterized by the quantity of built environment and human population density. This urbanization gradient was represented by four low-elevation sites along the coasts of Opunohu Bay and Cook's Bay. A cluster of four different mosquito traps were rotated between these sites in 24-hour intervals. A total of 874 individual mosquitoes were collected and *Aedes aegypti* and *Culex quinquefasciatus* were found to be more abundant in urban areas. *Aedes polynesiensis* was found to be more abundant in forested areas. When comparing abundance between the rainy and dry season, *Aedes aegypti* numbers were five times greater during the rainy season. *Culex quinquefasciatus* and *Aedes polynesiensis* numbers were less than they were in the dry season. These findings suggest that the effects of urbanization were species-specific and dependent on a nexus of socioecological factors, with significant correlations between mosquito abundance, urbanization, and seasonal climate conditions. The study also showed that different sampling methods designed to capture flying adult mosquitoes differed in both efficacy and selectivity, suggesting that trapping systems must be considered in establishing a sound assessment of mosquito community composition.

Major, Year, and Department: Molecular Environmental Biology, 3rd Year, Department of Environmental Sciences, Policy, and Management

INTRODUCTION

The persistence and emergence of mosquito-borne diseases is a significant public health concern.¹ In the face of climate change, rising temperatures and precipitation levels provide increasingly favorable environmental conditions in which mosquitoes may thrive and reproduce.² These conditions facilitate longer transmission seasons, increase biting frequency, and shift spatial patterns, which ultimately amplify the emergence of mosquito-borne diseases in novel host populations.³

Similarly, new pathways for invasion created by an increasingly globalized network of travel and trade have made it possible for many pathogens and their respective disease vectors, the transmitters of the pathogen, to occupy novel human-modified habitats.^{3,4} As the rate of introduction of new invasive alien species seems to be higher than ever before, it is critical to understand the factors that promote the colonization and coexistence of invasive disease-transmitting organisms. The resulting impacts of such invasions are of particular importance in the tropical regions of Sub-Saharan Africa, Central and South America, and South and Southeast Asia, where low-income communities are often disproportionately affected by the burden of mosquito-borne illnesses due to a number of socioecological factors.⁵

One such socioecological factor of considerable significance includes anthropogenic land-use change.⁶ According to the Intergovernmental Science-Policy Platform on Biodiversity and Ecosystem Services (IPBES), land-use change caused by human activity today has had the largest relative negative impact on the environment since that of the 1970s.⁷ Aside from agricultural ex-

pansion—the most common form of land-use change—the doubling of urbanization in urban areas since 1992 and the expansion of infrastructure are major contributors to the decline of biodiversity worldwide.⁷ These changes continue to impact the global ecosystem as the human population continues to grow, so it is essential to understand how different community assemblages react following these disturbances.

While the effect of habitat modifications on local ecosystems and community assemblages of disease vectors has been studied, there remains a lack of understanding of these effects specifically on remote oceanic islands.⁶ Previous studies on isolated island systems have analyzed larval mosquito distributions, interspecific competition, and the container types that are preferred oviposition sites for gravid females.^{8,9,10} However, little is known about how urban modification on islands affects the community structure of mosquitoes. This is especially important considering the rate at which human populations are growing in certain island communities. For instance, in the Commune of Moorea-Maiao, French Polynesia, the population experienced a 13% growth rate from 2002 to 2007, according to the French Polynesia Statistical Institute (2007).

The island of Moorea, French Polynesia is home to nine species of mosquitoes, including *Toxorhynchites amboinensis*, a species introduced in 1975 as a biocontrol agent against *Aedes* mosquitoes.^{8,11} Two of the nine species are more recent introductions and breed in artificial containers: *Aedes aegypti*, which is the primary disease vector of Dengue virus, and *Culex quinquefasciatus*, which is a significant vector of West Nile Virus in other parts of the world.^{10,11} *Aedes polynesiensis*, which was most likely introduced

by the early Polynesians, can carry *Wuchereria bancrofti*, a parasitic worm that causes lymphatic filariasis and elephantiasis, or the condition of extreme swelling and hardening of one's limbs.¹¹ The remaining six species are more elusive and are not recognized as salient disease vectors in Moorea.¹¹

Disease vector mosquitoes are especially important to study considering the history of infectious disease outbreaks that have occurred in French Polynesia in the past.¹² In 1996 and 1997, a dengue virus serotype 2 (DENV-2) outbreak arose in this region of the South Pacific. From October 2013 to April 2014, a Zika outbreak occurred, resulting in 66% of the entire population of French Polynesia to have been infected, followed by a number of microcephaly cases.¹² The Society Islands are currently undergoing another dengue outbreak, with at least 2400 cases of dengue fever between Tahiti and Moorea alone as of early January 2020.¹³ The outbreak history in French Polynesia prompts further research on not only mosquitoes and their disease vector capabilities, but also how the man-made environment contributes to the prolongation of mosquito-borne disease epidemics on remote oceanic islands.

The objective of this study is to understand the role of urbanization—in terms of the relative increase in both the human population and the built environment within a geographical location—on the community composition of mosquitoes, using the island of Moorea as a study system. This study aims to determine the relative abundance and distribution of existing mosquito species in Moorea by utilizing a system of mosquito traps to monitor host-seeking *Aedes* mosquitoes. Each trap within this system, incorporating a combination of two different trap types equipped with or without a human scent imitator, was then assessed upon its effectivity and selectivity in terms of capturing flying adult mosquitoes, regardless of sex. Due to inconsistent capture methodologies in previous Moorea-based studies, different trapping combinations were evaluated to the most effective method for future studies.^{8,9,10}

We postulate that the species introduced more recently to the island will predominate urban areas, characterized by more dense human populations and a more extensively built environment that supports a greater amount of artificial breeding sites. This may be due to a multitude of factors, but considering the absence of large mammalian species outside of human-inhabited areas on the island, urban areas may be the only, not just probable, sites for mosquitoes to find a blood meal.

METHODS

Study Site

This study took place on the island of Moorea, French Polynesia. We sampled four different coastal, low-elevation sites along the northern stretch of the island (Fig. 1). The climate of this region is characterized by high temperatures and humidity throughout the year, with daily temperatures averaging a maximum of 30°C and a minimum of 20°C. The rainy season occurs from November to March while the dry season extends from April to October.

At each of the four sampling sites, we captured flying adult mosquitoes using different combinations of Biogents (BG) mosquito traps for professional use and BG-Lures, which are small

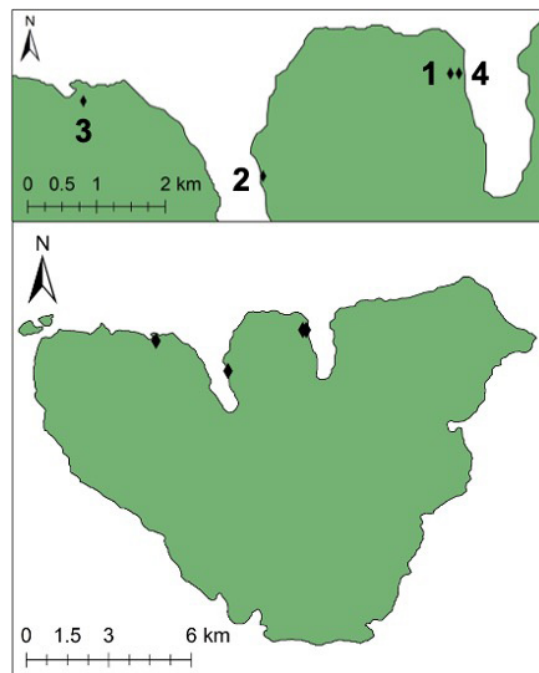


Figure 1: (Bottom) Outline of Moorea, French Polynesia marked with sampling sites. (Above) Close-up of sampling sites marked in order of increasing degrees of urbanization (1 least, 4 most urbanized).

cartridges of chemicals that emit artificial human skin odors and are effective for up to five months. The four trap-lure combinations used in this study were as follows: a BG-Sentinel Trap (BG), a BG-Sentinel Trap with a BG-Lure cartridge (BGL), an EVS-style BG-Pro Trap (MTS), and an EVS-style BG-Pro Trap with a BG-Lure cartridge (MTSL).

We deployed each trapping system onto tree branches or supporting artificial structures marked with colored tape, indicating which trap-lure combination was placed there. All four trapping systems were placed together at the same sampling site and hung approximately 1 meter off of the ground. Traps were placed approximately 30 meters apart from each other at the modified forest site and the Gump Research Station, and approximately 10 meters apart at the rural house sites due to limited space.

Mosquito abundance across a rural-to-urban gradient

For the first part of the study, we treated the four trapping systems as one large, conglomerate trap that was rotated between the four sites. Traps, lures, and batteries were provided by the Institut Louis Malardé (ILM) based in Tahiti. The sites were selected to represent an urbanization gradient. Degree of urbanization was established based upon visual inspection of factors including vegetation type and coverage, land-use, quantity and categorization of built environment, number of human inhabitants, and local human population density (Table 1). Each site was sampled three times and traps were deployed for 24-hours starting and ending at the same time between 0900 and 1600 h from October 14, 2019 to November 9, 2019 (Table 1).

After every 24-hour treatment, labeled catch nets were transferred to a small cardboard box, and then frozen along with their

Site	Description	GPS Location	Sampling Dates
1. Lowland, modified forest	Abandoned coconut plantation, <i>Hibiscus spp.</i> , <i>Coco nucifera</i> , high canopy coverage, large quantity of dead organic matter (DOM), no human inhabitants, bisected by a road.	17°29'25.9"S 149°49'40.9"W	Oct. 14-15, 20-21, 27-28
2. Rural House: Opunohu	Rural residential house in cul-de-sac-like neighborhood, <i>Hibiscus spp.</i> , <i>M. indica</i> , assortment of house plants and pottery, intermediate canopy coverage, ~5-10 human inhabitants, dogs present, children present, built environment consists of living quarters and roads.	17°30'14.0"S 149°51'09.0"W	Oct. 16-17, 22-23 Nov. 8-9
3. Rural House: Papeto'ai	Rural residential house in a relatively dense, "city" setting, <i>Hibiscus spp.</i> , <i>M. indica</i> , assortment of house plants and pottery, intermediate canopy coverage, ~2-3 human inhabitants, built environment consists of living quarters, nearby road, outhouses, and cement platform in yard.	17°29'38.8"S 149°52'33.5"W	Oct. 29-30 Nov. 3-4, 6-7
4. Gump Research Station	Coastal research station with well-manicured lawn, <i>Hibiscus spp.</i> , <i>C. nucifera</i> , <i>G. taitensis</i> , <i>A. altillis</i> , little to no canopy coverage, ~25-30 human inhabitants, high population density, dogs present, lots of built environment including dormitories, dry and wet laboratories, boating dock, and road.	17°29'25.7"S 149°49'36.8"W	Oct. 17-18, 21-22, 28-29

Table 1: Table of sampling sites and their descriptions.

contents for at least 1 hour. Individual mosquitoes were then identified by species and sexed using identification keys provided by the ILM. The abundance of non-mosquito species caught was also recorded. Although *A. aegypti* larvae were discovered at the Gump Research Station, larvae and rafts were not included in the study due to difficulty in finding them elsewhere.

Seasonal comparison of mosquito abundance

In order to assess mosquito abundance in dry and wet conditions, trapping systems were set around the Gump Research Station during the rainy season on November 12, 13, and 14. Catch abundance during this period was compared to that of the urbanization study data collected at the Gump Research Station during

the dry season in mid-October. The sampling approach was identical to that of examining a rural-to-urban gradient. Specifically, all four trap-lure combinations were considered one large, conglomerate trap in order to avoid trap bias and were deployed for 24-hour treatments, starting and ending at the same time between 0900 and 1100 h.

Trap-Lure system evaluation

This study evaluated the different trap-lure combinations in their effectiveness and selectivity in capturing mosquitoes of different species and sex by comparing abundance data for each trap-lure combination taken from both the rural-to-urban gradient and seasonal comparison traps.

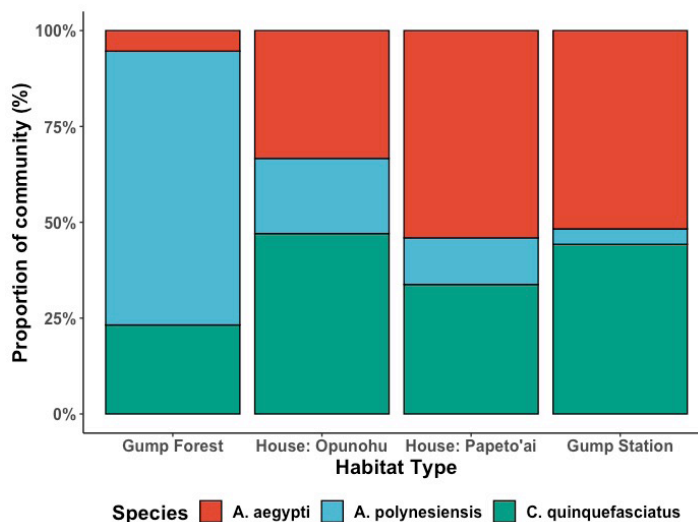


Figure 2: Overall mosquito community composition by species at each sampling site, organized from least to most urbanized.

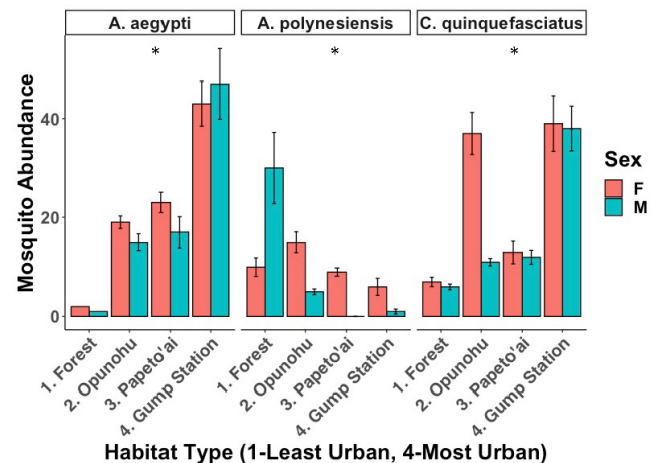


Figure 3: The effect of urbanization on mosquito abundance. The abundance of *A. aegypti* and *C. quinquefasciatus* are higher in urban areas, whereas the abundance of *A. polynesiensis* is higher in forested areas. Error bars show standard deviation (SD).

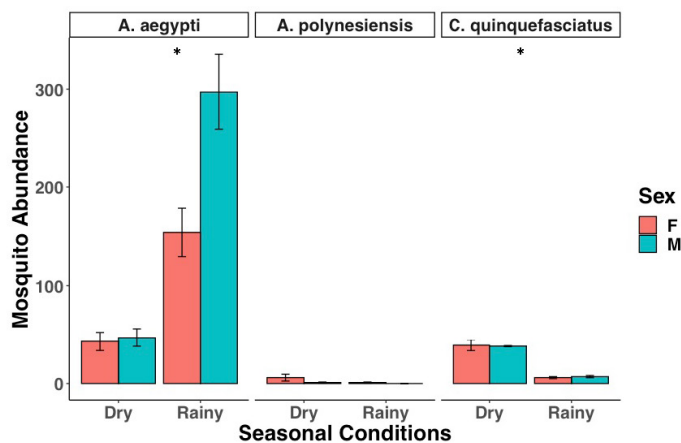


Figure 4: Mosquito abundance during the transition from the dry season to the rainy season. *A. aegypti* abundance increases while *A. polynesiensis* and *C. quinquefasciatus* decrease. Error bars show standard deviation (SD).

Statistical Analysis

To test for the differences in mosquito abundance, species richness, and sex across different urban habitat sites and captured via different trap-lure combinations, we used the χ^2 (Chi-Square) Test of Independence and Generalized Linear Mixed-Effects Models (GLMM) with Poisson link function in the R package lme4 (Bates et al., 2015). ANOVA tests were used to compare GLMM's with and without the variable of interest. All statistical analyses were conducted using RStudio version 1.2.1335 (RStudio, Inc © 2009), with $\alpha = 0.05$. Voucher specimens were deposited in the Essig Museum of Entomology (University of California, Berkeley, 94720).

RESULTS

Throughout the course of this study, we captured, identified, and sexed a total of 874 mosquitoes (Fig. 2). We observed *A. aegypti*, *A. polynesiensis*, and *C. quinquefasciatus* at all four sampling sites. Of all the caught mosquito species, *A. aegypti* ($n=618$) was the most abundant mosquito species, followed by *C. quinquefasciatus* ($n=176$), and *A. polynesiensis* ($n=77$). While *T. amboinensis* ($n=2$) and *Culex roseni* ($n=1$) were both present, these mosquito species were disregarded due to small sample sizes. Of all the mosquitoes captured at the lowland, modified forest region between the Gump Research Station and its hillside bungalows, *A. polynesiensis* (71.43%) was the most observed species, followed by *C. quinquefasciatus* (23.21%), and *A. aegypti* (5.36%). Of the mosquitoes trapped at the rural house located along the south eastern stretch of Opunohu Bay, *C. quinquefasciatus* was the most observed species (47.06%), followed by *A. aegypti* (33.33%), and *A. polynesiensis* (19.61%). At both the Gump Research Station and the rural house located along the outskirts of Papeto'ai, the most observed mosquito species were *A. aegypti* (house: 54.05%; station: 84.56%), followed by *C. quinquefasciatus* (house: 33.78%; station: 14.04%), and *A. polynesiensis* (house: 12.16%; station: 1.40%).

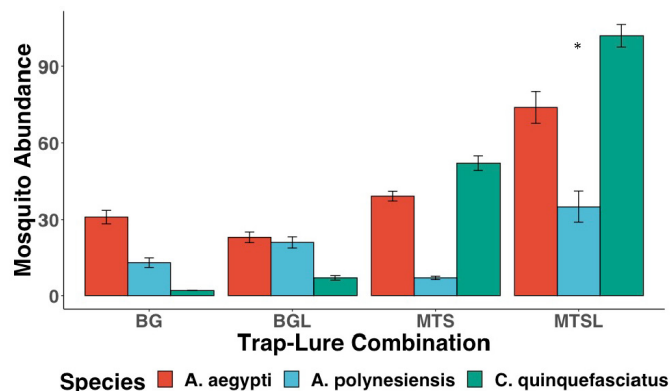


Figure 5: The evaluation of different mosquito trapping systems. The trap-lure combination that caught the greatest number of mosquitoes was the MTS trap equipped with a BG-Lure. Error bars show standard deviation (SD).

Mosquito abundance across a rural-to-urban gradient

The χ^2 Test of Independence showed significant differences between the abundance of the three different mosquito species and habitat type (Fig. 3, $\chi^2=138.31$, $df=6$, $p<0.001$). An ANOVA comparing two GLMMs, one with and one without habitat type, also showed significant differences (glmer, $df=3$, $p<0.001$). These tests determined that *A. aegypti* numbers increased significantly with increasing urbanization (glmer, $df=3$, $p<0.001$), as did the abundance of *C. quinquefasciatus* (glmer, $df=3$, $p<0.001$). On the other hand, *A. polynesiensis* numbers significantly decreased across this gradient (glmer, $df=3$, $p<0.001$).

Seasonal comparison of mosquito abundance

Significantly fewer mosquitoes were caught during the dry season than the rainy season (Fig. 4, glmer, $df=1$, $p<0.001$). The number of *A. aegypti* caught during the rainy season ($n=451$) was five times greater than that of the dry season ($n=90$) (glmer, $df=1$, $p<0.001$). The number of *A. polynesiensis* (dry: $n=7$; wet: $n=1$) decreased during the transition from dry to wet conditions, but did not display a significant difference (glmer, $df=1$, $p>0.05$). The number of *C. quinquefasciatus* (dry: $n=32$; wet: $n=13$) significantly decreased during this transition (glmer, $df=1$, $p<0.001$).

Trap-Lure system evaluation

The χ^2 Test of Independence showed significant differences between the abundance of the three different mosquito species and the trapping system used to capture the mosquitoes (Fig. 5, $\chi^2=64.491$, $df=6$, $p<0.001$). An ANOVA test comparing two GLMMs, one with and one without trap type, also showed significant differences (glmer, $df=3$, $p<0.001$). The MTSL trap-lure combination caught the largest total abundance of all three of the common mosquito species ($n=211$), followed by the MTS ($n=98$) and BGL traps ($n=51$). The BG trap caught the least mosquitoes overall ($n=46$). GLMMs indicated that the abundance of *A. aegypti* captured was significantly greater in the MTSL trapping system

(glmer, $df=3$, $p<0.001$). These tests also indicated that the abundance of both *C. quinquefasciatus* captured (glmer, $df=3$, $p<0.05$), and *A. polynesiensis* captured (glmer, $df=3$, $p<0.05$) were greater in the MTSL trapping system.

DISCUSSION

In terms of the community composition of mosquitoes found on Moorea, *A. aegypti*, *C. quinquefasciatus*, and *A. polynesiensis* were the most abundant mosquito species on the island, respectively. Other species, such as *T. amboinensis* and *C. roseni*, were rarely seen. Some mosquito species known to have inhabited the Society Islands, including *C. annulirostris*, *C. atriceps*, *C. kesseli*, and *A. edgari*, were unobserved.¹¹

The results of this study indicate that there is a correlation between the degree of urbanization and the abundance of mosquitoes. This correlation is specific to each species and in the context of remote, oceanic, tropical island systems. We found that *A. aegypti* and *C. quinquefasciatus*, two container-breeding species introduced most recently to Moorea, were more abundant in urban settings and increased in number along a rural-to-urban gradient. In contrast, we found that *A. polynesiensis*, introduced by early Polynesians, was more abundant in forest settings and decreased in number along the urbanization gradient. There was a greater abundance of mosquitoes overall in locations characterized by a higher human population density and a greater quantity of built environment. This supports the initial hypothesis made that *A. aegypti*, known to oviposit in artificial containers, and *C. quinquefasciatus* predominate urban areas due to a more extensive built environment that supports a greater amount of artificial breeding sites.

Mosquito abundance was also found to be correlated with environmental factors, such as seasonal climate conditions. During the transition from the dry season to the rainy season between October and November at the Gump Research Station, *A. aegypti* abundance rose five-fold while the abundance of *C. quinquefasciatus* and *A. polynesiensis* declined. It was also at this time towards the start of the rainy season that we found *A. aegypti* and *C. quinquefasciatus* mosquito larvae at two separate locations, both within the premises of the Gump Research Station. While larval sampling was disregarded in this study, future studies should incorporate larval sampling in addition to using trapping systems designed to catch flying adults. This will create a better and more comprehensive approach to studying the ecology of disease vector mosquitoes.

The results of the trap design evaluation showed that of the four different trap-lure combinations, the EVS-style BG-Pro trap equipped with a BG-Lure cartridge, or MTSL trap, was the most effective trap design for all of the observed mosquito species. The next best trapping system was the MTS trap without a lure, followed by the BG trap with a lure, and lastly the BG trap without a lure. These findings are consistent with the unpublished raw data released by Biogents in 2019 of a number of Latin square studies around the world indicating that the BG-Pro trap has been more effective than the BG-Sentinel when it comes to capturing *A. aegypti* mosquitoes.¹⁴ This may in fact be due to the improvements

in suction design and product quality on the BG-Pro, Biogents' newest development.

These trapping systems were far more efficient than those used in a previous trap-assessment study on Moorea, given that this past study was only able to capture 82 mosquitoes over 14 days, as opposed to the 874 mosquitoes caught in this study over 15 sampling days.¹⁵ Despite its small sample size, the results of the previous study aligned with ours in that *A. aegypti* and *C. quinquefasciatus* were caught most often at urban sites.¹⁵

A majority of the other mosquito-related studies on Moorea determined that rain-filled, rat-chewed coconuts were the most common oviposit sites for gravid female mosquitoes to lay their eggs.^{8,9} In terms of species eradication efforts and disease outbreak control programs, this would suggest that the removal of fallen coconut debris would be the most effective treatment.^{8,9} However, this solution may not be as effective as initially expected. Due to the understanding that *A. aegypti* predominate urban sites and *A. polynesiensis* predominate forest sites, the proposed removal of fallen coconut debris could potentially reduce natural breeding grounds of *A. polynesiensis*. Furthermore, backed by Pocock's conclusions that *A. polynesiensis* larvae outcompete *A. aegypti* larvae in natural containers when food is present, this decrease of natural containers would then reduce the pressure of spatial competition on *A. aegypti*, allowing this disease vector species to spread, which would only advance the spread of dengue fever in the South Pacific.¹⁰

It must be taken into account that disease vector distribution within this island system differs in continental systems. The trend established by this study that mosquito abundance is higher in urban areas than natural areas in tropical islands largely contrasts with the findings of similar studies in continental settings.⁷ In multiple regions of Europe including Spain, Belgium, and the Netherlands, mosquito communities are lower in abundance in urban and rural areas and higher in natural areas.^{16,17,18} However, it is recognized in these studies that these results are species-specific responses. The mosquito species that favor artificial habitats regardless of being in continental or island systems include *Anopheles*, *Culex*, and *Aedes*—three major disease vector mosquito species.⁶ This coincides with the findings of this study that disease vector mosquitoes, *C. quinquefasciatus* and *A. aegypti* in particular, had a positive relationship with urbanization.

Urbanization, however, is not the only form of anthropogenic land-use change influencing the distribution of disease vector mosquitoes and other invasive species. Increased trade, agricultural expansion, and other human population dynamics associated with urbanization have led to an increase in invasive species worldwide by 40% since 1980.⁷ This leaves nearly one-fifth of the Earth's surface at risk for biological invasions, putting economies as well as human health at risk.⁷ Therefore, it is more than necessary to be cognizant of this multitude of factors that come into play in this highly interconnected, globalized ecosystem.

Studying the factors that influence the abundance and distribution of mosquito vectors of human diseases also provides valuable information for public health management officials to identify priority areas for epidemiologic surveillance and vector control programs in preparation against large epidemics or outbreaks.

Having a better understanding of the associations between urbanization and the abundance and distribution of disease vector mosquitoes in Moorea can increase our awareness of how human-related changes such as urbanization are ultimately impacting the ecosystem and human health in the greater region of Oceania.

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Monitoring biodiversity and water pollution via high-throughput eDNA metabarcoding

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ABSTRACT

Traditionally, monitoring of water pollution via detection of bioindicators has been a tedious and time-consuming task that involves the use of stereomicroscope and morphological dichotomous keys. In contrast, with high-throughput eDNA metabarcoding, the identification of a bioindicator is carried out *in silico* in a cost- and time-effective manner. This study focuses on demonstrating the application of Illumina's MiSeq-based high-throughput amplicon sequencing to eDNA samples to characterize three freshwater ecosystems with substantial anthropogenic impacts in Berkeley, California. The method captures information that is indicative of the predator-prey relationships between arthropods and rotifers commonly found in freshwater ecosystems among all three habitats sampled. Furthermore, calculation of community pollution values revealed that Strawberry Creek possesses the worst water quality and ecosystem health across the three habitats sampled. A cost-benefit analysis demonstrated that the normalized cost per bioinformatic sample of high-throughput eDNA metabarcoding is estimated at 18.918 minutes and \$10 in terms of total time and cost, respectively. This is to be put in contrast with the at least 10-fold increase in requirements of 250 minutes and \$85 per traditional stereomicroscopic sample. We conclude with a novel quantitative approach that reproduces the same success regarding biodiversity documentation of a freshwater ecosystem elucidated by stereomicroscopic approaches using high-throughput eDNA metabarcoding in a cost- and time-efficient manner.

Major, Year, and Department: Computer Science and Microbial Biology, 3rd year, Departments of Integrative Biology and Statistics.

INTRODUCTION

Freshwater ecosystems are extremely vulnerable to anthropogenic impacts in a world of increasing human interventions. Species under such ecological environments are constantly threatened by the risk of an extinction event that results in drastic decline in biodiversity in the ecosystem. Thus, it is crucial to document the biodiversity of the ecosystem before it is permanently transformed beyond recovery. Living organisms shed traces of DNA in forms such as skin, scales, hair, and mucus from external interactions with the habitat, which accumulate in their surroundings and can be extracted as environmental DNA (eDNA). For clarification, we hereby adopt Thomsen and Willerslev's definition of eDNA when the term was first coined as: genetic material obtained directly

from environmental samples (soil, sediment, water, etc.) without any obvious signs of biological source material.⁴⁴ These traces of life provide information to predict and simulate the potential interspecies interactions within an ecosystem.

Under the prior approach, individual specimens must be carefully observed under the stereomicroscope prior to evaluating each of the criteria in the dichotomous keys. The use of dichotomous keys is oftentimes a complex and error-prone process that requires extensive knowledge into the clade that the specimen resides in. Additionally, establishing alternative methods for monitoring and documenting freshwater biodiversity is of high interest at the moment due to the decline in experienced taxonomic experts.^{49,50} The technology of next generation sequencing (NGS) differs from the current generation of serial DNA sequencing platforms in that it is capable of performing sequencing on millions of small DNA fragments in parallel.⁴⁶ This advancement has allowed researchers to evaluate millions of amplicon reads all within the setting of one experiment. Subsequently, the sample pool of amplicon reads is computationally aligned to known genomic sequences of bioindicators from the reference genome sequence database to determine the abundance of each bioindicator present within the sample. The bioinformatic approach for bioindicator monitoring is no longer susceptible to observer errors or limitations of the dichotomous keys, thus yielding a more precise measurement of water pollution than the traditional morphological approach.

One novel technique to approach biodiversity documentation is DNA metabarcoding, which aims to survey multispecies and high-level taxon biodiversity using typically degraded eDNA sampled under unknown ecological environments without the need of additional information regarding organismal composition in advance.⁴⁵ This is to be put in contrast with the standard method for DNA barcoding that focuses on identification of one predetermined species. In a DNA metabarcoding study, eDNA samples are first amplified by polymerase chain reaction (PCR). DNA metabarcoding utilizes universal primers that are complementary to highly conserved loci, thus allowing the sequence reads to be mapped back to the genome of most species when performing sequence alignments. With the rapid advancement of NGS, we hypothesize that documenting biodiversity by DNA metabarcoding requires less effort in terms of both cost and time than the stereomicroscopic approaches.

Past studies have struggled to augment the scale of the project due to PCR inhibition and the lack of techniques in producing high-throughput amplicon sequencing with universal primers on eDNA samples. In molecular biology, the throughput of a method is defined by the rate at which samples can be processed. More specifically, extensive eDNA metabarcoding studies in the past have

often succumbed to the presence of compounds that inhibit DNA polymerase activity, thus hindering the PCR amplification process. The build-up of organic matter in a freshwater ecosystem can be a consequence of soil run-off from the streamflow and dense vegetation. The accumulated organic matter indirectly inhibits PCR amplification by undergoing non-enzymatic decay during the eDNA extraction process, producing a range of PCR inhibitors, including complex polysaccharides, humic acid, and tannin compounds.¹⁵ In the present study, we were able to develop a thorough PCR amplification protocol to overcome DNA polymerase inhibitors present in the freshwater eDNA samples using an inhibitor-resistance DNA polymerase in combination with a “touchdown approach” in the PCR protocol, under which the annealing temperature is gradually reduced ($\Delta t = -1.5^\circ\text{C}$ per cycle) to maximize the specificity of PCR amplification.

Samples in the present study were collected among three freshwater habitats across the eastern region of the San Francisco Bay Area to represent freshwater ecosystems. Lake Anza serves as a recreational swimming reservoir in Tilden Regional Park, located in Berkeley, California (Figure 1). Lake Anza spans a total surface area of 1.56 square miles with most of its lake area sitting above ground. With significant human interventions, Lake Anza is most frequently visited by swimmers between months of May and September. On the other hand, located on the east side of the Berkeley Hills, San Pablo Reservoir covers a watershed of 23.37 square miles with a water tunnel running underground from the west side of the reservoir for further treatment and distribution to various households (Figure 1). Due to its functionality as a drinking water storage facility, swimming and wading are prohibited at San Pablo Reservoir with restricted allowance for fishing, boating,

and canoeing. Strawberry Creek covers a total area of 1.8 square miles with 60% of its water body flowing entirely underground.^{13,14} The hydrologic structure of the aboveground portion diverges into two affluents, the North Fork and the South Fork, through the University of California, Berkeley, campus (Figure 1).

Although it is difficult to directly compare the intensity and susceptibility of anthropogenic impacts at each of the three samples, Lake Anza is expected to be particularly prone to pollutants resulting from human interventions due to its immediate exposure to organic contaminants, including human feces, urine, and dead skin cells. San Pablo Reservoir is expected to have the lowest level of water pollution due to its rigorous prohibitions on various recreational water activities as a drinking water storage unit. It is uncertain how the water quality of Strawberry Creek will compare to samples from the other static water bodies due to the lack of prior study on correlations between stream flow rates and ecosystem health under the influence of human activities.

In this study, we hypothesize that eDNA metabarcoding via a high-throughput sequencing platform will effectively monitor biodiversity and water pollution under freshwater ecosystems in a more practical and cost-efficient manner over the traditional morphological approach. This is partly motivated by a related prior study by Šigut et al., looking at the performance of DNA metabarcoding and morphological approach in detecting interspecies interactions, specifically host-parasitoid interactions.⁵¹ Particularly, no significant differences in accuracy were found between the two methods regarding success of taxonomic identification.⁵¹ To evaluate the viability of our method, we sampled across three freshwater ecosystems in Berkeley, California. We first looked for the presence of the frequently found predator-prey relationships between



Figure 1: Geographical locations of Strawberry Creek, Lake Anza, and San Pablo Reservoir and the proximal habitats on the eastern region of the San Francisco Bay Area. Courtesy of the Google Maps, maps.google.com

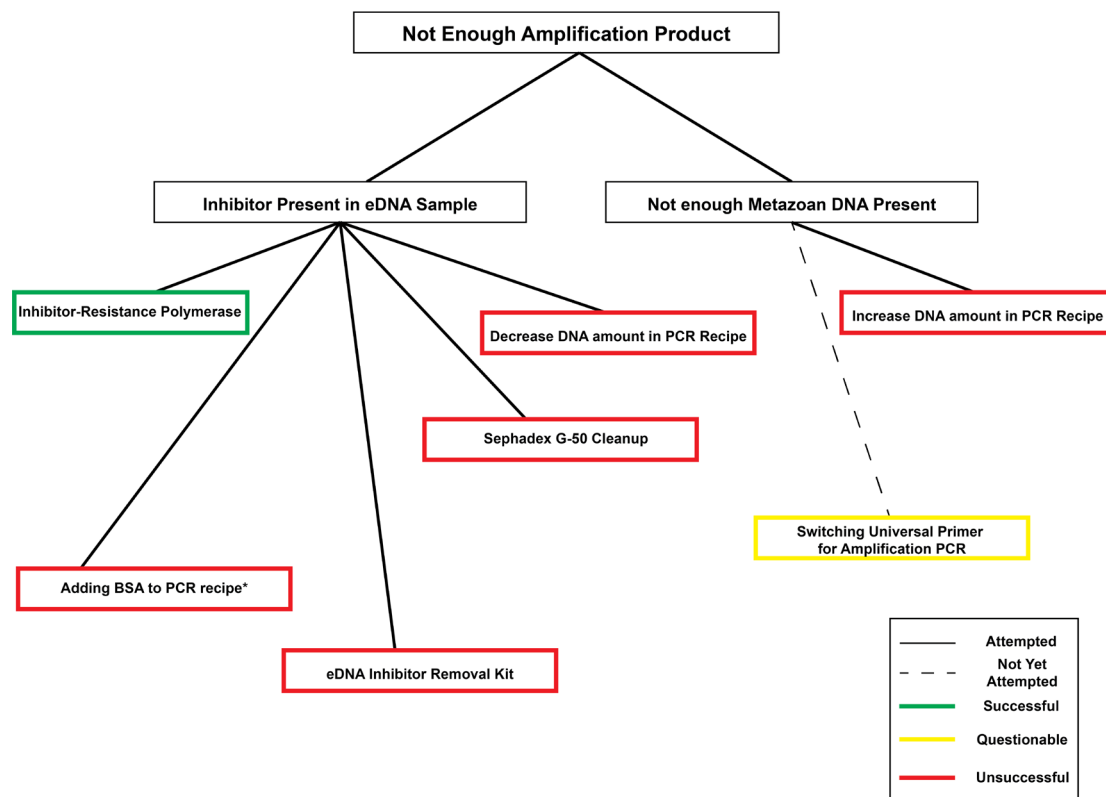


Figure 2: eDNA PCR Amplification Inhibitor Troubleshooting Guide: Different troubleshooting techniques were attempted depending on what is observed from the agarose gel electrophoresis. Leaf nodes are grouped by successful (green), questionable (yellow), and unsuccessful (red) attempts to rescue the inhibited eDNA samples during amplifications.

arthropods and rotifers, which would reveal whether our method is capable of capturing the interspecies interactions within a habitat. In addition, we were curious whether our method is capable of measuring the quality of water within a habitat by performing the traditional bioindicator experiment *in silico*. Thus, we computed the community pollution value across all three habitats to observe whether the trends in water pollution level agree with what we hypothesized based on the stringency of regulations on various recreational water activities at each sampling site. Further, through trial and error, we hereby also present a thorough PCR amplification protocol to overcome DNA polymerase inhibitors present in the freshwater eDNA samples (Figure 2). In closing, we conducted

a cost-benefit analysis to evaluate whether high-throughput eDNA metabarcoding is indeed more efficient in efforts of both time and cost compared to the stereomicroscopic approaches as expected (Figure 3).

MATERIALS AND METHODS

eDNA Sampling and Extraction

The samples were collected in three freshwater ecosystems in Berkeley, California. Lake Anza, San Pablo Reservoir, and downstream of Strawberry Creek were independently sampled

Figure 3: Cost and time estimates for identification of bioindicators by the bioinformatic and morphological approaches. Unit of time: minutes (min), unit of currency: U.S. dollars (\$)

Normalized Cost per sample	Bioinformatic	Morphological	Notes
Sample collection (min)	0.004198153	10	
DNA extraction (min)	0.125944584	-	
DNA amplification (min)	0.399874055	-	
Bioindicator identification (min)	-	240	
NGS library preparation (min)	0.251889169	-	
Sequencing cost (\$)	\$3	-	
Bioinformatics (min)	18.13602015	-	
Labor cost (\$)	\$6.43	\$85.00	\$0.34 per minute for average lab technician
Total time (min)	18.91792611	250	
Total cost (\$)	\$10	\$85.00	

to represent different levels of human interventions and distinct stream flow rates. Lake Anza and San Pablo Reservoir represent standing bodies of water while Strawberry Creek is distinguished by its fast-moving streamflow. Two 1 L samples were taken from one sampling site at each of the three water bodies, totaling up to six samples (2x San Pablo Reservoir, 2x Lake Anza, 1x Strawberry Creek, 1x distilled water as blank Control Sample). Six samples were kept isolated throughout the complete procedure of sample processing to avoid the contaminations of omnipresent human DNA from the researcher into the samples. eDNA samples were isolated from water samples using Sterivex GP 0.22 µm filters (Catalogue Number: SVGPL10RC) and BD Luer-Lok 50 mL syringes (Catalogue Number: 13-689-8) following protocol described from a similar prior study on eDNA monitoring of freshwater crayfish with minor modifications.¹ Extractions of DNA from the filters were performed with Qiagen DNeasy Blood & Tissue kit (Catalogue Number: 69504).

mtDNA-CO1 Amplification from eDNA Samples

Samples of extracted DNA were amplified using the universal mitochondrially encoded cytochrome c oxidase I (mtDNA-CO1) primers (Forward Primer [mlCO1intF]: GGWACWG-GWTGAACWGTWTAYCCYCC, Reverse Primer [jgHCO2198]: TANACYTCNGGRTGNCCRAARAAYCA) for metabarcoding animal and protist biodiversity. PCR amplification was performed in a total volume of 20 µl with 4 µl of 10 µM of each forward and reverse mtDNA-CO1 primers, 4 µl of Invitrogen Platinum GC Enhancer, 10 µl of Invitrogen Platinum II Hot-Start PCR Master Mix (2X) (Catalogue Number: 13000013), and 2 µl of the template DNA. To optimize the universality of our metabarcoding primers, we selected the degenerate mtDNA-CO1 primer set with some of the primer positions having multiple possible bases. The degeneracy of a primer sequence is defined by the number of unique sequence combinations it encodes.²⁶ Due to the degeneracy that arises from our primer sequences, PCR profile was modeled after a “touchdown approach” following supplied protocol of mtDNA-CO1 primers to minimize amplification of non-specific fragments with minor modifications.^{24,26} Under this touchdown approach of PCR reaction, the annealing temperature is gradually reduced ($\Delta t = -1.5^\circ\text{C}$ per cycle) to maximize the specificity of PCR amplification. We proceeded with 15min of denaturation at 95°C , followed by 13 initial cycles: 30s of denaturation at 94°C , annealing for 90s at 69.5°C ($\Delta t = -1.5^\circ\text{C}$ per cycle) and 90s of extension at 72°C , followed by another 40 subsequent cycles: 30s of denaturation at 94°C , annealing for 60s at 50°C and 90s of extension at 72°C , terminated by 10 min of extension at 72°C and constant storage at 4°C . The rationale behind the touchdown approach of the PCR profile can be illustrated as follows. At higher annealing temperature, molecules vibrate at a higher velocity, which makes it more difficult for primers to anneal to regions with low affinity and high number of mismatches, thus increasing the specificity of the annealing process. The purpose of dropping the annealing temperature per cycle is to make sure most of the primers in solution are properly incorporated into the PCR amplification reaction. With a lower annealing temperature, primers can now bind to regions of the template DNA with lower affinity despite the mis-

matches, increasing the likelihood of annealing. Performance of the amplification was evaluated by 1.5% agarose gel electrophoresis. A single clear band per lane indicates success of the PCR amplification, whereas the absence of band indicates failure of amplification of any product. Presence of multiple bands per lane represents amplification of mtDNA-CO1 gene along with non-specific markers.

NGS Library Preparation

Library preparation for Next-Generation Sequencing (NGS) was performed using KAPA Hyper Prep Kit (Catalogue Number: 07962363001). Individual amplicons of mtDNA-CO1 barcode were processed with end repairing and A-tailing, followed by adapter ligation, concluded by two cycles SPRI magnetic bead cleanup. Quality control for absence of contaminants was performed on both Qubit fluorometer and Agilent Bioanalyzer using 1 µl of final amplicon library product.

NGS Library Sequencing

The amplicon products were pooled into the library in equimolar concentration before performing paired-end sequencing on two separate runs of the Illumina MiSeq v3 Platform at the QB3 Vincent J. Coates Genomics Sequencing Laboratory (GSL) at the University of California, Berkeley.

Taxonomy Assignment

Taxonomy of the eDNA reads from NGS library sequencing was assigned using the Anacapa Toolkit, which monitors sample metabarcoding in three stages.⁷ First, the toolkit builds reference sequence libraries from Creating Reference libraries Using existing tools (CRUX). In the second stage, Dada2 was applied for quality control and assignment of Amplicon Sequence Variants (ASV), under which merging and dereplication are performed based on overlaps and sequence variations with elimination of chimeric reads. Lastly, the processed reads are assigned with taxonomy by running Bowtie 2 specific Bayesian Least Common Ancestor (BLCA).

Read Count Normalization

Batch effects occur when technical differences associated with the lab workflow contribute to the variations in experimental outcome. To avoid batch effects across different samples, taxonomic read counts are normalized by min-max normalization before relative abundances between species were computed. The feature scaling is given by the following formula:

$$X_{std} = \frac{X - X_{min}}{X_{max} - X_{min}}$$

$$X_{scaled} = X_{std} * (\max - \min) + \min$$

$$\text{where feature_range} = (\min, \max)$$

Community Pollution Value

Community Pollution Value (CPV), or the biotic index, is a metric measuring the tolerance of organisms residing within an ecosystem to pollution. The tolerance of each organism is measured by the Species Pollution Value (SPV) on a 0 to 10 scale, which is empirically determined with 0 being most sensitive to pollution and 10 being

most tolerant to pollution.²⁰ The organisms with SPVs assigned are also known as bioindicator species as they reflect the ecosystem health. The CPV for each sample is calculated by the sample mean of SPV with the following formula:

$$CPV = \frac{\sum_{i=1}^n SPV_i}{n}$$

where SPV_i = SPV of the i^{th} bioindicator species

n = number of species in a community

Cost-Benefit Analysis

A cost-benefit analysis for detection of bioindicator was conducted by replicating the procedure described by Fernández et al. in a similar study, looking at metabarcoding of eDNA samples collected from Nalón River, Spain.⁴³ The duration of each stage of the experiment is recorded following the methods as previously explained. The approximated runtime for bioinformatic analysis accounts for the entire pipeline of the Anacapa Toolkit. The labor cost is estimated as the total allocated time multiplied by the average salary of a laboratory technician in California, which is equivalent to \$0.34 per minute at the time of writing. The cost per bioinformatic sample is normalized by the number of unique species assigned in each sample to make comparable to the nature of the morphological approach. The estimated time per morphological sample was reproduced based on the data presented by Fernández et al.⁴³

RESULTS

In total, a grand total of 2,390 species were identified across all six samples. Based on the taxonomic assignments from amplicon-based next-generation sequencing, *Skistodiaptomus pallidus* and *Keratella cochlearis* are found to be the predominant species in Lake Anza (Figure 4). *Skistodiaptomus pallidus* is a copepod commonly found in freshwater ecosystems. It has been known to be an efficient omnivorous predator, mainly preying on specific rotifers and microzooplankton populations.^{11,35,36,41} *Keratella cochlearis* is a freshwater planktonic organism, particularly rotifers, species from the phylum Rotifera characterized by the wheel-like ciliated structure at the frontal end of their body.⁵ As previously mentioned, *Keratella cochlearis* is a preferred prey of *Skistodiaptomus pallidus*. Predator-prey interactions like this support the idea that our methodology of monitoring may suggest the interspecies dynamics among living organisms in Lake Anza (Figure 5). Reads from San Pablo Reservoir were primarily assigned to species from the Hypocreales, which is an order of fungi that are characterized by their vibrant colorations and perithecial shape of their spore-producing structures (Figure 4). Some Hypocreales found in our San Pablo Reservoir include *Fusarium*, *Calonectria colhounii*, and *Verticillium nonalfalfae*. Similar predation of arthropods on rotifers was also evident in our San Pablo Reservoir samples between *Skistodiaptomus pallidus* and *Keratella quadrata* (Figure 5). Reads

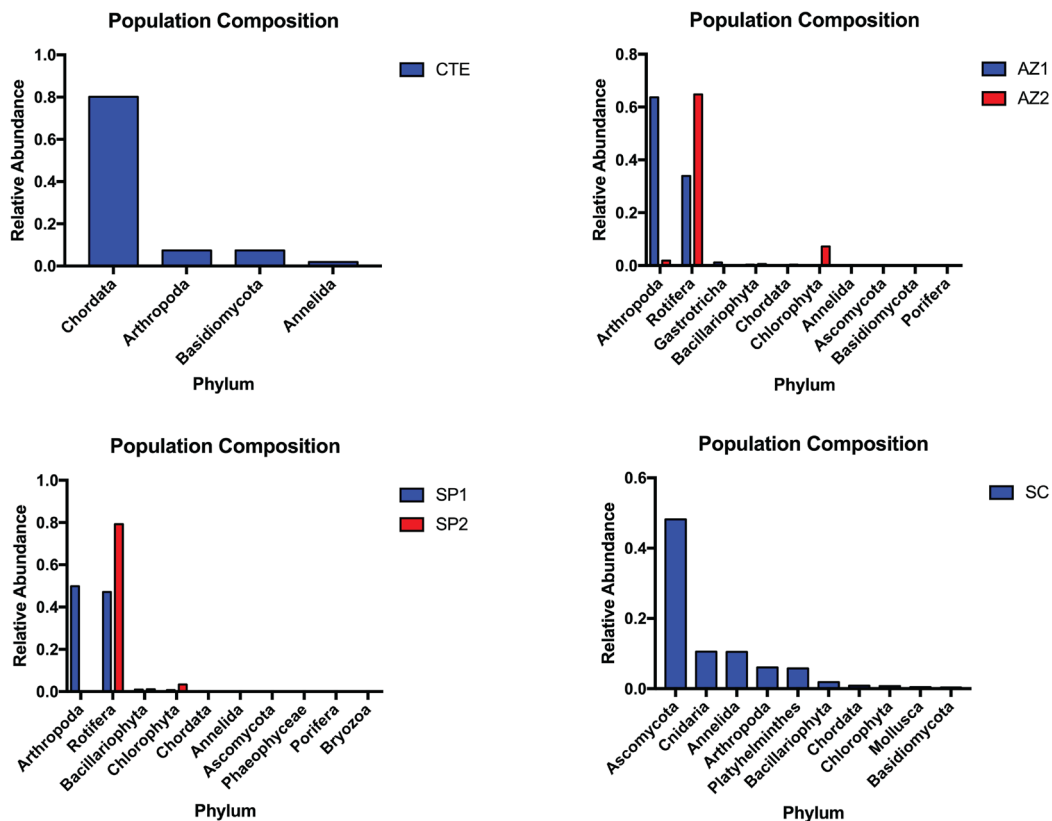


Figure 4: Population composition from each of the six samples (CTE: Control Sample, AZ: Lake Anza, SP: San Pablo, SC: Strawberry Creek). Representative phyla among the 10 largest relative abundances present in the amplicon samples are plotted.

from Strawberry Creek were mainly assigned to descendants of order Ploima, which is another order of rotifers commonly found in freshwater habitats (Figure 4). Unexpectedly, a number of reads was also assigned under phylum Arthropoda (Figure 5). Overall, the high quantities of eDNA assigned to both arthropods and rotifers may be indicative of a potential predator-prey relationship between the two organisms.

In addition to the expected biological interaction between freshwater predators and prey, we also found traces amount of eDNA assigned to species nonnative to the freshwater ecosystem in the Strawberry Creek sample. Some preeminent examples from this group of animals include human (*Homo sapiens*), black rat (*Rattus rattus*), house mouse (*Mus musculus*), fox squirrel (*Sciurus niger*), domestic dog (*Canis lupus*), and racoon (*Procyon lotor*). In particular, these are all land animals commonly found in high densities around the area of urbanization. This suggests amplification of non-target taxa due to the high sensitivity of our methodology and the universality of our primer design.⁴⁸

To evaluate our taxonomical classification quantitatively, we performed assessment of freshwater pollution by calculating community pollution value (CPV) for each of the three sample sites (Figure 6). The calculation of CPV is derived from the species pollution value (SPV) of various protozoan bioindicators present in the water sample. SPV of the protozoan species were computed in a previous study with associations between these species and various chemical parameters.²⁰ Water sample from Strawberry Creek showed the highest CPV among all samples, indicating the intensity of pollution at the sample collection site (Figure 6). Succeeding, the water quality of Lake Anza and San Pablo Reservoir also possessed a CPV slightly above the threshold value of 3.95, indicating a severely polluted water (Figure 6). It is important to note that the samples from San Pablo Reservoir were taken at the source water, thus do not directly reflect the water quality of the distributed drinking water after proceeding treatment.

A related prior study by Šigut et al. on performance of DNA metabarcoding and morphological approach in detecting interspecies host-parasitoid interactions has found no significant differences in accuracy between the taxonomic identification made by the two approaches.⁵¹ Here, we conducted a cost-benefit analysis from an economic standpoint to directly compare the effectiveness of high-throughput eDNA metabarcoding and stereomicroscopic approaches in the detection of bioindicators. As hypothesized, the bioinformatic approach requires less effort in terms of both cost and time (Figure 3). To make the analysis as practical as possible, we also accounted for the labor cost per sample by calculating the product of total allocated time per sample multiplied by the average salary for a laboratory technician in California, which corresponds to \$0.34 per minute at the time of writing. Without loss of generality, the total time per morphological sample was reproduced from data presented by Fernández et al. in a previous study on freshwater ecosystems of Nalón River, Spain.⁴³ As species need to be sorted individually per well sample in a morphological experiment, the cost per sample was normalized by the number of species assigned in each bioinformatic sample to make the two approaches more comparable. Across the board, the normalized cost per bioinformatic sample is estimated at 18.918 minutes and

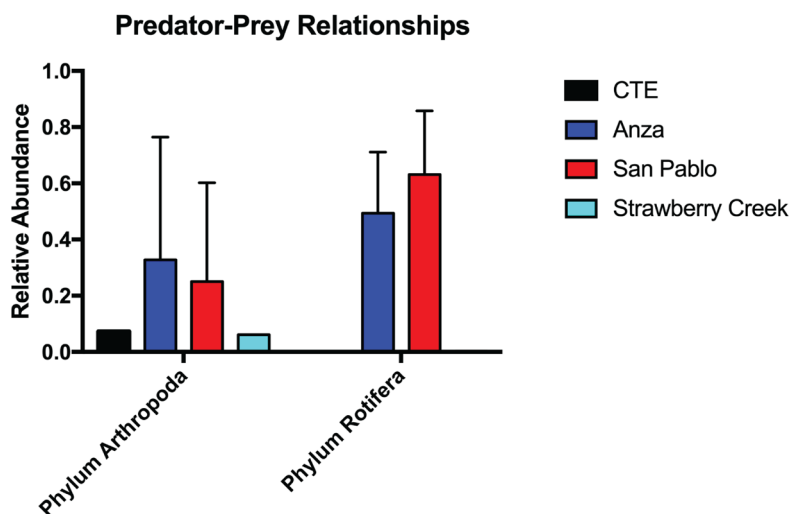


Figure 5: Predator-prey relationships between arthropods and rotifers found in Lake Anza and San Pablo Reservoir. Phylum Arthropoda: [CTE: 0.073, Anza: 0.328, San Pablo: 0.250, Strawberry Creek: 0.061]. Phylum Rotifera: [CTE: 0.000, Anza: 0.493, San Pablo: 0.632, Strawberry Creek: 0.003].

\$10 for total time and cost respectively (Figure 3). In contrast, each stereomicroscopic sample requires about 250 minutes and \$85 in terms of total time and cost, driving up the cost by at least 10 folds (Figure 3).

DISCUSSION

In this paper, we addressed the viability of high-throughput amplicon sequencing-based eDNA metabarcoding for monitoring biodiversity and water pollution under freshwater ecosys-

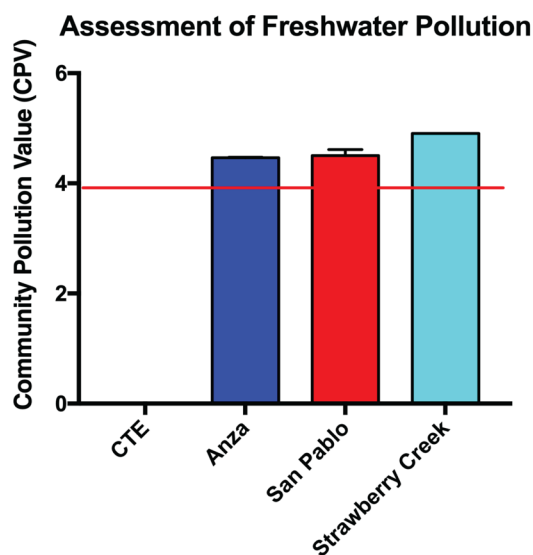


Figure 6: Assessment of freshwater pollution from each of the six samples based on biotic index of species pollution value (SPV) and community pollution value (CPV). Red line indicates the CPV threshold of 3.95 for severely polluted water. CPV: [CTE: 0.000, Anza: 4.465, San Pablo: 4.504, Strawberry Creek: 4.906].

tems. From the taxonomical assignments of bioindicator species through bioinformatic construction of phylogenetic trees, we were able to calculate the biotic indices for freshwater samples computationally in a more practical and cost-efficient manner. This is to be put in contrast with the qualitative and labor-intensive approach of manually identifying bioindicator species present in the water sample using stereomicroscope and morphological dichotomous keys, a method prone to confirmation bias and human error.³⁸ It would be challenging to replicate and capture the same scale of information regarding biodiversity with the traditional observation methods. However, with the rapid development of large-scale data collection, the importance of quality over quantity should still be emphasized. In other words, we must proceed with stringent quality control to ensure the integrity of reads and taxonomic assignments, including the use of Dada2 in taxonomy assignment and proper normalization of read counts as previously described. Despite promising results demonstrated in a related prior study by Šigut et al., further juxtaposition of the accuracy between the two freshwater biodiversity surveying procedures await to be carried out with additional experiments in future study.⁵¹

In the process of amplifying the amount of DNA sufficient enough to perform metabarcoding, we also developed a troubleshooting guide for high-fidelity and high-throughput PCR amplification of eDNA consisting of several inhibitors for common DNA polymerase, including but not limited to complex polysaccharides, heme, humic acid, and urea.^{2,21,28,32,37,39} We attempted numerous different variants in our PCR profile as well as parameters in our PCR amplification step (Figure 2). These attempts were differentiated by what was observed from the agarose gel electrophoresis (Figure 2). Overall, our combining use of GC enhancer and inhibitor-resistance polymerase with “touchdown approach” of PCR profile was able to obtain high yields of amplicon product on universal mtDNA-CO1 barcode primers.

Observed from the calculations of CPV among each of the three communities, our experimental outcome was somewhat consistent with what we hypothesized. Primarily, Lake Anza suffers from the severe magnitude of anthropogenic pollution through its direct exposure to organic contaminants due to its recreational purposes (Figure 6). Identified as most severely polluted water body, sample from Strawberry Creek downstream was tested to have a CPV that is above the extremity of Lake Anza water quality (Figure 6). The poor water quality from Strawberry Creek is likely a result of lenient management and recent urbanization along with historical improper pipeline connections and illicit dumping.^{13,14} Unexpectedly, the strict regulations of water quality at the San Pablo Reservoir were not reflected in our assessments of CPV, locating right around the same level as Lake Anza water quality (Figure 6). We suspect this to be a consequence of our current technique for sampling eDNA from freshwater ecosystems. Assuming DNA molecules are unevenly distributed due to preferential direction of water currents, our approach to replicate sampling from the same site might have been attributed to the unbalanced load of DNA in each collection of water samples. Our current replication procedure aims to keep replicate samples completely isolated and free of merging throughout the entire process of extraction, amplification, and sequencing to avoid the contaminations of omnipres-

ent human DNA from the researcher into the samples. Merging is an extraction technique where X replicate samples from the same sampling site are first combined and mixed thoroughly into one container before randomly re-sampling X “pseudo-replicates” to ensure even distribution of eDNA in each replicate sample collected.⁴⁷ Instead of making X isolated collections of 1 L sample to make X replicate samples, perhaps a feasible approach of replication to attempt in future experiments is to make one merged collection of X L sample that is subsequently filtered through X filters to avoid bias in the sample collection process.

Computationally, our current usage of mtDNA-CO1 barcode is restrained to preferably monitoring animal diversity with limited coverage on protist diversity. We aim to incorporate several other universal primer pairs as well to expand the capability of this cost-efficient biodiversity monitoring solution to metabarcoding studies of other communities of life, including the internal transcribed spacer region of rRNA (rRNA-ITS) for fungi and ribulose-1,5-bisphosphate carboxylase/oxygenase (Rubisco)-encoding RbcS gene for plants.^{30,40}

Through capturing of the predator-prey relationships between arthropods and rotifers and measurement of community pollution value across three freshwater ecosystems, we have demonstrated the effectiveness of high-throughput eDNA metabarcoding as a novel method for not only documenting the biodiversity of a freshwater ecosystem but also completing the experiment in a more effective manner as compared to the traditional morphological approach. Currently, it is of high demand to establish alternative methods capable of effective freshwater assessments due to decreasing numbers of taxonomic experts in the field of morphological identification.^{49,50}

Overall, it is unrealistic to expect either approach to make absolute inferences on species abundance under a freshwater ecosystem given the uneven distribution of matter in the water. However, high-throughput eDNA metabarcoding is capable of capturing meaningful information regarding the interspecies dynamics and monitoring quality of the ecosystem as it changes over time. More importantly, compared to the traditional morphological approach, the bioinformatic approach for documenting biodiversity and bioindicator monitoring no longer succumbs to the observer errors or limitations of the dichotomous keys. In addition, as suggested by the cost-benefit analysis, high-throughput eDNA metabarcoding emerges as a dependable, cost- and time-efficient solution to approach large-scale biodiversity documentation (Figure 3).

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